Prolog Logicbase

the fact checker behind Natural Knowledge
Book 4 of *Natural Knowledge - a collection*

Freddi Van Gogh

Prolog Logicbase: the fact checker behind Natural Knowledge

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First published 2026.

ISBN 979-8-14-019782-7 (paperback)

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How this book was made

The ideas, the questions and the decisions in this book are the author's. The author wrote the notes, the riffs and the rough drafts, and set out a recipe for each part. AI assistants turned that material into prose, checked it against sources and against the series' fact checker, and raised questions where the notes were wrong or unclear. The author answered them, made the calls, and edited the result.

In the terms publishers now use, the text is AI-generated: an AI drafted it, working from the author's own notes, research and writing as its source material. The cover was also made with an AI assistant, drawn in code.

A visual companion

Want to see these concepts in games and fun video clips? A collection of these and some more detail on visual subjects are further explored at:

https://freddi-van-gogh.github.io/Natural_Knowledge_git/data-vis-compendium/

The code can be found more quickly in the source repository, which also holds the books as PDFs:

https://github.com/freddi-van-gogh/Natural_Knowledge_git

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The fact checker

Tests the manuscript for statements that contradict each other, arithmetic that doesn't add up, and arguments that claim more than they've shown.

```
python3 factcheck/run.py          # needs SWI-Prolog: sudo apt install
    swi-prolog
```

The commands in this book are run from the source repository, which holds the text the checker reads and the Python scripts that run it, as well as the Prolog files printed here:

https://github.com/freddi-van-gogh/Natural_Knowledge_git

In one paragraph, for a ten-year-old

Prolog is a computer language for working things out. Most programs are recipes: do this, then this. Prolog is more like a detective with a notebook. You give it facts ("chapter 9 says the giraffe's nerve is four and a half metres long") and rules ("if two chapters give different numbers for the same thing, that's a problem"). Then you ask it a question ("is there a problem anywhere?") and it checks every fact against every rule, without getting bored, and tells you everything it finds. Tell it that all dogs are mammals and Rex is a dog, and it works out that Rex is a mammal by itself.

How it works

A Prolog program is a set of facts and rules (Horn clauses). Prolog answers a query by searching for a proof. It *unifies* the query's variables with facts and rule heads, and it *backtracks* through every way the rules can be satisfied. Here every solution is collected, so the search is exhaustive. It also assumes a *closed world*: anything neither stated nor derivable counts as false. So a step the book never supports shows up as missing, instead of being filled in the way a sympathetic reader fills it in.

The facts come in three layers:

Layer	Where	How it's made
The text	`generated/text.pl`	Automatic. Every sentence of both editions, with its numbers, dates, named authorities, chapter cross-references, certainty and hedge words, and its tier where the text marks one (HS boxes; the full edition's italic author passages).
The companion	`generated/glossary.pl`, `generated/notation.pl`	Automatic. *Modern Alchemy*'s 119 concepts, and its worked chains as relations: → produces, ⊂ kind of, ≈ analogous, ↑ amplifies, ↓ constrains. Plus the "whole arc", one step per arrow across its strands.
The propositions	`claims.pl`, `claims/*.pl`	By hand. What key sentences *mean*: values, tier, sources, dependencies, arithmetic, dates. Each claim quotes its sentence verbatim, and `generated/anchors.pl` records whether that quote is still in the book.

`generated/` is rebuilt on every run and is not committed. `refs.pl` there is built from the reference list in chapter 19 of *Natural Knowledge*.

What it reports

Severity	Checks
error	the same quantity stated incompatibly (also across editions) · arithmetic the text gets wrong · events out of order, or a stated gap that the dates contradict · an in-text citation year with no matching reference · a source missing from the reference list · a cross-reference to a chapter that doesn't exist · a heading that promises *n* items and lists a different number · circular support · a claim whose quote has left the text · a duplicate claim id or contradictory event dates
blocker	a settled claim resting, directly or through a chain, on a contested or author's-own premise · a settled claim resting on an acknowledged gap · a step in the companion's whole arc with no worked chain behind it (a path of up to four chains counts, and the path is printed under info) · an analogy (≈) also used as a mechanism (→) · a chain that skips a substrate (chemical ⟶ cultural) in one arrow
warning	numbers that drift between related passages · certainty words ("always", "proves") inside contested or author-marked passages · settled empirical claims with no source · sources whose support is not yet checked († in the reference list) · tier mismatches across editions · boxes that claim more scope than the edition's own rule allows
info	gaps the book itself acknowledges, and arc steps it describes as descriptive rather than mechanistic · coverage

Exit status is 1 when there are errors or blockers, and 3 when Prolog could not load a file. That second case would otherwise go unnoticed: a stray `/*` inside a comment opens a nested comment in SWI-Prolog and silently swallows the rest of the file.

Adding a claim

Find sentences that nothing covers yet:

```
python3 factcheck/extract.py --uncovered book:14-17 book-hs:14-16
```

Then write the claim in `claims.pl`, or in a file under `claims/`. The schema is in the header of `claims.pl`, and the existing claims are worked examples:

```
claim(c9_giraffe, book, 9, "about four and a half metres of nerve to span
   a gap of inches", settled, quantitative).
source(c9_giraffe, 'Wedel', 2012).
says(c9_giraffe, giraffe_laryngeal_nerve_path, metres, approx(4.5)).
```

Two rules matter more than the rest:

- **The tier is how the text presents the claim, not how you'd like it to be.** The checker's value is in catching the text overstating itself, and that only works if the encoding is faithful.
- **When both editions state the same fact, use the same subject and property**, so the two get compared.

Check a claims file's quotes without touching anything shared:

```
python3 factcheck/extract.py --check-quotes factcheck/claims/your-file.pl
```

What it cannot do

It cannot establish that anything is *true*. It tests whether the book agrees with itself and with the sources it names, and an error that is consistent everywhere will pass. The automatic layer is literal. It knows that two sentences share a number and four words, not that they are about the same thing, so its drift warnings are prompts for a reader rather than verdicts. And a hand-encoded claim is only as good as its encoding. A wrong tier or a missing dependency hides exactly what the checker exists to find.

The rules

From `factcheck/checks.pl`

Natural Knowledge — fact checker.

Loads the generated facts (what the text says) and claims.pl (what the text's propositions are, structured), then looks for:

```
error    — two statements that cannot both be true, arithmetic the text
           gets wrong, events out of order, citations that disagree with
           the reference list, references to chapters that don't exist
blocker  — a step in an argument that does not follow: a settled
           conclusion resting on a contested premise, circular support,
           a step in the book's arc that no chain supports, an analogy
           used as a mechanism, a chain that jumps tiers without prose
warning  — worth a look: numbers that drift between passages, certainty
           words inside contested or author-marked passages, settled
           claims with no source, sources whose content is unchecked
info     — coverage, and gaps the book itself acknowledges
```

Run through factcheck/run.py, which regenerates the facts first.

```
:- discontiguous check/3.
:- dynamic claim/6, says/4, source/3, depends_on/2, computes/3,
   event_year/3,
           order_claim/3, interval_claim/4, variant_ok/3, gap/3,
   rests_on_gap/2,
           arc_gap/3, arc_descriptive/3, cite_ok/3, about/2, drift_ok/3,
           prose_ok/3, common_knowledge/1, certainty_ok/2.
```

Claims are spread across claims.pl and claims/*.pl. Declaring them multifile is what lets a second file add clauses instead of silently replacing the first file's.

```
:- multifile claim/6, says/4, source/3, depends_on/2, computes/3,
   event_year/3,
             order_claim/3, interval_claim/4, variant_ok/3, gap/3,
   rests_on_gap/2,
             arc_gap/3, arc_descriptive/3, cite_ok/3, about/2,
   drift_ok/3,
             prose_ok/3, common_knowledge/1, certainty_ok/2.
```

```
:- dynamic sentence/6, number/3, year/2, keywords/2, certainty_word/2,
   hedged/1,
          xref/2, cites/3, heading_count/6, section_box/5, chapter/4,
   concept/5,
          substrate_rank/2, chain/6, chain_cycles/1, arc_step/3, ref/6,
   anchor/2.

load_all :-
    prolog_load_context(directory, Dir), !,
    forall(member(F, ['generated/text.pl', 'generated/glossary.pl',
   'generated/notation.pl',
                      'generated/refs.pl', 'generated/anchors.pl',
   'claims.pl']),
          ( atomic_list_concat([Dir, '/', F], Path), consult(Path) )),
    atomic_list_concat([Dir, '/claims/*.pl'], Pattern),
    expand_file_name(Pattern, More),
    forall(member(P, More), consult(P)).
:- load_all.
```

Values

```
interval(N, N, N)           :- number(N), !.
interval(approx(N), L, H)   :- !, L is N * 0.85, H is N * 1.15.
interval(range(L, H), L, H) :- !.
interval(at_least(N), N, inf)  :- !.
interval(more_than(N), N, inf) :- !.
interval(at_most(N), -inf, N)  :- !.

le(-inf, _) :- !.
le(_, inf)  :- !.
le(inf, _)  :- !, fail.
le(_, -inf) :- !, fail.
le(A, B)    :- A =< B.

within(X, V) :- interval(V, L, H), le(L, X), le(X, H).

compatible(V, V) :- !.
compatible(V1, V2) :-
    interval(V1, L1, H1), interval(V2, L2, H2), !,
    le(L1, H2), le(L2, H1).

tier_rank(settled, 3).
tier_rank(contested, 2).
tier_rank(author, 1).
```

Dependencies

```
reaches(A, B) :- reaches(A, B, [A]).
```

```
reaches(A, B, _)    :- depends_on(A, B).
reaches(A, B, Seen) :- depends_on(A, C), \+ memberchk(C, Seen),
   reaches(C, B, [C|Seen]).

where(Id, W) :- claim(Id, Ed, Ch, _, _, _), !, format(atom(W), '~w
   ch.~w', [Ed, Ch]).
where(_, '?').
```

Errors

```
check(error, anchor, M) :-
    anchor(Id, missing), where(Id, W),
    format(atom(M), '~w (~w): its quoted sentence is no longer in the
   text — the claim has drifted from the book', [Id, W]).

check(error, anchor, M) :-
    claim(Id, _, _, _, _, _), \+ anchor(Id, _),
    format(atom(M), '~w: the extractor could not read this claim (check
   its syntax in claims.pl)', [Id]).

check(error, duplicate, M) :-
    claim(Id, E1, C1, Q1, _, _), claim(Id, E2, C2, Q2, _, _), (E1-C1-Q1)
   @< (E2-C2-Q2),
    format(atom(M), 'claim id ~w is used twice (~w ch.~w and ~w ch.~w);
   every claim needs its own id', [Id, E1, C1, E2, C2]).

check(error, chronology, M) :-
    event_year(E, Y1, B1), event_year(E, Y2, B2), Y1 < Y2,
    format(atom(M), 'event ~w is dated ~w (~w) and ~w (~w)', [E, Y1, B1,
   Y2, B2]).

check(error, conflict, M) :-
    says(A, S, P, V1), says(B, S, P, V2), A @< B,
    \+ compatible(V1, V2), \+ variant_ok(A, B, _), \+ variant_ok(B, A,
   _),
    where(A, WA), where(B, WB),
    format(atom(M), '~w: ~w says ~w = ~w, but ~w (~w) says ~w', [S, WA,
   P, V1, B, WB, V2]).

check(error, arithmetic, M) :-
    computes(Id, Expr, Stated), Got is Expr, \+ within(Got, Stated),
   where(Id, W),
    format(atom(M), '~w (~w): the text states ~w, but ~w = ~4g', [Id, W,
   Stated, Expr, Got]).

check(error, chronology, M) :-
    order_claim(Id, E1, E2), event_year(E1, Y1, _), event_year(E2, Y2,
   _), Y1 >= Y2, where(Id, W),
```

```
    format(atom(M), '~w (~w): the text puts ~w (~w) before ~w (~w)', [Id,
   W, E1, Y1, E2, Y2]).

check(error, chronology, M) :-
    interval_claim(Id, E1, E2, Stated), event_year(E1, Y1, _),
   event_year(E2, Y2, _),
    Gap is Y2 - Y1, \+ within(Gap, Stated), where(Id, W),
    format(atom(M), '~w (~w): the text implies ~w years between ~w and
   ~w; the dates give ~w', [Id, W, Stated, E1, E2, Gap]).

check(error, citation, M) :-
    setof(S-Name-Y, cites(S, Name, Y), All), member(S-Name-Y, All),
    once((ref(_, _, _, As, _, _), memberchk(Name, As))),
    \+ (ref(_, _, Y, As2, _, _), memberchk(Name, As2)),
    \+ cite_ok(Name, Y, _),
    sentence(S, Ed, Ch, Line, _, _),
    format(atom(M), '~w ch.~w line ~w: "~w … ~w" — no reference by ~w
   dated ~w', [Ed, Ch, Line, Name, Y, Name, Y]).

check(error, source, M) :-
    source(Id, Name, Y), \+ ref(_, Name, Y, _, _, _), where(Id, W),
    format(atom(M), '~w (~w): cites ~w ~w, which is not in the reference
   list', [Id, W, Name, Y]).

check(error, xref, M) :-
    xref(S, N), sentence(S, Ed, Ch, Line, _, _), \+ chapter(Ed, N, _, _),
    format(atom(M), '~w ch.~w line ~w: refers to chapter ~w, which this
   edition does not have', [Ed, Ch, Line, N]).

check(error, heading, M) :-
    heading_count(Ed, Ch, Line, H, Stated, Found), Found > 0, Stated =\=
   Found,   % 0 = not itemized; nothing to count
    format(atom(M), '~w ch.~w line ~w: "~w" promises ~w, the section has
   ~w', [Ed, Ch, Line, H, Stated, Found]).

check(error, cycle, M) :-
    claim(A, _, _, _, _, _), reaches(A, A), where(A, W),
    format(atom(M), '~w (~w) supports itself through its own
   dependencies', [A, W]).

check(error, concept, M) :-
    about(Id, C), \+ concept(C, _, _, _, _),
    format(atom(M), '~w: refers to companion concept ~w, which does not
   exist', [Id, C]).

check(error, concept, M) :-
    ( chain(N, _, _, L, R, _), append(L, R, Syms) ; arc_step(N, X, Y),
   Syms = [X, Y] ),
    member(Sym, Syms), \+ concept(_, _, Sym, _, _),
```

```
    format(atom(M), 'notation chain ~w uses symbol ~w, which the glossary
   does not define', [N, Sym]).
```

Blockers

```
check(blocker, tier, M) :-
    claim(A, _, _, _, TA, _), reaches(A, B), claim(B, _, _, _, TB, _),
    tier_rank(TA, RA), tier_rank(TB, RB), RA > RB, where(A, WA), where(B,
   WB),
    format(atom(M), '~w (~w, ~w) rests on ~w (~w, ~w): the conclusion is
   stated more firmly than its premise', [A, WA, TA, B, WB, TB]).

check(blocker, gap, M) :-
    claim(A, _, _, _, settled, _),
    ( rests_on_gap(A, G) ; reaches(A, B), rests_on_gap(B, G) ), gap(G,
   Desc, _), where(A, W),
    format(atom(M), '~w (~w) is stated as settled but rests on an open
   gap: ~w', [A, W, Desc]).
```

An arc step X ⟶ Y is supported when worked “produces” chains lead from X to Y. The arc compresses chains, so a path is allowed, capped at four links so that generic concepts cannot connect everything to everything. The shortest path is reported (info), which keeps the support auditable rather than assumed.

```
step(X, Y, N) :- chain(N, _, produces, L, R, _), memberchk(X, L),
   memberchk(Y, R).

arc_path(X, Y, Max, Path) :- between(1, Max, D), path_of(D, X, Y, [X],
   Path), !.
path_of(1, X, Y, _, [N]) :- step(X, Y, N).
path_of(D, X, Y, Seen, [N|Rest]) :- D > 1, step(X, Z, N), \+ memberchk(Z,
   Seen),
    D1 is D - 1, path_of(D1, Z, Y, [Z|Seen], Rest).

arc_supported(X, Y) :- arc_path(X, Y, 4, _).

formula(N, F) :-
    chain(N, _, _, [X], R, _), chain_cycles(N), !, atomic_list_concat(R,
   ' + ', RS),
    format(atom(F), '~w ↺ ~w → ~w', [X, X, RS]).
formula(N, F) :-
    chain(N, _, _, L, R, _), atomic_list_concat(L, ' + ', LS),
   atomic_list_concat(R, ' + ', RS),
    format(atom(F), '~w → ~w', [LS, RS]).

check(blocker, arc, M) :-
```

```
    arc_step(I, X, Y), \+ arc_supported(X, Y), \+ arc_gap(X, Y, _), \+
   arc_descriptive(X, Y, _),
    concept(_, TX, X, _, _), concept(_, TY, Y, _, _),
    format(atom(M), 'step ~w of the whole arc, ~w → ~w (~w → ~w), has no
   worked chain behind it and is not declared a gap', [I, X, Y, TX, TY]).

check(blocker, analogy, M) :-
    chain(N1, _, analogous, [X], [Y], _),
    ( step(X, Y, _) ; step(Y, X, _) ),          % direct only: a path is
   not a mechanism claim
    format(atom(M), 'chain ~w marks ~w ≈ ~w as analogy only, but another
   chain treats one as producing the other', [N1, X, Y]).
```

The notation's rule: "Crossing a tier boundary demands prose. A chain from CH to MC has smuggled in an analogy." What changes across a boundary is the substrate — chemical, biological, cultural — so tiers are grouped by substrate. PH, CH and PL are one physical substrate; a chain that stays inside it is chemistry throughout.

```
substrate_group(ph, physical). substrate_group(ch, physical).
   substrate_group(pl, physical).
substrate_group(bi, biological). substrate_group(mc, cultural).
group_rank(physical, 1). group_rank(biological, 2). group_rank(cultural,
   3).
group_of(Sym, G, R) :- concept(_, _, Sym, T, _), substrate_group(T, G),
   group_rank(G, R).

check(blocker, tier_crossing, M) :-
    chain(N, Sec, produces, L, [Out], _), \+ prose_ok(L, Out, _),
    group_of(Out, GO, RO), member(In, L), group_of(In, GI, RI), abs(RO -
   RI) >= 2,
    format(atom(M), 'chain ~w (~w): ~w (~w) → ~w (~w) skips a substrate in
   one arrow — the notation calls this a smuggled analogy', [N, Sec, In,
   GI, Out, GO]).

check(warning, tier_crossing, M) :-
    chain(N, Sec, produces, L, [Out], _), \+ prose_ok(L, Out, _),
    group_of(Out, GO, RO), member(In, L), group_of(In, GI, RI), abs(RO -
   RI) =:= 1,
    format(atom(M), 'chain ~w (~w): ~w (~w) → ~w (~w) crosses a substrate;
   say where the prose that carries it is (prose_ok/2)', [N, Sec, In, GI,
   Out, GO]).
```

Warnings

```
check(warning, drift, M) :-
    number(S1, V1, U), U \== none, keywords(S1, K1),
```

```
    number(S2, V2, U), S1 @< S2, V1 =\= V2, keywords(S2, K2),
    Diff is abs(V1 - V2) / max(abs(V1), abs(V2)), Diff > 0.15,
    intersection(K1, K2, Common), length(Common, NC), NC >= 4,
    sentence(S1, E1, C1, L1, _, T1), sentence(S2, E2, C2, L2, _, T2),
    \+ (drift_ok(F1, F2, _), (sub_string(T1, _, _, _, F1), sub_string(T2,
   _, _, _, F2) ; sub_string(T1, _, _, _, F2), sub_string(T2, _, _, _,
   F1))),
    format(atom(M), '~w ~w (~w ch.~w l.~w) vs ~w ~w (~w ch.~w l.~w),
   shared words ~w', [V1, U, E1, C1, L1, V2, U, E2, C2, L2, Common]).

check(warning, certainty, M) :-
    certainty_word(S, W), sentence(S, Ed, Ch, Line, T, Text),
   memberchk(T, [contested, author]),
    \+ (certainty_ok(Frag, _), sub_string(Text, _, _, _, Frag)),
    sub_string(Text, 0, 90, _, Start),
    format(atom(M), '~w ch.~w line ~w: "~w" inside a passage marked ~w —
   "~w…"', [Ed, Ch, Line, W, T, Start]).

check(warning, unsourced, M) :-
    claim(Id, _, _, _, settled, Kind), memberchk(Kind, [empirical,
   quantitative, historical]),
    \+ source(Id, _, _), \+ common_knowledge(Id),
    \+ depends_on(Id, _),          % a claim built on other encoded
   claims inherits their support
    where(Id, W),
    format(atom(M), '~w (~w): a settled ~w claim with no source attached
   and no encoded premise', [Id, W, Kind]).

check(warning, unchecked_source, M) :-
    source(Id, Name, Y), ref(_, Name, Y, _, _, dagger), where(Id, W),
    format(atom(M), '~w (~w) rests on ~w ~w, whose support for it is not
   yet checked (†)', [Id, W, Name, Y]).

check(warning, tier_mismatch, M) :-
    says(A, S, P, _), says(B, S, P, _), A @< B,
    claim(A, _, _, _, TA, _), claim(B, _, _, _, TB, _), TA \== TB,
   where(A, WA), where(B, WB),
    format(atom(M), '~w/~w is ~w in ~w but ~w in ~w', [S, P, TA, WA, TB,
   WB]).

check(warning, tier_marking, M) :-
    claim(Id, Ed, Ch, Q, Tier, _), Tier \== settled,
    prefix40(Q, Q40),
    sentence(_, Ed, Ch, _, ST, Text), sub_string(Text, _, _, _, Q40),
    ST == unmarked, Ed == 'book-hs',
    format(atom(M), '~w: encoded as ~w, but the HS text shows it unboxed
   (which the HS edition says means settled)', [Id, Tier]).

check(warning, box_scope, M) :-
```

```
    section_box(Ed, Ch, Line, Tier, Heading),
    format(atom(M), '~w ch.~w line ~w: a ~w box claims to cover its whole
   section ("~w"), but the edition''s stated rule is that a box covers
   only what is inside it', [Ed, Ch, Line, Tier, Heading]).
```

Info

```
check(info, arc, M) :-
    arc_step(I, X, Y), arc_path(X, Y, 4, Path),
    findall(F, (member(N, Path), formula(N, F)), Fs),
   atomic_list_concat(Fs, '; then ', Via),
    format(atom(M), 'arc step ~w, ~w → ~w, is carried by: ~w', [I, X, Y,
   Via]).

check(info, gap, M) :-
    arc_step(I, X, Y), \+ arc_supported(X, Y), arc_gap(X, Y, G), gap(G,
   Desc, Where),
    format(atom(M), 'arc step ~w, ~w → ~w: unsupported, and acknowledged
   by the book (~w): ~w', [I, X, Y, Where, Desc]).

check(info, gap, M) :-
    arc_step(I, X, Y), \+ arc_supported(X, Y), arc_descriptive(X, Y,
   Where),
    format(atom(M), 'arc step ~w, ~w → ~w: descriptive rather than
   mechanistic, as the book itself says (~w)', [I, X, Y, Where]).

prefix40(Q, P) :- string_length(Q, L), ( L =< 40 -> P = Q ; sub_string(Q,
   0, 40, _, P) ).

covered(S) :- sentence(S, Ed, Ch, _, _, Text), claim(_, Ed, Ch, Q, _, _),
              prefix40(Q, Q40), sub_string(Text, _, _, _, Q40), !.

stats :-
    aggregate_all(count, sentence(_, book, _, _, _, _), SF),
    aggregate_all(count, sentence(_, 'book-hs', _, _, _, _), SH),
    aggregate_all(count, (sentence(S, _, _, _, _, _), once((number(S, _,
   _) ; year(S, _)))), SN),
    aggregate_all(count, (sentence(S, _, _, _, _, _), once((number(S, _,
   _) ; year(S, _))), covered(S)), SC),
    aggregate_all(count, claim(_, _, _, _, _, _), NC),
    aggregate_all(count, depends_on(_, _), ND),
    aggregate_all(count, concept(_, _, _, _, _), NK),
    aggregate_all(count, chain(_, _, _, _, _, _), NCh),
    aggregate_all(count, ref(_, _, _, _, _, _), NR),
    aggregate_all(count, (sentence(_, _, _, _, T, _), T \== unmarked),
   NT),
    Pct is round(100 * SC / max(SN, 1)),
    Tot is SF + SH,
```

```
    format('~nCaptured: ~w sentences (full ~w, HS ~w), ~w carrying
   numbers or dates, ~w in explicitly tier-marked passages.~n', [Tot, SF,
   SH, SN, NT]),
    format('Encoded:  ~w structured claims, ~w dependencies; ~w companion
   concepts; ~w notation chains; ~w reference facts.~n', [NC, ND, NK,
   NCh, NR]),
    format('Coverage: ~w of ~w number- or date-bearing sentences (~w%)
   are tied to a structured claim.~n', [SC, SN, Pct]).
```

Report

```
severity_order([error, blocker, warning, info]).

report(Max) :-
    severity_order(Sevs),
    forall(member(Sev, Sevs),
           ( findall(Check-Msg, check(Sev, Check, Msg), L0), sort(L0, L),
             length(L, N),
             ( N =:= 0 -> format('~n~w: none~n', [Sev])
             ; format('~n~w: ~w~n', [Sev, N]),
               ( Sev == warning, N > Max ->
                   length(Show, Max), append(Show, _, L),
                   forall(member(C-Msg, Show), format('  [~w] ~w~n', [C,
   Msg])),
                   Rest is N - Max, format('  … and ~w more (run with --
   all to see them)~n', [Rest])
               ; forall(member(C-Msg, L), format('  [~w] ~w~n', [C,
   Msg])) ) ) )),
    stats.

main :-
    current_prolog_flag(argv, Argv),
    ( memberchk('--all', Argv) -> Max = 100000 ; Max = 25 ),
    report(Max),
    ( (check(error, _, _) ; check(blocker, _, _)) -> halt(1) ; halt(0) ).
```

The claims

From `factcheck/claims.pl`

Structured claims: the manuscript's propositions, in a form logic can test.

The generated facts capture every sentence; this file says what some of them *mean*. Each claim quotes its sentence exactly (whitespace and Markdown emphasis ignored). If the book changes and the quote no longer appears, the checker reports the claim as drifted rather than silently testing stale text.

```
claim(Id, Edition, Chapter, "verbatim quote", Tier, Kind).
    Tier: settled | contested | author — as the text presents it, not as
   we
          would like it to be. The HS edition marks tiers with boxes; the
   full
          edition in prose.
    Kind: quantitative | historical | empirical | generalization |
   interpretive
says(Id, Subject, Property, Value).
    Value: N | approx(N) | range(Lo, Hi) | at_least(N) | more_than(N) |
   at_most(N) | atom
    Two claims with the same Subject and Property must have compatible
   Values.
source(Id, FirstAuthor, Year).          must exist in book/19's reference
   list
depends_on(Id, OtherId).                Id's argument uses OtherId as a
   premise
computes(Id, Expression, StatedValue).  arithmetic the text performs or
   implies
event_year(Event, Year, Basis).         dated anchors for chronology
   checks
order_claim(Id, Earlier, Later).        the text puts Earlier first
interval_claim(Id, From, To, Years).    the text states the gap between
   them
gap(GapId, Description, WhereTheBookSaysSo).
rests_on_gap(Id, GapId).
arc_gap(X, Y, GapId) / arc_descriptive(X, Y, Where).   annotate Modern
   Alchemy's arc
variant_ok(Id1, Id2, Why) / cite_ok(Name, Year, Why) /
   certainty_ok(Fragment, Why)
```

```
prose_ok(ChainInputs, ChainOutput, Where) / common_knowledge(Id) /
   about(Id, ConceptId)

:- discontiguous claim/6, says/4, source/3, depends_on/2, computes/3,
   event_year/3,
                order_claim/3, interval_claim/4, variant_ok/3, gap/3,
   rests_on_gap/2,
                arc_gap/3, arc_descriptive/3, cite_ok/3, about/2,
   drift_ok/3,
                prose_ok/3, common_knowledge/1, certainty_ok/2.
```

Acknowledged gaps

```
gap(heredity_origin,
    "how a chemical system first came to copy a pattern reliably enough
   for differences to accumulate is not known",
    'book chs. 4, 8, 16, 17, 20; HS ch. 1').
```

Since 2026-11-03 the arc has two strands, following the author's capture reading in ch. 20. Copying joins the protocell through Pc + H ⟶ NS, so the gap now sits on the copying strand itself, at RN ⟶ H.

```
arc_gap('RN', 'H', heredity_origin).

arc_descriptive('L', 'CT', 'book ch. 16: "Shared vocabulary rather than
   shared mechanism, above language"').
arc_descriptive('CT', 'Re', 'book ch. 16: "Shared vocabulary rather than
   shared mechanism, above language"').

prose_ok(['H', 'Va', 'Sl'], 'NS', 'book chs. 8–9 carry heredity into
   selection; the crossing is the subject of both').
prose_ok(['Pc', 'H'], 'NS', 'book ch. 20: a copier taken into a
   protocell, after the coupling measured in chs. 5 and 17').
prose_ok(['Mx', 'Sp', 'NS'], 'PR', 'book ch. 13: perception and
   representation built by nervous systems, from the retina up').
prose_ok(['Mx', 'Sp', 'NS'], 'PR', 'book ch. 16: "through
   multicellularity, specialization, nervous systems, and the capacity to
   build and use internal models"').

claim(c16_break, book, 16, "How a chemical system first came to copy a
   pattern reliably enough for differences to accumulate across
   generations is not known.", settled, empirical).
says(c16_break, heredity_origin, known, false).
claim(c4_open_ended, book, 4, "no demonstrated autocatalytic system
   evolves open-endedly.", settled, empirical).
says(c4_open_ended, open_ended_chemical_evolution, demonstrated, false).
```

Chapter 3: the trial count

```
claim(c3_table_day, book, 3, "Suppose one such cycle completes every six
   minutes.", settled, supposition).   % an explicit supposition; the
   text then shows the result does not depend on it
computes(c3_table_day, 24 * 60 / 6, 240).
computes(c3_table_day, 24 * 60 / 6 * 365, approx(88000)).
computes(c3_table_day, 24 * 60 / 6 * 365 * 100, approx(8.8e6)).
computes(c3_table_day, 24 * 60 / 6 * 365 * 1.0e7, approx(1.0e12)).
claim(c3_hourly, book, 3, "Make the cycle an hour instead and the trial
   count falls by a factor of ten, to around 9 × 10<sup>10</sup>.",
   settled, quantitative).
computes(c3_hourly, 24 * 365 * 1.0e7, approx(9.0e10)).
claim(c3_daily, book, 3, "Make it a full day and it falls by a factor of
   240, to around 4 × 10<sup>9</sup>.", settled, quantitative).
computes(c3_daily, 365 * 1.0e7, approx(4.0e9)).
common_knowledge(c3_hourly).
common_knowledge(c3_daily).

claim(c3hs_hourly, 'book-hs', 3, "Say the pool only resets once an hour.
   That cuts the total by ten — about ninety billion.", settled,
   quantitative).
computes(c3hs_hourly, 24 * 365 * 1.0e7, approx(9.0e10)).
claim(c3hs_daily, 'book-hs', 3, "Say once a day. That cuts it by 240 —
   about four billion.", settled, quantitative).
computes(c3hs_daily, 365 * 1.0e7, approx(4.0e9)).
common_knowledge(c3hs_hourly).
common_knowledge(c3hs_daily).
says(c3_hourly, trials_hourly_cycle_ten_million_years, count,
   approx(9.0e10)).
says(c3hs_hourly, trials_hourly_cycle_ten_million_years, count,
   approx(9.0e10)).
says(c3_daily, trials_daily_cycle_ten_million_years, count,
   approx(4.0e9)).
says(c3hs_daily, trials_daily_cycle_ten_million_years, count,
   approx(4.0e9)).
```

Chapter 4

```
claim(c4_formose, book, 4, "known since Butlerov described it in 1861",
   settled, historical).
source(c4_formose, 'Butlerow', 1861).
cite_ok('Butlerov', 1861, 'the reference spells it Butlerow, the German
   transliteration').
```

```
claim(c4_peptide, book, 4, "In 1996 a research group reported a peptide
   (a short protein-like chain) 32 amino acids long", settled,
   empirical).
source(c4_peptide, 'Lee', 1996).
says(c4_peptide, ghadiri_peptide, residues, 32).
claim(c4_nucleic, book, 4, "Comparable systems had already been built
   from short nucleic acid strands a decade earlier", settled,
   historical).
source(c4_nucleic, 'Kiedrowski', 1986).
event_year(nucleic_acid_replicator, 1986, 'von Kiedrowski 1986').
event_year(peptide_replicator, 1996, 'Lee et al. 1996').
interval_claim(c4_nucleic, nucleic_acid_replicator, peptide_replicator,
   approx(10)).

claim(c4_kauffman, book, 4, "Stuart Kauffman argued in 1986 that such
   collectively autocatalytic sets should appear almost unavoidably",
   contested, empirical).
source(c4_kauffman, 'Kauffman', 1986).
claim(c4_status, book, 4, "Whether real prebiotic chemistry was rich
   enough, and whether the catalysts available were specific enough, is
   not settled.", contested, empirical).
depends_on(c4_status, c4_kauffman).
```

Before 2026-11-03 this read “Self-sustaining reaction networks are chemistry, demonstrated in chapter 4.” and depended on c4_status, which chapter 4 marks as not settled. The checker flagged the settled-on-contested dependency; the text now separates what is shown from what is open.

```
claim(c16_lower_half, book, 16, "chapter 4 shows autocatalysis
   demonstrated, and small collectively self-sustaining networks built in
   the laboratory", settled, empirical).
depends_on(c16_lower_half, c4_peptide).
```

A plain statement that a question is open is settled as presented, in both editions (d_c16hs_prebiotic_open). c4_status stands for the open question itself, which is why it is contested and why this claim no longer rests on it (re-encoded 2026-11-03). It is empirical, like its HS twin, and shares that twin’s sources (fix_origins.pl).

```
claim(c16_prebiotic_open, book, 16, "Whether prebiotic chemistry was rich
   enough to produce them unaided is, as chapter 4 says, not settled.",
   settled, empirical).
claim(c16_no_new_principle, book, 16, "Nothing in this stretch requires a
   new principle at any step; what remains open is whether early Earth
   supplied the conditions", settled, interpretive).
depends_on(c16_no_new_principle, c16_lower_half).
```

Chapter 6

Corrected on 2026-11-03. The text had Margulis "submitting the paper … since 1964", but her own account (Margulis 1995) says she wrote it in 1966, and the paper is marked "Received 8 June 1966". No source gives 1964, so that date is gone from the text, along with the event and the order claim that rested on it.

```
claim(c6_margulis, book, 6, "recalled that the paper that set out the
   modern case was turned down by around fifteen journals", settled,
   historical).
source(c6_margulis, 'Margulis', 1995).
event_year(margulis_publication, 1967, 'Sagan 1967').
says(c6_margulis, endosymbiosis_paper_rejections, count, approx(15)).
cite_ok('Margulis', 1967, 'she published as Lynn Sagan; the reference is
   Sagan 1967').
```

```
claim(c6_mito, book, 6, "The human mitochondrial genome carries thirty-
   seven.", settled, quantitative).
source(c6_mito, 'Anderson', 1981).
says(c6_mito, human_mitochondrial_genes, count, 37).
```

Chapter 8

```
claim(c8_codons, book, 8, "Four possible nucleotides in three positions
   gives 4 × 4 × 4 combinations — 64 codons.", settled, quantitative).
computes(c8_codons, 4 ** 3, 64).
common_knowledge(c8_codons).
claim(c8hs_codons, 'book-hs', 8, "Four letters in three slots gives 64
   possible triplets.", settled, quantitative).
computes(c8hs_codons, 4 ** 3, 64).
common_knowledge(c8hs_codons).
claim(c8hs_meanings, 'book-hs', 8, "64 triplets, 21 meanings.", settled,
   quantitative).
computes(c8hs_meanings, 20 + 1, 21).
common_knowledge(c8hs_meanings).
```

```
claim(c8_million, book, 8, "they found roughly one in a million random
   codes performed better than the real one", settled, quantitative).
claim(c8hs_million, 'book-hs', 8, "Only about one in a million random
   codes did better than the real one.", settled, quantitative).
source(c8_million, 'Freeland', 1998).
source(c8hs_million, 'Freeland', 1998).
says(c8_million, codes_beating_standard_code, fraction, approx(1.0e-6)).
says(c8hs_million, codes_beating_standard_code, fraction,
   approx(1.0e-6)).
```

Chapter 9

```
claim(c9_giraffe, book, 9, "about four and a half metres of nerve to span
   a gap of inches", settled, quantitative).
claim(c9hs_giraffe, 'book-hs', 9, "About four and a half metres of nerve,
   to cross a gap of inches.", settled, quantitative).
source(c9_giraffe, 'Wedel', 2012).
source(c9hs_giraffe, 'Wedel', 2012).
says(c9_giraffe, giraffe_laryngeal_nerve_path, metres, approx(4.5)).
says(c9hs_giraffe, giraffe_laryngeal_nerve_path, metres, approx(4.5)).

claim(c9hs_eyes, 'book-hs', 9, "surveys put it somewhere above forty,
   depending on how strictly you count", settled, quantitative).
source(c9hs_eyes, 'Salvini-Plawen', 1977).
says(c9hs_eyes, independent_eye_origins, count, more_than(40)).

claim(c9_multiple_origins, book, 9, "Here the author would go further:
   that early Earth, running an enormous number of parallel chemical
   experiments", author, interpretive).
```

Chapter 10

```
claim(c10_golden, book, 10, "converges on about 137.5 degrees", settled,
   quantitative).
claim(c10hs_golden, 'book-hs', 10, "The angle between seeds settles at
   about 137.5 degrees", settled, quantitative).
computes(c10_golden, 360 * (1 - 1 / ((1 + sqrt(5)) / 2)), range(137.0,
   138.0)).
computes(c10hs_golden, 360 * (1 - 1 / ((1 + sqrt(5)) / 2)), range(137.0,
   138.0)).
common_knowledge(c10_golden).
common_knowledge(c10hs_golden).
says(c10_golden, phyllotaxis_divergence_angle, degrees, approx(137.5)).
says(c10hs_golden, phyllotaxis_divergence_angle, degrees, approx(137.5)).

claim(c10_douady, book, 10, "Douady and Couder produced the same spirals
   in 1992", settled, empirical).
claim(c10hs_douady, 'book-hs', 10, "in 1992, researchers dropped
   magnetized droplets into a dish", settled, empirical).
source(c10_douady, 'Douady', 1992).
source(c10hs_douady, 'Douady', 1992).
says(c10_douady, magnetic_droplet_phyllotaxis, year, 1992).
says(c10hs_douady, magnetic_droplet_phyllotaxis, year, 1992).

claim(c10_thompson, book, 10, "D'Arcy Wentworth Thompson worked this out
   in detail in 1917", settled, historical).
```

```
source(c10_thompson, 'Thompson', 1917).

claim(c10_schumann, book, 10, "These are the Schumann resonances,
   predicted in 1952 and confirmed experimentally within the decade.",
   settled, historical).
source(c10_schumann, 'Schumann', 1952).
event_year(schumann_prediction, 1952, 'Schumann 1952').
event_year(schumann_confirmation, 1960, 'Balser & Wagner 1960').
interval_claim(c10_schumann, schumann_prediction, schumann_confirmation,
   range(0, 10)).
claim(c10_lightning, book, 10, "roughly fifty strikes somewhere on the
   planet every second", settled, quantitative).
source(c10_lightning, 'Christian', 2003).
says(c10_lightning, global_lightning_rate, per_second, approx(50)).

claim(c10_spiral_principle, book, 10, "a spiral is what extension looks
   like when something must keep a fixed relationship to where it has
   already been", settled, generalization).
claim(c10hs_spiral_principle, 'book-hs', 10, "A spiral is what growing
   looks like when something has to keep its connection to where it's
   already been.", settled, generalization).
```

Corrected on 2026-11-03. The excavator (Glory 2008) describes three strands, not two. The cord lay on a palaeosol dated 17,190 ± 135 radiocarbon years BP, which calibrates to about 21,000 years (Ducasse & Langlais 2019). The HS “at least 17,000 years” stays true. The subject is no longer “oldest_laid_rope”: neither edition claims it is the oldest, and older plied cord exists (Abri du Maras, 41–52 ka).

```
claim(c10_rope, book, 10, "What appears to be three-strand laid rope,
   about twenty-one thousand years old, was found in the cave at
   Lascaux.", settled, historical).
claim(c10hs_rope, 'book-hs', 10, "People have made twisted rope for at
   least 17,000 years.", settled, historical).
says(c10_rope, twisted_rope_making, years_attested, approx(21000)).
says(c10hs_rope, twisted_rope_making, years_attested, at_least(17000)).

claim(c10_nucleosomes, book, 10, "146 base pairs at a time and about
   thirty million times in each human cell", settled, quantitative).
claim(c10hs_nucleosomes, 'book-hs', 10, "about thirty million in each
   cell", settled, quantitative).
source(c10_nucleosomes, 'Luger', 1997).
says(c10_nucleosomes, nucleosomes_per_human_cell, count, approx(3.0e7)).
says(c10hs_nucleosomes, nucleosomes_per_human_cell, count,
   approx(3.0e7)).
```

```
claim(c10_supercoil, book, 10, "Circular DNA from a virus was found
   twisted on itself in 1965, and the writhing number was defined in
   1971", settled, historical).
source(c10_supercoil, 'Vinograd', 1965).
event_year(supercoiled_dna_found, 1965, 'Vinograd et al. 1965').
event_year(writhe_defined, 1971, 'Fuller 1971').
order_claim(c10_supercoil, supercoiled_dna_found, writhe_defined).

claim(c10hs_coil_rule, 'book-hs', 10, "The twist-and-coil rule is real
   mathematics, partly worked out because of DNA.", settled, historical).
source(c10hs_coil_rule, 'Fuller', 1971).
claim(c10hs_storage, 'book-hs', 10, "A coil is the shape of something
   long that's meant to be paid out later, in order.", author,
   interpretive).
depends_on(c10hs_storage, c10hs_spiral_principle).
depends_on(c10hs_storage, c10hs_coil_rule).

claim(c10_newgrange, book, 10, "The passage tomb at Newgrange in Ireland,
   built around 3100 BCE", settled, historical).
```

O'Kelly 1982 (†), the excavation report, could not be read, and the text doesn't name it. Since 2026-11-03 the claim rests on Smyth et al. 2025 (fix_spirals.pl). O'Kelly stays in the reading list, still marked unchecked. Corrected on 2026-11-03 from ICOMOS 2006 (Murray & Viramontes, p. 5): "principally spirals, dot figures and scale models of architectural features". The 750–1200 dates belonged to the site, not the carvings, and INAH dates the city's height earlier, so the dates are gone.

```
claim(c10_plazuelas, book, 10, "More than 1,400 petroglyphs at Las
   Plazuelas in central Mexico are mainly spirals, dot figures, and
   carved models of buildings", settled, empirical).
source(c10_plazuelas, 'ICOMOS', 2006).

claim(c10_starry, book, 10, "Vincent van Gogh painted The Starry Night in
   June 1889, from the window of his room in the asylum at Saint-Rémy-de-
   Provence, where he had admitted himself the month before.", settled,
   historical).
common_knowledge(c10_starry).
event_year(van_gogh_admitted, 1889.35, 'Saint-Paul-de-Mausole, 8 May
   1889').
event_year(starry_night_painted, 1889.45, 'mid-June 1889').
event_year(rosse_whirlpool_drawing, 1845, 'Nasim 2010: discovery and
   first sketch, 1845').
order_claim(c10_starry, van_gogh_admitted, starry_night_painted).
claim(c10_rosse, book, 10, "tracing the image to Lord Rosse's 1845
   drawing of the Whirlpool galaxy", contested, historical).
```

Whitney 1986 (†) could not be read. Since 2026-11-03 the claim rests on Whitney's own 1992 letter, read in full (fix_spirals.pl), and on Nasim 2010 for the date of the drawing. The 1986 paper stays in the reading list, still marked unchecked.

```
order_claim(c10_rosse, rosse_whirlpool_drawing, starry_night_painted).
claim(c10_turbulence, book, 10, "In 2024 a team of fluid dynamicists
   tried measuring the painting instead of interpreting it.", settled,
   empirical).
source(c10_turbulence, 'Ma', 2024).
claim(c10_degas, book, 10, "and obtained the same slope", settled,
   empirical).
source(c10_degas, 'Bourgault', 2025).
claim(c10_ortiz, book, 10, "the Cuban scholar Fernando Ortiz argued in
   1947", contested, interpretive).
source(c10_ortiz, 'Ortiz', 1947).

claim(c10_potter, book, 10, "A comment from the author, not a finding:
   the potter's wheel may be the plainest case of a person working with a
   spiral rather than drawing one.", author, interpretive).
claim(c10hs_potter, 'book-hs', 10, "Not a fact, only a way of seeing it:
   a potter's wheel lets a person work with a spiral instead of drawing
   one.", author, interpretive).
claim(c10_newgrange_reading, book, 10, "Here the author would go further.
   The pairing reads as a record of the two ways a bounded space gets
   filled.", author, interpretive).
```

Chapter 11

```
claim(c11_ice, book, 11, "The longest continuous Antarctic record reaches
   back about 800,000 years, across eight glacial cycles", settled,
   quantitative).
source(c11_ice, 'EPICA community members', 2004).
says(c11_ice, longest_antarctic_ice_record, years, approx(800000)).

claim(c11_smolin_proposal, book, 11, "proposed by Lee Smolin in 1992",
   settled, historical).          % the theory is contested; that he
   proposed it is not
source(c11_smolin_proposal, 'Smolin', 1992).
claim(c11_smolin_limit, book, 11, "which Smolin put at about 1.6 times
   the mass of the Sun", settled, historical).
source(c11_smolin_limit, 'Brown', 1994).
says(c11_smolin_limit, cns_neutron_star_limit, solar_masses,
   approx(1.6)).
event_year(cns_proposed, 1992, 'Smolin 1992').
event_year(kaon_calculations, 1994, 'Brown & Bethe 1994').
event_year(first_two_solar_mass_pulsar, 2010, 'Demorest et al. 2010').
```

```
order_claim(c11_smolin_limit, cns_proposed, kaon_calculations).
order_claim(c11_smolin_limit, kaon_calculations,
   first_two_solar_mass_pulsar).

claim(c11_stakes, book, 11, "a neutron star heavier than about 1.5 solar
   masses would be troubling for the theory, and one of 2.5 would
   certainly refute it", settled, historical).
source(c11_stakes, 'Smolin', 2007).
certainty_ok("would certainly refute it", 'reports what Smolin wrote, in
   his words').
claim(c11_measured, book, 11, "A pulsar of nearly two solar masses was
   measured in 2010; one at roughly 2.08 followed, and another at around
   2.35 after that.", settled, quantitative).
source(c11_measured, 'Demorest', 2010).
claim(c11_exceeded, book, 11, "The limit he committed to has been
   exceeded several times.", settled, quantitative).
computes(c11_exceeded, min(1.97, min(2.08, 2.35)), more_than(1.6)).
depends_on(c11_exceeded, c11_measured).
depends_on(c11_exceeded, c11_smolin_limit).
claim(c11_not_reached, book, 11, "The higher figure he named as certain
   refutation has not yet been reached.", settled, quantitative).
computes(c11_not_reached, max(1.97, max(2.08, 2.35)), at_most(2.5)).
depends_on(c11_not_reached, c11_measured).
depends_on(c11_not_reached, c11_stakes).
common_knowledge(c11_exceeded).
common_knowledge(c11_not_reached).
```

Chapter 12

```
claim(c12_definitions, book, 12, "One compilation counts over a hundred
   and twenty distinct published definitions of life", settled,
   quantitative).
claim(c12hs_definitions, 'book-hs', 12, "One survey counted over a
   hundred and twenty published definitions of life.", settled,
   quantitative).
source(c12_definitions, 'Trifonov', 2011).
source(c12hs_definitions, 'Trifonov', 2011).
says(c12_definitions, published_definitions_of_life, count,
   more_than(120)).
says(c12hs_definitions, published_definitions_of_life, count,
   more_than(120)).
computes(c12_definitions, 123, more_than(120)).

claim(c12_malformed, book, 12, "\"When did life begin\" is malformed in
   the same way \"when did it become a heap\" is malformed.", settled,
   interpretive).
```

Chapter 13

```
claim(c13_retina, book, 13, "A human retina holds something like ninety-
   six million photoreceptors.", settled, quantitative).
claim(c13hs_retina, 'book-hs', 13, "There are about ninety-six million of
   them in one retina.", settled, quantitative).
source(c13_retina, 'Curcio', 1990).
source(c13hs_retina, 'Curcio', 1990).
says(c13_retina, photoreceptors_per_retina, count, approx(9.6e7)).
says(c13hs_retina, photoreceptors_per_retina, count, approx(9.6e7)).
claim(c13_fibres, book, 13, "The optic nerve carrying their output to the
   brain contains roughly one million fibres.", settled, quantitative).
claim(c13hs_fibres, 'book-hs', 13, "The optic nerve has about one million
   fibres.", settled, quantitative).
says(c13_fibres, optic_nerve_fibres, count, approx(1.0e6)).
says(c13hs_fibres, optic_nerve_fibres, count, approx(1.0e6)).
claim(c13_ratio, book, 13, "Around a hundred receptors feed each one",
   settled, quantitative).
claim(c13hs_ratio, 'book-hs', 13, "So roughly a hundred sensors feed each
   outgoing wire.", settled, quantitative).
computes(c13_ratio, 9.6e7 / 1.0e6, approx(100)).
computes(c13hs_ratio, 9.6e7 / 1.0e6, approx(100)).
depends_on(c13_ratio, c13_retina).
depends_on(c13_ratio, c13_fibres).
depends_on(c13hs_ratio, c13hs_retina).
depends_on(c13hs_ratio, c13hs_fibres).
common_knowledge(c13_fibres).
common_knowledge(c13hs_fibres).
common_knowledge(c13_ratio).
common_knowledge(c13hs_ratio).
```

Chapter 14

```
claim(c14_boas, book, 14, "Franz Boas, in 1911, noted four distinct
   roots", settled, historical).
claim(c14hs_boas, 'book-hs', 14, "In 1911, an anthropologist noted four
   root words", settled, historical).
source(c14_boas, 'Boas', 1911).
source(c14hs_boas, 'Boas', 1911).
says(c14_boas, boas_snow_roots, count, 4).
says(c14hs_boas, boas_snow_roots, count, 4).
claim(c14_whorf, book, 14, "Benjamin Lee Whorf, in a 1940 article,
   repeated the observation with seven, three of which he did not
   source.", settled, historical).
```

```
claim(c14hs_whorf, 'book-hs', 14, "A later writer repeated it with seven,
   three of which he didn't source.", settled, historical).
source(c14_whorf, 'Whorf', 1940).
source(c14hs_whorf, 'Whorf', 1940).
computes(c14_whorf, 7 - 3, 4).          % the four he did source are
   Boas's four
computes(c14hs_whorf, 7 - 3, 4).
depends_on(c14_whorf, c14_boas).
depends_on(c14hs_whorf, c14hs_boas).
event_year(boas_handbook, 1911, 'Boas 1911').
event_year(whorf_article, 1940, 'Whorf 1940').
event_year(martin_traces_history, 1986, 'Martin 1986').
event_year(pullum_essay, 1989, 'Pullum 1989').
order_claim(c14_whorf, boas_handbook, whorf_article).
claim(c14_martin, book, 14, "Laura Martin traced this history in 1986,
   and Geoffrey Pullum's 1989 essay popularizing her case", settled,
   historical).
source(c14_martin, 'Martin', 1986).
order_claim(c14_martin, martin_traces_history, pullum_essay).

claim(c14_seaice, book, 14, "Surveys across roughly ten Inuit and Yupik
   dialects have recorded on the order of seventy terms for ice
   conditions in one Alaskan dialect, around forty snow descriptors in
   Central Siberian Yupik, and some fifty in a Canadian Inuit dialect.",
   settled, quantitative).
claim(c14hs_seaice, 'book-hs', 14, "Surveys across about ten Inuit and
   Yupik dialects recorded something like seventy terms for ice
   conditions in one Alaskan dialect, around forty snow terms in another
   language, about fifty in a third.", settled, quantitative).
source(c14_seaice, 'Krupnik', 2010).
source(c14hs_seaice, 'Krupnik', 2010).
says(c14_seaice, alaskan_ice_terms, count, approx(70)).
says(c14hs_seaice, alaskan_ice_terms, count, approx(70)).
says(c14_seaice, siberian_yupik_snow_terms, count, approx(40)).
says(c14hs_seaice, siberian_yupik_snow_terms, count, approx(40)).
says(c14_seaice, canadian_inuit_snow_terms, count, approx(50)).
says(c14hs_seaice, canadian_inuit_snow_terms, count, approx(50)).
says(c14_seaice, dialects_surveyed, count, approx(10)).
says(c14hs_seaice, dialects_surveyed, count, approx(10)).

claim(c14_russian, book, 14, "In a 2007 study, Russian speakers were
   faster at telling two blues apart", settled, empirical).          % a
   report of what the study found
claim(c14hs_russian, 'book-hs', 14, "In a 2007 experiment, Russian
   speakers were faster at telling two blues apart", settled, empirical).
source(c14_russian, 'Winawer', 2007).
source(c14hs_russian, 'Winawer', 2007).
```

```
claim(c14_replication, book, 14, "And a 2020 study attempting to
   replicate it did not find the effect.", settled, empirical).
claim(c14hs_replication, 'book-hs', 14, "And then a 2020 attempt to
   repeat that study didn't find the effect", settled, empirical).
source(c14_replication, 'Martinovic', 2020).
source(c14hs_replication, 'Martinovic', 2020).
event_year(russian_blues_study, 2007, 'Winawer et al. 2007').
event_year(russian_blues_replication, 2020, 'Martinovic et al. 2020').
order_claim(c14_replication, russian_blues_study,
   russian_blues_replication).
```

Corrected on 2026-11-03: Pirahã has three quantity words (hói, hoí, baágiso; Everett 2005, Frank 2008), and two of them were glossed as numerals. The glosses were Everett's own (Everett 2008), so "the same researcher" holds. Sources are in fix_culture.pl.

```
claim(c14_number, book, 14, "two of whose quantity words were initially
   glossed as one and two", settled, empirical).
claim(c14hs_number, 'book-hs', 14, "two of whose quantity words were
   first translated as \"one\" and \"two,\"", settled, empirical).
source(c14_number, 'Frank', 2008).
says(c14_number, piraha_words_glossed_as_numerals, count, 2).
says(c14hs_number, piraha_words_glossed_as_numerals, count, 2).
```

What the results mean. Until 2026-11-03 the HS edition covered these with one box reading "Everything in this section", against its own rule that a box covers only what is inside it. The interpretations now have their own boxes; the reports stand unboxed.

```
claim(c14hs_colour_meaning, 'book-hs', 14, "The original reading was that
   the language was being used live, during the task, rather than having
   permanently rewired anyone's eyes.", contested, interpretive).
says(c14hs_colour_meaning, russian_blues_interpretation, reading,
   live_recruitment).
depends_on(c14hs_colour_meaning, c14hs_russian).
claim(c14_colour_meaning, book, 14, "which the authors read as the
   language being recruited during the discrimination itself rather than
   having permanently restructured perception", contested,
   interpretive).   % was "which indicates", stated as settled; aligned
   with the HS box 2026-11-03
says(c14_colour_meaning, russian_blues_interpretation, reading,
   live_recruitment).
depends_on(c14_colour_meaning, c14_russian).
claim(c14hs_number_meaning, 'book-hs', 14, "The difficulty doesn't seem
   to be seeing or matching exact quantity.", contested, interpretive).
depends_on(c14hs_number_meaning, c14hs_number).
```

```
claim(c14hs_narrow_version, 'book-hs', 14, "The narrow version: words you
   use constantly get recruited while you do mental work", contested,
   interpretive).
depends_on(c14hs_narrow_version, c14hs_colour_meaning).
depends_on(c14hs_narrow_version, c14hs_replication).

claim(c14_wcs, book, 14, "Across the 110 languages of the World Color
   Survey, warm colours are communicated more precisely than cool ones",
   settled, empirical).
source(c14_wcs, 'Gibson', 2017).
claim(c14_hanunoo, book, 14, "Among the Hanunóo of the Philippines, the
   term that covers green also means fresh", settled, empirical).
source(c14_hanunoo, 'Conklin', 1955).
claim(c14hs_landscape, 'book-hs', 14, "A landscape fills a language with
   words for the differences that can hurt you or feed you", settled,
   generalization).
depends_on(c14hs_landscape, c14hs_seaice).
```

Chapter 15

```
claim(c15_nixta_equipment, book, 15, "dating to roughly 1500–1200 BCE,
   which would put it well over three thousand years ago", settled,
   historical).
computes(c15_nixta_equipment, 1200 + 2026, more_than(3000)).
```

Corrected on 2026-11-03 (Santini et al. 2022): the identification is microscopic, comes from a single site, and dates to the Late and Terminal Classic, around the eighth and ninth centuries CE.

```
claim(c15_nixta_direct, book, 15, "first reported from the Maya site of
   San Bartolo and dating to the Late and Terminal Classic, around the
   eighth and ninth centuries CE", settled, historical).
source(c15_nixta_direct, 'Santini', 2022).

claim(c15_pellagra, book, 15, "between 1906 and 1940 it affected more
   than three million people and killed over a hundred thousand",
   settled, quantitative).
claim(c15hs_pellagra, 'book-hs', 15, "between 1906 and 1940 it affected
   more than three million people and killed over a hundred thousand",
   settled, quantitative).
source(c15_pellagra, 'Bollet', 1992).
source(c15hs_pellagra, 'Bollet', 1992).
says(c15_pellagra, us_pellagra_cases_1906_1940, count, more_than(3.0e6)).
says(c15hs_pellagra, us_pellagra_cases_1906_1940, count,
   more_than(3.0e6)).
says(c15_pellagra, us_pellagra_deaths_1906_1940, count,
   more_than(1.0e5)).
```

```
says(c15hs_pellagra, us_pellagra_deaths_1906_1940, count,
   more_than(1.0e5)).
claim(c15_vitamin, book, 15, "and the specific vitamin was identified in
   1937", settled, historical).
claim(c15hs_vitamin, 'book-hs', 15, "The specific vitamin wasn't
   identified until 1937.", settled, historical).
source(c15_vitamin, 'Elvehjem', 1937).
source(c15hs_vitamin, 'Elvehjem', 1937).
says(c15_vitamin, niacin_identified, year, 1937).
says(c15hs_vitamin, niacin_identified, year, 1937).
event_year(goldberger_trial, 1915, 'Goldberger et al. 1915').
event_year(niacin_identified, 1937, 'Elvehjem et al. 1937').
order_claim(c15_vitamin, goldberger_trial, niacin_identified).
```

Chapter 17

```
claim(c17_gould, book, 17, "Gould's image, from Full House (1996), is a
   random walk beside a wall", settled, interpretive).
source(c17_gould, 'Gould', 1996).
```

Exemptions

```
certainty_ok("never revisiting what it has already laid down", 'describes
   edge growth by definition, not a claim of certainty').
certainty_ok("never reaches a resting state", 'the author''s own reading,
   marked as such, of what recursion means').
certainty_ok("became impossible to change", 'states the content of the
   frozen-accident hypothesis inside the box that marks it as argued
   about').
```

Claims: chapters 0–5

From `factcheck/claims/a_ch00-05.pl`

```
:- discontiguous claim/6, says/4, source/3, depends_on/2, computes/3,
   event_year/3,
                order_claim/3, interval_claim/4, variant_ok/3, gap/3,
   rests_on_gap/2,
                arc_gap/3, arc_descriptive/3, cite_ok/3, about/2,
   drift_ok/3,
                prose_ok/2, common_knowledge/1, certainty_ok/2.
```

Without this, SWI-Prolog (9.0.4) treats a second file that defines claim/6, says/4, ... as a redefinition and wipes the clauses claims.pl already loaded ("Redefined static procedure"), even though checks.pl declares them dynamic. Declaring them multifile makes this file, and every claims file loaded after it, add to claims.pl instead of replacing it. Harmless when loaded alone.

```
:- multifile claim/6, says/4, source/3, depends_on/2, computes/3,
   event_year/3,
             order_claim/3, interval_claim/4, variant_ok/3, gap/3,
   rests_on_gap/2,
             arc_gap/3, arc_descriptive/3, cite_ok/3, about/2,
   drift_ok/3,
             prose_ok/2, common_knowledge/1, certainty_ok/2.
```

Structured claims for full-edition chapters 0-5 and HS chapters 0-5. Schema: see the header of factcheck/claims.pl.

```
Not encoded (number-bearing sentences left uncovered on purpose):
  - book ch.0 front matter (licence, attribution): not propositions; and
   since
    00-front-matter.md and 00-preface.md share chapter number 0,
   extract.py's
    chapter_texts() keeps only the preface, so no front-matter quote can
   anchor.
  - Sentences under 40 characters cannot be matched by checks.pl's
   covered/1
```

(it takes the first 40 characters of a quote): book ch.5 "The first five
improve the chemistry."; HS ch.1 "Three completely different machines.";
HS ch.2 "Two molecules ...", "Two puppies ...", "Two retellings ...",
"All eight are descriptions.", "Three rows, one heading."; HS ch.3
"Four billion tries."
- HS ch.1 "How this book checks itself" is skipped (being written separately).

Chapter 1 (full)

```
claim(a_c01_same_word, book, 1, "The same word can be correctly applied
   to a chemical reaction network and to a language without the two
   sharing any machinery.", settled, generalization).
says(a_c01_same_word, shared_vocabulary, establishes_shared_mechanism,
   false).

claim(a_c01_three_selections, book, 1, "Selection in a population of
   molecules, in a population of organisms, and in a population of ideas
   are three unrelated processes that a single word happens to fit.",
   settled, interpretive).
says(a_c01_three_selections, selection_mechanisms_compared, count, 3).

claim(a_c01_largest_gap, book, 1, "The largest one is named in chapter 8:
   how chemistry first came to copy a pattern reliably enough for
   selection to accumulate across generations is not known, and nothing
   here narrows it.", settled, empirical).
says(a_c01_largest_gap, heredity_origin, known, false).

claim(a_c01_three_voices, book, 1, "Presenting all three in the same
   confident voice would be the easiest way to mislead, and the hardest
   to detect.", settled, interpretive).
says(a_c01_three_voices, statement_tiers, count, 3).       % settled /
   disputed / the author's own

claim(a_c01_three_code_accounts, book, 1, "It is why chapter 8 gives
   three competing accounts of why the genetic code is error-tolerant and
   declines to pick, rather than presenting the most popular one as
   settled.", settled, interpretive).
says(a_c01_three_code_accounts, genetic_code_error_tolerance_accounts,
   count, 3).   % book ch.8 l.41 and HS ch.8 l.31 both give three

claim(a_c01_borrowing_limits, book, 1, "The limits of the borrowing
   should be stated plainly, because the two enterprises settle questions
   by completely different means.", settled, interpretive).
says(a_c01_borrowing_limits, talmud_and_this_book,
   settle_questions_by_same_means, false).
```

Courts and legislatures (added 2026-11-06 at the author's request: "don't cite any specific country to avoid any citation adds"). General statements about common institutional practice, so common_knowledge rather than a source; the HS twins below say the same.

```
claim(a_c01_courts_dissent, book, 1, "In many courts a judge who
   disagrees with a ruling can publish a dissenting opinion beside it,
   and a later court sometimes adopts reasoning that first appeared in
   dissent.", settled, generalization).
says(a_c01_courts_dissent, dissenting_opinions, sometimes_adopted_later,
   true).
common_knowledge(a_c01_courts_dissent).
claim(a_c01_precedent, book, 1, "Courts that follow precedent explain
   each decision against the earlier cases it follows or departs from, so
   the reasons travel with the ruling.", settled, generalization).
common_knowledge(a_c01_precedent).
claim(a_c01_legislative_record, book, 1, "Many legislatures keep a
   verbatim record of their debates, with the amendments that failed and
   how members voted, so the case against a law stays on the record after
   the law passes.", settled, generalization).
says(a_c01_legislative_record, legislatures, keep_record_of_debates,
   true).
common_knowledge(a_c01_legislative_record).
claim(a_c01_courts_limits, book, 1, "These institutions settle questions
   by authority and by vote rather than by experiment, so the limits
   stated above apply to them too.", settled, interpretive).
says(a_c01_courts_limits, courts_legislatures_and_this_book,
   settle_questions_by_same_means, false).

claim(a_c01_order_departure, book, 1, "Chapters 5, 6, and 7 follow
   boundaries as a theme — compartments, then compartments inside
   compartments, then an organism and the organisms living on it — which
   means chapters 6 and 7 describe things that already require heredity,
   while chapter 8, where heredity arrives, comes after them.", settled,
   interpretive).
says(a_c01_order_departure, book_chapter_order, strictly_chronological,
   false).   % agrees with book ch.8 l.65
```

Chapter 1 (HS)

```
claim(a_c00hs_three_kinds, 'book-hs', 0, "Every book like this mixes
   three different kinds of statement.", settled, interpretive).
says(a_c00hs_three_kinds, statement_tiers, count, 3).
```

```
claim(a_c01hs_shared_question, 'book-hs', 1, "That question is worth
   asking in all three cases. The answer is different every time, and the
   answer is the part that matters.", settled, interpretive).
says(a_c01hs_shared_question, selection_mechanisms_compared, count, 3).

claim(a_c01hs_largest_gap, 'book-hs', 1, "There is a big one, and chapter
   8 is where you hit it: nobody knows how chemistry first started
   copying itself reliably.", settled, empirical).
says(a_c01hs_largest_gap, heredity_origin, known, false).

claim(a_c01hs_three_code_accounts, 'book-hs', 1, "In chapter 8 there are
   three competing explanations for something, and I don't pick.",
   settled, interpretive).
says(a_c01hs_three_code_accounts, genetic_code_error_tolerance_accounts,
   count, 3).

claim(a_c01hs_borrowing_limits, 'book-hs', 1, "The two things are not the
   same, and I want to be clear about that. Those texts settle questions
   by authority and tradition.", settled, interpretive).
says(a_c01hs_borrowing_limits, talmud_and_this_book,
   settle_questions_by_same_means, false).
```

Courts and parliaments, the HS twins of the a_c01_ claims above (2026-11-06).

```
claim(a_c01hs_courts_dissent, 'book-hs', 1, "Sometimes, years later,
   another court ends up agreeing with the dissent.", settled,
   generalization).
says(a_c01hs_courts_dissent, dissenting_opinions,
   sometimes_adopted_later, true).
common_knowledge(a_c01hs_courts_dissent).
claim(a_c01hs_dissent_published, 'book-hs', 1, "In many courts, a judge
   who disagrees with a decision can write a dissent, explaining why, and
   it's published right next to the ruling.", settled, generalization).
common_knowledge(a_c01hs_dissent_published).
claim(a_c01hs_legislative_record, 'book-hs', 1, "Many parliaments keep a
   word-for-word record of their debates, so the arguments against a law
   are still there after it passes.", settled, generalization).
says(a_c01hs_legislative_record, legislatures, keep_record_of_debates,
   true).
common_knowledge(a_c01hs_legislative_record).
claim(a_c01hs_courts_limits, 'book-hs', 1, "They settle things by
   authority and by vote, though, not by experiment, so the same limits
   apply to them too.", settled, interpretive).
says(a_c01hs_courts_limits, courts_legislatures_and_this_book,
   settle_questions_by_same_means, false).
```

Chapter 2 (full)

```
claim(a_c02_word_warning, book, 2, "A word that correctly applies to a
   chemical reaction and to a language has not shown that the two have
   anything in common beyond being describable.", settled,
   generalization).
says(a_c02_word_warning, shared_vocabulary, establishes_shared_mechanism,
   false).

claim(a_c02_rhyme, book, 2, "That abstraction is useful for noticing that
   two situations rhyme. It is useless for establishing that they work
   alike, and the step from one to the other is where arguments of this
   kind usually fail.", settled, interpretive).
says(a_c02_rhyme, shared_vocabulary, establishes_shared_mechanism,
   false).

claim(a_c02_three_rows, book, 2, "Three rows under the heading
   \"selection.\" The processes share no machinery.", settled,
   interpretive).
says(a_c02_three_rows, selection_mechanisms_compared, count, 3).

claim(a_c02_shared_question, book, 2, "What the shared term buys is the
   question — is something being varied, and is something treating the
   variants unequally? — which is worth asking in all three cases.",
   settled, interpretive).
says(a_c02_shared_question, selection_mechanisms_compared, count, 3).

claim(a_c02_companion, book, 2, "These concepts and roughly a hundred
   more are set out in Modern Alchemy, the companion volume, with
   permanent identifiers, one symbol each, and a notation for combining
   them.", settled, quantitative).
says(a_c02_companion, companion_concepts_beyond_chapter2, count,
   approx(100)).
computes(a_c02_companion, 119 - 16, approx(100)).   % 119 concepts in
   generated/glossary.pl less the 16 terms this chapter defines
   (snapshot; update if either changes)
common_knowledge(a_c02_companion).                  % a statement about
   the book's own companion; no outside source applies
```

Chapter 2 (HS)

```
claim(a_c02hs_eight_words, 'book-hs', 2, "Eight words do most of the work
   in this book.", settled, interpretive).
says(a_c02hs_eight_words, hs_core_vocabulary_terms, count, 8).   % the
   chapter lists eight bolded terms
```

```
claim(a_c02hs_rhyme, 'book-hs', 2, "That's useful for noticing that two
   situations rhyme. It is useless for showing they work the same way.",
   settled, interpretive).
says(a_c02hs_rhyme, shared_vocabulary, establishes_shared_mechanism,
   false).
```

Chapter 3 (full)

```
claim(a_c03_three_engines, book, 3, "Early Earth had three largely
   independent sources. Each pushed in one direction, and none of them
   was going to run out.", settled, empirical).
says(a_c03_three_engines, early_earth_energy_sources, count, 3).   % the
   Sun, the planet's interior, the Moon
claim(a_c03_three_together, book, 3, "The three together matter more than
   any one of them.", settled, interpretive).
depends_on(a_c03_three_together, a_c03_three_engines).
```

Added 2026-11-07 from the author's sticky note ("SOX / NOX / sulfates and silica ... taken by the winds"): three settled sentences drafted and sourced, then the author's own last sentence (contested).

```
claim(a_c03_volcanic_material, book, 3, "Volcanoes also supplied raw
   material: sulfur dioxide and hydrogen sulfide, which dissolve in water
   as sulfite and sulfide, and silica-rich hot-spring water that crusts
   surfaces with sinter.", settled, empirical).
source(a_c03_volcanic_material, 'Ranjan', 2018).
source(a_c03_volcanic_material, 'Djokic', 2017).
says(a_c03_volcanic_material, volcanic_sulfur_gases_in_water, forms,
   sulfite_and_sulfide).
says(a_c03_volcanic_material, hot_spring_silica, deposits_as, sinter).
claim(a_c03_lightning_nitrogen, book, 3, "The same lightning, like that
   of ordinary storms, turned nitrogen from the air into nitrogen oxides,
   which end up in water as nitrite and, mostly, nitrate.", settled,
   empirical).
source(a_c03_lightning_nitrogen, 'Navarro-González', 2001).
source(a_c03_lightning_nitrogen, 'Aroskay', 2024).
source(a_c03_lightning_nitrogen, 'Ranjan', 2019).
says(a_c03_lightning_nitrogen, prebiotic_dissolved_nitrogen_oxides,
   dominant_form, nitrate).
claim(a_c03_ash_and_sulfate, book, 3, "Coarse ash blankets the ground
   near the vent; sulfur gas lofted high becomes sulfate droplets that
   stay aloft for about a year, spreading on the wind.", settled,
   empirical).
source(a_c03_ash_and_sulfate, 'Robock', 2000).
```

```
says(a_c03_ash_and_sulfate, volcanic_sulfate_aerosol_residence, years,
   approx(1)).
```

The author's own sentence. Ranjan et al.'s models agree with its second half: shallow lakes could hold useful sulfite, and nitrite and nitrate were likely scarce in the ocean, if not in some ponds.

```
claim(a_c03_amounts_contested, book, 3, "Whether these compounds were at
   concentrations to matter is still contested, although one can imagine
   smaller pools and lakes accumulating meaningful levels more so than
   the ocean.", contested, empirical).
source(a_c03_amounts_contested, 'Ranjan', 2018).
source(a_c03_amounts_contested, 'Ranjan', 2019).
says(a_c03_amounts_contested,
   prebiotic_pool_sulfur_and_nitrogen_anion_levels, known, false).

claim(a_c03_rounding_error, book, 3, "Ten million years is a rounding
   error in Earth's history.", settled, quantitative).
says(a_c03_rounding_error, pool_trial_window, years, 1.0e7).
says(a_c03_rounding_error, ten_million_years, share_of_earth_history,
   negligible).
common_knowledge(a_c03_rounding_error).
```

Added 2026-11-07 from the author's sticky note (the "mega-cycle" count); the author chose the Earth's-age sentence alone. 4.54e9 / 1e7 = 454.

```
claim(a_c03_earth_spans, book, 3, "Earth is about 4.54 billion years old,
   so its history holds about 450 ten-million-year spans.", settled,
   quantitative).
source(a_c03_earth_spans, 'Dalrymple', 2001).
says(a_c03_earth_spans, earth_age, years, approx(4.54e9)).
computes(a_c03_earth_spans, 4.54e9 / 1.0e7, approx(450)).

claim(a_c03_invented, book, 3, "The six-minute figure is invented, and
   the honest thing is to say so and then show that it does not matter.",
   settled, supposition).
says(a_c03_invented, supposed_pool_cycle, minutes, 6).
```

The italic note that follows (l.48) says "This way of framing the problem is the author's", and ch.1 lists "the reading of early Earth as a vast parallel trial" among the author's own; so author, though the extractor sees no marker here.

```
claim(a_c03_parallel_picture, book, 3, "So the picture is not one slow
   process grinding away for four billion years. It is an extremely large
   number of short, parallel, independent, cheap experiments, nearly all
   of which failed within minutes, running simultaneously across an
   entire planet.", author, interpretive).
says(a_c03_parallel_picture, origin_of_life_search, framing,
   many_short_parallel_trials).
depends_on(a_c03_parallel_picture, c3_table_day).
depends_on(a_c03_parallel_picture, c3_daily).

claim(a_c03_components_established, book, 3, "The underlying components —
   wet-dry cycling, spontaneous assembly of amphiphiles, the role of
   mineral surfaces — are well established and are taken up again in
   chapters 4, 5, and 17.", settled, empirical).
source(a_c03_components_established, 'Rajamani', 2008).
source(a_c03_components_established, 'Damer', 2020).
says(a_c03_components_established, prebiotic_cycling_and_self_assembly,
   established, true).

claim(a_c03_selection_bias, book, 3, "This is selection bias, and it is
   unusually severe in this case, because the surviving lineage is the
   only one that produced observers capable of asking about it.",
   settled, interpretive).
claim(a_c03_probability_unknown, book, 3, "How probable life was, given
   early Earth's conditions, remains genuinely unknown.", settled,
   empirical).
says(a_c03_probability_unknown, probability_of_life_given_early_earth,
   known, false).
depends_on(a_c03_probability_unknown, a_c03_selection_bias).
```

Chapter 3 (HS)

```
claim(a_c03hs_six_minutes, 'book-hs', 3, "Now suppose the pool gets
   stirred and resettles every six minutes.", settled, supposition).
says(a_c03hs_six_minutes, supposed_pool_cycle, minutes, 6).
computes(a_c03hs_six_minutes, 24 * 60 / 6, 240).
computes(a_c03hs_six_minutes, 24 * 60 / 6 * 365, approx(88000)).
computes(a_c03hs_six_minutes, 24 * 60 / 6 * 365 * 100, approx(8.8e6)).
computes(a_c03hs_six_minutes, 24 * 60 / 6 * 365 * 1.0e7,
   approx(1.0e12)).   % "about a trillion"

claim(a_c03hs_nothing, 'book-hs', 3, "Ten million years is nothing in
   Earth's history.", settled, quantitative).
says(a_c03hs_nothing, pool_trial_window, years, 1.0e7).
says(a_c03hs_nothing, ten_million_years, share_of_earth_history,
   negligible).
common_knowledge(a_c03hs_nothing).
```

```
claim(a_c03hs_made_up, 'book-hs', 3, "I made up the six minutes, and I
   should say so, because you should always be suspicious of a number
   someone picked.", settled, supposition).
says(a_c03hs_made_up, supposed_pool_cycle, minutes, 6).

claim(a_c03hs_cut_ten, 'book-hs', 3, "That cuts the total by ten — about
   ninety billion.", settled, quantitative).
computes(a_c03hs_cut_ten, (24 * 60 / 6) / 24, 10).
computes(a_c03hs_cut_ten, 24 * 365 * 1.0e7, approx(9.0e10)).
says(a_c03hs_cut_ten, trials_hourly_cycle_ten_million_years, count,
   approx(9.0e10)).
common_knowledge(a_c03hs_cut_ten).

claim(a_c03hs_cut_240, 'book-hs', 3, "That cuts it by 240 — about four
   billion.", settled, quantitative).
computes(a_c03hs_cut_240, (24 * 60 / 6) / 1, 240).
computes(a_c03hs_cut_240, 365 * 1.0e7, approx(4.0e9)).
says(a_c03hs_cut_240, trials_daily_cycle_ten_million_years, count,
   approx(4.0e9)).
common_knowledge(a_c03hs_cut_240).
```

Unboxed, so settled by the HS edition's own rule, although the "My own idea" box right after it (l.42) says this framing is the author's. The full edition marks the same framing as the author's (a_c03_parallel_picture).

```
claim(a_c03hs_parallel_picture, 'book-hs', 3, "So the picture isn't one
   slow process grinding away forever. It's an absurd number of short,
   cheap, parallel experiments, almost all of which failed within
   minutes, running everywhere at the same time.", author,
   interpretive).   % boxed as the author's since 2026-11-03, as the full
   edition marks it
says(a_c03hs_parallel_picture, origin_of_life_search, framing,
   many_short_parallel_trials).
depends_on(a_c03hs_parallel_picture, a_c03hs_six_minutes).
```

Moved out of the "My own idea" box on 2026-11-03, so settled, as in the full edition.

```
claim(a_c03hs_components_solid, 'book-hs', 3, "The pieces underneath it
   are solid: wet-dry cycles really do drive chemistry, those split-
   personality molecules really do assemble on their own, and both have
   been demonstrated in labs many times.", settled, empirical).
source(a_c03hs_components_solid, 'Rajamani', 2008).
source(a_c03hs_components_solid, 'Damer', 2020).
says(a_c03hs_components_solid, prebiotic_cycling_and_self_assembly,
   established, true).
```

Chapter 4 (full)

Both claims below were rewritten on 2026-11-03 at the author's direction. The window used to be "minutes to hours, from chapter 3", which is ch. 3's invented cycle time inside a framing ch. 3 marks as the author's. An environment is now treated as a concept that can be drawn at any scale, and the window as "enough time for something to happen". Line 19 was qualified to match, since some environments (deep pore water) do hold still for ten thousand years.

```
claim(a_c04_ten_thousand_years, book, 4, "Reactions that are
   thermodynamically allowed but take ten thousand years at ambient
   temperature might as well not happen in most places, because few
   conditions hold still that long.", settled, generalization).

claim(a_c04_reachable, book, 4, "It changes which possibilities are
   reachable within the time a given environment stays put.", settled,
   generalization).
claim(a_c04_environment_concept, book, 4, "It names a concept people
   made, not a unit nature hands over.", settled, interpretive).
says(a_c04_environment_concept, environment, kind, human_concept).
claim(a_c04_window, book, 4, "the only measure that matters is whether
   there was enough time for something to happen", settled,
   interpretive).
says(a_c04_window, relevant_timescale, measure,
   enough_time_for_something_to_happen).

claim(a_c04_template_ligation, book, 4, "Presented with two shorter
   fragments corresponding to its own halves, it binds them against
   itself and accelerates their joining into another complete copy.",
   settled, empirical).
source(a_c04_template_ligation, 'Lee', 1996).
says(a_c04_template_ligation, ghadiri_peptide_fragments, count, 2).

claim(a_c04_self_repro, book, 4, "Self-reproduction is a chemical
   property available to fairly simple molecules, and not something that
   had to wait for life to invent it.", settled, generalization).
depends_on(a_c04_self_repro, c4_peptide).
depends_on(a_c04_self_repro, c4_nucleic).

claim(a_c04_theory_solid, book, 4, "The theoretical result is solid: in a
   sufficiently rich network, self-sustaining subsets are expected.",
   settled, generalization).
source(a_c04_theory_solid, 'Hordijk', 2004).
says(a_c04_theory_solid, collective_autocatalysis_in_rich_networks,
   expected_in_theory, true).
```

```
claim(a_c04_convection, book, 4, "Past a threshold, the liquid abruptly
   organizes itself into a tiling of convection cells, often hexagonal,
   each circulating in a fixed direction.", settled, empirical).
says(a_c04_convection, benard_convection_cells, typical_sides, 6).
common_knowledge(a_c04_convection).

claim(a_c04_prigogine, book, 4, "Ilya Prigogine's work on these systems,
   recognized with a Nobel Prize in 1977, is the reason the phrase
   exists.", settled, historical).
source(a_c04_prigogine, 'Prigogine', 1978).
says(a_c04_prigogine, prigogine_nobel_prize, year, 1977).
event_year(a_prigogine_nobel, 1977, 'Nobel Prize in Chemistry 1977').
```

Rewritten on 2026-11-03 at the author's direction. "Steps 1 through 5 are supported" rested on collectively autocatalytic sets, whose prebiotic relevance the same chapter calls not settled. The text now separates what is directly demonstrated from what has been shown only in the laboratory, and calls the sequence a conjecture.

```
claim(a_c04_steps_shown, book, 4, "Steps 1 through 3 are directly
   demonstrated", settled, interpretive).
depends_on(a_c04_steps_shown, c4_formose).
claim(a_c04_steps_lab_only, book, 4, "Steps 4 and 5 have been shown only
   in the laboratory, in systems that are small and mostly designed
   rather than found", settled, empirical).
depends_on(a_c04_steps_lab_only, c4_peptide).
claim(a_c04_conjecture, book, 4, "So the sequence as a whole is a
   conjecture, and stays a conjecture until the chemistry is better
   understood.", contested, interpretive).
depends_on(a_c04_conjecture, c4_status).
claim(a_c04_step6, book, 4, "Step 6 is where the account becomes
   genuinely incomplete.", settled, interpretive).

claim(a_c04_three_limits, book, 4, "Three honest limits, since a chapter
   that presents only the encouraging results is not worth much.",
   settled, interpretive).
says(a_c04_three_limits, chapter4_stated_limits, count, 3).
```

Chapter 4 (HS)

Full edition: formaldehyde "plausibly abundant on the early Earth". Since 2026-11-03 the HS sentence separates presence (almost certain) from amount (uncertain), matching that hedge.

```
claim(a_c04hs_formose, 'book-hs', 4, "This is real, it's been known since
   1861, and it uses a chemical the early Earth almost certainly made,
   though maybe not in the amounts the reaction needs.", settled,
   historical).
source(a_c04hs_formose, 'Butlerow', 1861).
says(a_c04hs_formose, formose_reaction_described, year, 1861).

claim(a_c04hs_peptide, 'book-hs', 4, "Better: in 1996 researchers built a
   small protein, 32 building blocks long, that makes copies of itself.",
   settled, empirical).
source(a_c04hs_peptide, 'Lee', 1996).
says(a_c04hs_peptide, ghadiri_peptide, residues, 32).

claim(a_c04hs_template_ligation, 'book-hs', 4, "Give it the two halves of
   itself and it grabs them, holds them against its own body, and helps
   them join into a complete new copy.", settled, empirical).
source(a_c04hs_template_ligation, 'Lee', 1996).
says(a_c04hs_template_ligation, ghadiri_peptide_fragments, count, 2).
```

“How long is long enough?” was added on 2026-11-03 at the author’s direction. It is the full edition’s environment point (a_c04_environment_concept, a_c04_window) in second person, using the same subjects so that the two editions get compared.

```
claim(a_c04hs_environment_concept, 'book-hs', 4, "An environment isn't
   something nature hands you; it's an idea people use, and you can draw
   it at any size.", settled, interpretive).
says(a_c04hs_environment_concept, environment, kind, human_concept).
claim(a_c04hs_window, 'book-hs', 4, "There's no single right timescale,
   only the question of whether there was enough time for something to
   happen.", settled, interpretive).
says(a_c04hs_window, relevant_timescale, measure,
   enough_time_for_something_to_happen).
```

QT45 (Gianni et al. 2026), added on 2026-11-03 at the author’s request. Checked against the published structured abstract (Science; a copy saved by the author) and the accepted manuscript (PMC7618777). It is 45 nt, selected from random pools and then evolved and engineered further. It works with triplet substrates in mildly alkaline eutectic ice. It makes the (-) strand at 94.1% per-nucleotide fidelity, and a copy of itself with defined substrates, both at about 0.2% in 72 days, in separate reactions. At the observed fidelities the paper’s quasispecies model leaves under 0.5% of variants active.

```
event_year(qt45_reported, 2026, 'Gianni et al. 2026').
```

```
claim(a_c04_qt45, book, 4, "In 2026 a group reported an RNA enzyme only
   45 nucleotides long that copies RNA templates.", settled, empirical).
source(a_c04_qt45, 'Gianni', 2026).
says(a_c04_qt45, qt45_ribozyme, length_nt, 45).
claim(a_c04_qt45_strands, book, 4, "it can make both its complementary
   strand and a copy of itself, which are the two halves of a replication
   cycle", settled, empirical).
source(a_c04_qt45_strands, 'Gianni', 2026).
claim(a_c04_qt45_limits, book, 4, "with yields of about 0.2 per cent
   after 72 days and about 6 per cent of letters copied wrongly",
   settled, quantitative).
source(a_c04_qt45_limits, 'Gianni', 2026).
says(a_c04_qt45_limits, qt45_ribozyme, yield_after_72_days,
   approx(0.002)).
says(a_c04_qt45_limits, qt45_ribozyme, error_rate_per_letter,
   approx(0.06)).
claim(a_c04_qt45_reading, book, 4, "Its discoverers note that 45 letters
   approaches the lengths unaided chemistry may be able to string
   together", settled, interpretive).
source(a_c04_qt45_reading, 'Gianni', 2026).
claim(a_c04hs_qt45, 'book-hs', 4, "The newest one, reported in 2026, is
   different in one way: it didn't start as a design.", settled,
   historical).
source(a_c04hs_qt45, 'Gianni', 2026).
claim(a_c04hs_qt45_strands, 'book-hs', 4, "It's only 45 letters long, and
   it can build both strands a copy needs", settled, empirical).
source(a_c04hs_qt45_strands, 'Gianni', 2026).
says(a_c04hs_qt45_strands, qt45_ribozyme, length_nt, 45).
claim(a_c04hs_qt45_limits, 'book-hs', 4, "it gets about one letter in
   seventeen wrong, and after 72 days only about two copies in a thousand
   are finished", settled, quantitative).
source(a_c04hs_qt45_limits, 'Gianni', 2026).
says(a_c04hs_qt45_limits, qt45_ribozyme, yield_after_72_days,
   approx(0.002)).
says(a_c04hs_qt45_limits, qt45_ribozyme, error_rate_per_letter,
   approx(0.059)).
```

Moved out of the “Still argued about” box on 2026-11-03, so settled, as the full edition calls the same theoretical result “solid” (a_c04_theory_solid).

```
claim(a_c04hs_theory_shown, 'book-hs', 4, "It's been shown that in a big
   enough, messy enough mixture, loops like this should turn up almost
   unavoidably.", settled, generalization).
source(a_c04hs_theory_shown, 'Hordijk', 2004).
says(a_c04hs_theory_shown, collective_autocatalysis_in_rich_networks,
   expected_in_theory, true).
```

```
claim(a_c04hs_convection, 'book-hs', 4, "Then past a certain temperature
   the whole thing suddenly organizes itself into a tidy grid of
   circulating cells, often six-sided, like a honeycomb.", settled,
   empirical).
says(a_c04hs_convection, benard_convection_cells, typical_sides, 6).
common_knowledge(a_c04hs_convection).

claim(a_c04hs_three_limits, 'book-hs', 4, "Three honest problems, because
   a chapter with only good news isn't worth much.", settled,
   interpretive).
says(a_c04hs_three_limits, chapter4_stated_limits, count, 3).
```

Chapter 5 (full)

```
claim(a_c05_amphiphile_ends, book, 5, "Amphiphiles have two ends with
   opposite preferences: one that associates readily with water, one that
   does not.", settled, empirical).
says(a_c05_amphiphile_ends, amphiphile_ends, count, 2).
common_knowledge(a_c05_amphiphile_ends).

claim(a_c05_bilayer, book, 5, "The structure that does contain something
   is a bilayer vesicle — two sheets of amphiphiles back to back, tails
   meeting in the middle, water-facing heads on both outer surfaces, the
   whole thing curved around and closed.", settled, empirical).
source(a_c05_bilayer, 'Israelachvili', 1976).
says(a_c05_bilayer, bilayer_vesicle_wall, sheets, 2).

claim(a_c05_film, book, 5, "That arrangement encloses a volume of water,
   separated from the water outside by a film two molecules thick.",
   settled, quantitative).
says(a_c05_film, bilayer_membrane, thickness_in_molecules, 2).
common_knowledge(a_c05_film).

claim(a_c05_two_tails, book, 5, "Those with two tails, which cannot taper
   to a point, pack into sheets instead, and sheets close into
   vesicles.", settled, empirical).
source(a_c05_two_tails, 'Israelachvili', 1976).
says(a_c05_two_tails, double_tailed_amphiphiles, packing,
   bilayer_sheets).

claim(a_c05_phospholipid_tails, book, 5, "The phospholipids in every cell
   membrane have two tails.", settled, empirical).
says(a_c05_phospholipid_tails, membrane_phospholipid_tails, count, 2).
common_knowledge(a_c05_phospholipid_tails).   % textbook generalization
   ("every" admits exceptions such as four-chain cardiolipin)
```

```
claim(a_c05_gradient, book, 5, "It can hold a difference in the
   concentration and electrical charge of ions between its two sides — an
   electrochemical gradient.", settled, empirical).
common_knowledge(a_c05_gradient).

claim(a_c05_osmotic, book, 5, "The consequence, observed in the
   laboratory, is that vesicles with more encapsulated content grow — and
   they grow at the expense of emptier vesicles nearby, which shrink as
   their membrane molecules migrate.", settled, empirical).
source(a_c05_osmotic, 'Chen', 2004).
says(a_c05_osmotic, osmotic_vesicle_competition, demonstrated, true).
claim(a_c05_selection_without_life, book, 5, "This is as close as
   chemistry comes to selection without life, and it arrives early, from
   the geometry of amphiphilic molecules and the behaviour of water.",
   settled, interpretive).
depends_on(a_c05_selection_without_life, a_c05_osmotic).

claim(a_c05_vents, book, 5, "Some proposals for the origin of life invert
   the usual order here and start from gradients that already existed —
   the natural differences in acidity and chemistry across the mineral
   walls of alkaline hydrothermal vents, where the geology supplies the
   gradient and the first membranes only have to take advantage of it.",
   contested, empirical).
source(a_c05_vents, 'Martin', 2003).
source(a_c05_vents, 'Lane', 2010).
says(a_c05_vents, vent_gradients_before_membranes, proposed, true).
```

Chapter 5 (HS)

```
claim(a_c05hs_bilayer, 'book-hs', 5, "The structure that does have an
   inside is a double layer: two sheets of these molecules back to back,
   greasy tails pointed at each other in the middle, water-friendly ends
   facing out on both faces.", settled, empirical).
source(a_c05hs_bilayer, 'Israelachvili', 1976).
says(a_c05hs_bilayer, bilayer_vesicle_wall, sheets, 2).

claim(a_c05hs_film, 'book-hs', 5, "Curl that around and close it up and
   you have a bag — a pocket of water, sealed off from the water outside
   by a film two molecules thick.", settled, quantitative).
says(a_c05hs_film, bilayer_membrane, thickness_in_molecules, 2).
common_knowledge(a_c05hs_film).

claim(a_c05hs_two_tails, 'book-hs', 5, "Two tails, and they can't taper
   to a point, so they form sheets instead — and sheets close into
   bags.", settled, empirical).
source(a_c05hs_two_tails, 'Israelachvili', 1976).
```

```
says(a_c05hs_two_tails, double_tailed_amphiphiles, packing,
   bilayer_sheets).

claim(a_c05hs_phospholipid_tails, 'book-hs', 5, "The molecules in your
   cell membranes have two tails.", settled, empirical).
says(a_c05hs_phospholipid_tails, membrane_phospholipid_tails, count, 2).
common_knowledge(a_c05hs_phospholipid_tails).

claim(a_c05hs_gradient, 'book-hs', 5, "A membrane can also hold a
   difference between its two sides — more of some charged particle
   inside than outside.", settled, empirical).
common_knowledge(a_c05hs_gradient).
```

Boxed “Still argued about” on 2026-11-03, so contested, as the full edition presents it as a proposal.

```
claim(a_c05hs_vents, 'book-hs', 5, "Some researchers flip the order
   entirely and start from differences that were already there — the
   natural gaps in acidity across the rock walls of deep-sea vents, where
   the planet supplies the gradient and the first membranes only have to
   take advantage of it.", contested, empirical).
source(a_c05hs_vents, 'Martin', 2003).
source(a_c05hs_vents, 'Lane', 2010).
says(a_c05hs_vents, vent_gradients_before_membranes, proposed, true).
```

Claims: chapters 6–9

From `factcheck/claims/b_ch06-09.pl`

```
:- discontiguous claim/6, says/4, source/3, depends_on/2, computes/3,
   event_year/3,
                order_claim/3, interval_claim/4, variant_ok/3, gap/3,
   rests_on_gap/2,
                arc_gap/3, arc_descriptive/3, cite_ok/3, about/2,
   drift_ok/3,
                prose_ok/3, common_knowledge/1, certainty_ok/2.
```

Structured claims for chapters 6-9 of both editions (book/ and book-hs/). Schema and conventions: the header of ../claims.pl. Tiers are as the text presents them; comments say where that presentation looks wrong. Quotes under 40 characters are whole sentences that are themselves that short.

Chapter 6

```
claim(b_c06_transitions_proposed, book, 6, "John Maynard Smith and Eörs
   Szathmáry proposed in 1995 that the history of life is punctuated by a
   small number of major transitions", settled, historical).   % that
   they proposed it is settled; the list is disputed (b_c06_critics)
source(b_c06_transitions_proposed, 'Maynard Smith', 1995).
```

```
claim(b_c06_eight_entries, book, 6, "Their list has eight entries.",
   settled, quantitative).
source(b_c06_eight_entries, 'Maynard Smith', 1995).
says(b_c06_eight_entries, major_transitions_list, entries, 8).
```

the next sentence names them: compartments; chromosomes; DNA/protein division of labour; eukaryotes; sexual populations; multicellularity; colonies of non-reproducers; societies organized by language

```
computes(b_c06_eight_entries, 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1, 8).
```

```
claim(b_c06_critics, book, 6, "Critics have argued that the eight entries
   lack theoretical unity — that they were assembled because each is a
   large change rather than because they share a mechanism.", contested,
   interpretive).
source(b_c06_critics, 'O'Malley', 2016).

claim(b_c06_signature_survives, book, 6, "The signature is the part that
   survives the dispute, and it is the useful part.", settled,
   interpretive).
claim(b_c06hs_signature_agreed, 'book-hs', 6, "The signature is the part
   that survives the argument.", settled, interpretive).   % moved out of
   the HS "Still argued about" box on 2026-11-03, and worded as the full
   edition words it
says(b_c06_signature_survives, major_transitions_signature,
   survives_list_dispute, true).
says(b_c06hs_signature_agreed, major_transitions_signature,
   survives_list_dispute, true).
```

2026-11-04: the sentence now names what independence is exchanged for, "a dependency on cooperation". The claim is unchanged: what is given up is the capacity to continue alone.

```
claim(b_c06_given_up, book, 6, "Something is given up every time, and it
   is always the same thing: the capacity to continue alone, exchanged
   for a dependency on cooperation.", settled, generalization).
depends_on(b_c06_given_up, b_c06_signature_survives).
```

The Margulis rejection history. Corrected on 2026-11-03: "since 1964" had no source. Margulis 1995 says she wrote the paper in 1966, and Sagan 1967 is marked "Received 8 June 1966". The full edition now says she "recalled" about fifteen rejections; it was published in 1967 (c6_margulis).

```
claim(b_c06_sagan, book, 6, "Lynn Margulis — publishing in 1967 under the
   name Lynn Sagan — recalled that the paper that set out the modern case
   was turned down by around fifteen journals", settled, historical).
source(b_c06_sagan, 'Sagan', 1967).
source(b_c06_sagan, 'Margulis', 1995).
says(b_c06_sagan, endosymbiosis_modern_case_paper, published, 1967).
```

Until 2026-11-03 the HS text read "When this was proposed properly in 1967, it was rejected by around fifteen journals before one took it", which put the rejections after publication. It now has them before "one published it, in 1967".

```
claim(b_c06hs_rejected, 'book-hs', 6, "The paper that made the case
   properly was rejected by around fifteen journals before one published
   it, in 1967.", settled, historical).
source(b_c06hs_rejected, 'Margulis', 1995).
says(b_c06hs_rejected, endosymbiosis_paper_rejections, count,
   approx(15)).
says(b_c06hs_rejected, endosymbiosis_modern_case_paper, published, 1967).

claim(b_c06hs_two_reasons, 'book-hs', 6, "That's worth remembering for
   two reasons, and people usually only take the first.", settled,
   interpretive).
says(b_c06hs_two_reasons, margulis_episode_lessons, count, 2).
claim(b_c06hs_vindicated, 'book-hs', 6, "Also: a theory being rejected
   fifteen times and later vindicated is a real pattern, and it tells you
   nothing about any particular rejected theory.", settled,
   generalization).
claim(b_c06_declined_correctly, book, 6, "Most ideas that fifteen
   journals decline are declined correctly.", settled, generalization).
claim(b_c06hs_declined_correctly, 'book-hs', 6, "Most ideas that fifteen
   journals turn down were turned down correctly.", settled,
   generalization).
says(b_c06_declined_correctly, ideas_declined_by_many_journals,
   mostly_declined_correctly, true).
says(b_c06hs_declined_correctly, ideas_declined_by_many_journals,
   mostly_declined_correctly, true).

claim(b_c06hs_fission, 'book-hs', 6, "They don't get built from scratch —
   they split in two, like bacteria.", settled, empirical).
common_knowledge(b_c06hs_fission).

claim(b_c06_mito_ancestor, book, 6, "The ancestral bacterium would have
   carried something on the order of a few thousand genes.", settled,
   quantitative).
claim(b_c06hs_mito_ancestor, 'book-hs', 6, "The original bacterium would
   have carried a few thousand genes.", settled, quantitative).
source(b_c06_mito_ancestor, 'Timmis', 2004).
source(b_c06hs_mito_ancestor, 'Timmis', 2004).
says(b_c06_mito_ancestor, mitochondrial_ancestor_genes, count,
   range(2000, 9000)).
says(b_c06hs_mito_ancestor, mitochondrial_ancestor_genes, count,
   range(2000, 9000)).

claim(b_c06hs_thirty_seven, 'book-hs', 6, "Your mitochondria have thirty-
   seven.", settled, quantitative).
source(b_c06hs_thirty_seven, 'Anderson', 1981).
says(b_c06hs_thirty_seven, human_mitochondrial_genes, count, 37).
```

```
claim(b_c06_genes_migrated, book, 6, "The rest were not discarded — the
   great majority migrated to the host cell's nucleus, where they
   remain", settled, empirical).
source(b_c06_genes_migrated, 'Timmis', 2004).

claim(b_c06_integration, book, 6, "Integration at this level is not
   cooperation between two parties; it is the construction of a single
   system that can no longer be separated into the things it was made
   from.", settled, interpretive).
claim(b_c06hs_not_two_partners, 'book-hs', 6, "That's not two partners
   cooperating.", settled, interpretive).
says(b_c06_integration, mitochondrial_integration, kind,
   single_inseparable_system).
says(b_c06hs_not_two_partners, mitochondrial_integration, kind,
   single_inseparable_system).
depends_on(b_c06_integration, b_c06_mito_ancestor).
depends_on(b_c06_integration, c6_mito).
depends_on(b_c06_integration, b_c06_genes_migrated).
depends_on(b_c06hs_not_two_partners, b_c06hs_mito_ancestor).
depends_on(b_c06hs_not_two_partners, b_c06hs_thirty_seven).

claim(b_c06_combination, book, 6, "They can arise by combination — two
   evolutionary histories merging into one.", settled, generalization).
depends_on(b_c06_combination, b_c06_integration).
```

Corrected on 2026-11-03. The head start is over animal multicellularity, since red algae are older (Aktipis 2015). The HS edition now excludes cells with no nucleus, which by count are most of them (Sender 2016).

```
claim(b_c06_billion_years, book, 6, "They carry machinery for division
   that predates animal multicellularity by a billion years.", settled,
   quantitative).
claim(b_c06hs_billion_years, 'book-hs', 6, "Every one of them that has a
   nucleus still carries the machinery to divide — machinery that's a
   billion years older than animals.", settled, quantitative).
says(b_c06_billion_years,
   division_machinery_head_start_on_multicellularity, years,
   approx(1.0e9)).
says(b_c06hs_billion_years,
   division_machinery_head_start_on_multicellularity, years,
   approx(1.0e9)).

claim(b_c06hs_cells, 'book-hs', 6, "Your body is a deal between about
   thirty trillion cells.", settled, quantitative).   % no entry in
   book/19 gives a cell count; the extractor has no "trillion" scale and
   records 30
says(b_c06hs_cells, human_body_cells, count, approx(3.0e13)).
```

```
claim(b_c06_cancer, book, 6, "Cancer is what the arrangement looks like
   when the checks fail.", settled, interpretive).
source(b_c06_cancer, 'Aktipis', 2015).
claim(b_c06_suppression, book, 6, "The general shape is this: a higher
   level of organization persists only while it suppresses competition at
   the level below it, and the suppression is never complete.", settled,
   generalization).
depends_on(b_c06_suppression, b_c06_cancer).
```

Chapter 7

```
claim(b_c07_skin_sites, book, 7, "Two sites on the same person can differ
   more from each other than the same site differs between two people.",
   settled, empirical).
claim(b_c07hs_skin_sites, 'book-hs', 7, "Two spots on your own body can
   differ from each other more than the same spot differs between you and
   a stranger.", settled, empirical).
source(b_c07_skin_sites, 'Grice', 2011).
source(b_c07hs_skin_sites, 'Grice', 2011).
says(b_c07_skin_sites, skin_microbiome_site_variation,
   within_person_can_exceed_between_people, true).
says(b_c07hs_skin_sites, skin_microbiome_site_variation,
   within_person_can_exceed_between_people, true).

claim(b_c07_into_occupied, book, 7, "Skin as it now exists — layered,
   cornified, chemically selective — evolved into that environment.",
   settled, historical).
claim(b_c07_two_follow, book, 7, "Two things follow that are worth
   stating plainly.", settled, interpretive).   % the skin sets
   conditions rather than assembling its microbiome; the barrier is a
   process
says(b_c07_two_follow, skin_barrier_consequences, count, 2).
depends_on(b_c07_two_follow, b_c07_into_occupied).

claim(b_c07_two_records, book, 7, "So a single molecule carries two
   records at once: what selection could never permit to change, and what
   it never had reason to protect.", settled, interpretive).
claim(b_c07hs_two_records, 'book-hs', 7, "So one molecule carries two
   records at once: what could never be allowed to change, and what there
   was never any reason to protect.", settled, interpretive).
says(b_c07_two_records, rrna_16s_gene, records_carried, 2).
says(b_c07hs_two_records, rrna_16s_gene, records_carried, 2).
```

Was “form a domain of life” (1977); “domain” is the 1990 term. Corrected 2026-11-03.

```
claim(b_c07_woese, book, 7, "Carl Woese used this gene in 1977 to
   establish that archaea are a primary branch of life distinct from
   bacteria", settled, historical).   % Woese & Fox 1977 said "primary
   kingdom"; the rank "domain" is Woese, Kandler & Wheelis 1990
source(b_c07_woese, 'Woese', 1977).
```

Winogradsky columns (added 2026-11-06 at the author's request; both editions). Sources, read in the open-access texts: Rundell et al. 2014 (PLoS ONE 9: e104134) for the column, the oxygen and sulfide gradients, cellulose breakdown, ferrous sulfide, the green and red-violet zones, and their absence in that survey's columns; Esteban et al. 2015 (PLOS ONE 10: e0134588) for blackening "within the first week", orange "most likely iron oxides", the light-dependent surface biofilm with its cyanobacteria, and rare taxa becoming abundant.

```
claim(b_c07_winogradsky, book, 7, "A Winogradsky column, named for the
   microbiologist Sergei Winogradsky, who devised it, is a clear tube
   filled with sediment enriched with a source of cellulose, such as leaf
   litter, and a source of sulfate, then left in the light.", settled,
   empirical).
source(b_c07_winogradsky, 'Rundell', 2014).
source(b_c07_winogradsky, 'Esteban', 2015).
claim(b_c07_oxygen_falls, book, 7, "At the top there is oxygen, and it
   falls away with depth, so the mud below is anoxic, without oxygen.",
   settled, empirical).
source(b_c07_oxygen_falls, 'Rundell', 2014).
claim(b_c07_cellulose, book, 7, "Decomposers break the cellulose into
   simpler carbon compounds that other microbes ferment.", settled,
   empirical).
source(b_c07_cellulose, 'Rundell', 2014).
claim(b_c07_blackening, book, 7, "Down in the anoxic mud, bacteria
   respire sulfate in place of oxygen and release hydrogen sulfide, which
   combines with iron to form ferrous sulfide; the mud turns black, in
   one study within the first week.", settled, empirical).
source(b_c07_blackening, 'Rundell', 2014).
source(b_c07_blackening, 'Esteban', 2015).
says(b_c07_blackening, winogradsky_blackening, within_weeks, 1).
claim(b_c07_sulfide_up, book, 7, "The sulfide diffuses upwards towards
   the oxygen, so each depth offers a different mix, and a different
   niche.", settled, empirical).
source(b_c07_sulfide_up, 'Rundell', 2014).
claim(b_c07_bands, book, 7, "Green sulfur bacteria typically form a green
   zone in the lower middle of the column, and purple bacteria a red-
   violet zone above it.", settled, empirical).
source(b_c07_bands, 'Rundell', 2014).
```

```
says(b_c07_bands, winogradsky_green_zone, below_red_violet_zone, true).
claim(b_c07_orange, book, 7, "Near the top, patches of orange appear,
   most likely iron oxides.", settled, empirical).
source(b_c07_orange, 'Esteban', 2015).
claim(b_c07_surface_film, book, 7, "On the inside of the tube, a biofilm
   grows that is less diverse than the mud behind it, with cyanobacteria
   among its commonest members; in columns kept in the dark, no visible
   surface community forms.", settled, empirical).
source(b_c07_surface_film, 'Esteban', 2015).
says(b_c07_surface_film, winogradsky_surface_biofilm, needs_light, true).
claim(b_c07_bands_typical, book, 7, "The textbook bands are typical, not
   guaranteed: in one survey of columns, distinct green and red-violet
   zones did not appear.", settled, empirical).
source(b_c07_bands_typical, 'Rundell', 2014).
says(b_c07_bands_typical, winogradsky_bands, always_appear, false).
claim(b_c07_rare_abundant, book, 7, "Organisms that are rare in the
   starting mud can become abundant once conditions suit them, so the
   column does not import a community; it selects one from what was
   already there.", settled, empirical).
source(b_c07_rare_abundant, 'Esteban', 2015).
says(b_c07_rare_abundant, winogradsky_rare_taxa, can_become_abundant,
   true).
```

The author's "mini holobiont, in a way, if logical to say so": logical only as a labelled analogy, since the chapter's own definition requires a host.

```
claim(b_c07_not_holobiont, book, 7, "A column is not a holobiont in the
   sense used above, because there is no host.", author, interpretive).
claim(b_c07_holobiont_analogy, book, 7, "But the whole tube behaves a
   little like one: a self-contained system whose members live on one
   another's by-products, with a boundary, a division of labour, and
   conditions that its own residents keep remaking.", author,
   interpretive).
depends_on(b_c07_holobiont_analogy, b_c07_not_holobiont).
```

HS twins: "A jar of mud" (ch. 7).

```
claim(b_c07hs_jar, 'book-hs', 7, "Science classes do this: fill a clear
   jar with pond mud, mix in some dead leaves and a little calcium
   sulfate, and leave it in the light.", settled, empirical).
source(b_c07hs_jar, 'Esteban', 2015).
common_knowledge(b_c07hs_jar).
claim(b_c07hs_blackening, 'book-hs', 7, "In one study, the bottom turned
   black within the first week.", settled, empirical).
source(b_c07hs_blackening, 'Esteban', 2015).
says(b_c07hs_blackening, winogradsky_blackening, within_weeks, 1).
```

```
claim(b_c07hs_sulfide, 'book-hs', 7, "Down there, where there's no
   oxygen, bacteria use sulfate the way you use oxygen, and give off
   hydrogen sulfide, the gas that smells of rotten eggs.", settled,
   empirical).
source(b_c07hs_sulfide, 'Rundell', 2014).
claim(b_c07hs_iron, 'book-hs', 7, "It reacts with iron in the mud and
   makes a black mineral.", settled, empirical).
source(b_c07hs_iron, 'Rundell', 2014).
claim(b_c07hs_bands, 'book-hs', 7, "Over the following weeks, bands of
   colour can appear in between: green, and higher up, reddish purple,
   made by bacteria that use that gas for photosynthesis.", settled,
   empirical).
source(b_c07hs_bands, 'Rundell', 2014).
says(b_c07hs_bands, winogradsky_green_zone, below_red_violet_zone, true).
claim(b_c07hs_orange, 'book-hs', 7, "Near the top, patches of rusty
   orange can show up within days.", settled, empirical).
source(b_c07hs_orange, 'Esteban', 2015).
claim(b_c07hs_film, 'book-hs', 7, "A film you can see grows on the inside
   of the jar, but only in the light.", settled, empirical).
source(b_c07hs_film, 'Esteban', 2015).
says(b_c07hs_film, winogradsky_surface_biofilm, needs_light, true).
claim(b_c07hs_not_always, 'book-hs', 7, "The bands don't always appear.",
   settled, empirical).
source(b_c07hs_not_always, 'Rundell', 2014).
says(b_c07hs_not_always, winogradsky_bands, always_appear, false).
claim(b_c07hs_rare, 'book-hs', 7, "In one study, most of the film was
   made of kinds that had been rare in the mud at the start.", settled,
   empirical).
source(b_c07hs_rare, 'Esteban', 2015).
says(b_c07hs_rare, winogradsky_rare_taxa, can_become_abundant, true).
```

Chapter 8

```
claim(b_c08_one_molecule, book, 8, "A single class of molecule doing both
   jobs is an attractive way to connect chemical activity with heredity,
   because it removes the need for two separate inventions.", settled,
   interpretive).   % the appeal of the RNA world, not its truth; the
   hypothesis itself is presented as contested

claim(b_c08_codon_length, book, 8, "Three nucleotides make a codon.",
   settled, quantitative).
claim(b_c08hs_read_three, 'book-hs', 8, "They're read three at a time.",
   settled, quantitative).
claim(b_c08hs_four_letters, 'book-hs', 8, "DNA is written in four
   letters.", settled, quantitative).
says(b_c08_codon_length, codon, nucleotides, 3).
says(b_c08hs_read_three, codon, nucleotides, 3).
```

```
says(b_c08hs_four_letters, dna_alphabet, letters, 4).
common_knowledge(b_c08_codon_length).
common_knowledge(b_c08hs_read_three).
common_knowledge(b_c08hs_four_letters).

claim(b_c08_meanings, book, 8, "Those 64 specify 20 standard amino acids
   plus a stop instruction. More codons than meanings means the code is
   redundant", settled, quantitative).
claim(b_c08hs_amino_acids, 'book-hs', 8, "Those 64 triplets specify 20
   amino acids — the building blocks of proteins — plus a \"stop here\"
   instruction.", settled, quantitative).
says(b_c08_meanings, standard_genetic_code, amino_acids, 20).
says(b_c08hs_amino_acids, standard_genetic_code, amino_acids, 20).
computes(b_c08_meanings, 64 - (20 + 1), more_than(0)).
common_knowledge(b_c08_meanings).
common_knowledge(b_c08hs_amino_acids).

claim(b_c08_third_four, book, 8, "For many amino acids, all four
   nucleotides in the third position of the codon specify the same amino
   acid.", settled, empirical).
claim(b_c08hs_third_four, 'book-hs', 8, "For a lot of amino acids, all
   four options in the third slot mean the same thing.", settled,
   empirical).
says(b_c08_third_four, fourfold_degenerate_third_position,
   synonymous_nucleotides, 4).
says(b_c08hs_third_four, fourfold_degenerate_third_position,
   synonymous_nucleotides, 4).
common_knowledge(b_c08_third_four).
common_knowledge(b_c08hs_third_four).

claim(b_c08_wobble, book, 8, "The flexibility of this position is
   sometimes described through the wobble in how transfer RNA pairs with
   the third base, a mechanism proposed by Crick in 1966.", settled,
   historical).   % that Crick proposed it in 1966 is not in dispute
source(b_c08_wobble, 'Crick', 1966).

claim(b_c08_first_two, book, 8, "The first two positions carry most of
   the meaning.", settled, empirical).
claim(b_c08hs_first_two, 'book-hs', 8, "The first two letters carry most
   of the meaning.", settled, empirical).
says(b_c08_first_two, codon_positions_carrying_most_meaning, count, 2).
says(b_c08hs_first_two, codon_positions_carrying_most_meaning, count, 2).
common_knowledge(b_c08_first_two).
common_knowledge(b_c08hs_first_two).

claim(b_c08_compression, book, 8, "That arrangement is what information
   compression looks like when it is working well: discard what can be
   discarded without losing the message, and protect what cannot.",
   settled, interpretive).
```

```
depends_on(b_c08_compression, b_c08_third_four).
depends_on(b_c08_compression, b_c08_first_two).

claim(b_c08_freeland, book, 8, "Freeland and Hurst compared it in 1998
   against large numbers of randomly generated alternative codes, scoring
   each by how much damage a copying or reading error would do.",
   settled, empirical).
claim(b_c08hs_freeland, 'book-hs', 8, "In 1998 two researchers compared
   the real code against enormous numbers of randomly scrambled
   alternative codes, scoring each by how much damage a typical error
   would do.", settled, empirical).
source(b_c08_freeland, 'Freeland', 1998).
source(b_c08hs_freeland, 'Freeland', 1998).
says(b_c08_freeland, freeland_hurst_code_study, year, 1998).
says(b_c08hs_freeland, freeland_hurst_code_study, year, 1998).
says(b_c08_freeland, freeland_hurst_code_study, authors, 2).
says(b_c08hs_freeland, freeland_hurst_code_study, authors, 2).
```

“The figure” is the one-in-a-million result (c8_million). Both editions say plainly that it depends on the model and has been argued over (the HS caveat was added on 2026-11-03), so it is settled as presented. The claim is about the figure’s robustness, not about its value.

```
claim(b_c08_figure_argued, book, 8, "The figure depends on which chemical
   property is weighted and which model of error is assumed, and it has
   been argued over since.", settled, interpretive).
says(b_c08_figure_argued, codes_beating_standard_code, figure_robustness,
   model_dependent).
claim(b_c08hs_figure_argued, 'book-hs', 8, "The exact figure depends on
   how you score the damage, and it's been argued over ever since.",
   settled, interpretive).
says(b_c08hs_figure_argued, codes_beating_standard_code,
   figure_robustness, model_dependent).
claim(b_c08hs_held_up, 'book-hs', 8, "What has held up is the main
   result: the real code is far better than a typical random code.",
   settled, empirical).
source(b_c08hs_held_up, 'Freeland', 1998).
depends_on(b_c08hs_held_up, c8hs_million).
```

QT45 at the heredity gap, added on 2026-11-03; sources and checks as for a_c04_qt45. The accepted manuscript (PMC7618777) says “How this capacity for heredity and evolution first emerged is unknown.”

```
claim(b_c08_qt45, book, 8, "The nearest approach so far is the 45-
   nucleotide RNA described in chapter 4", settled, empirical).
source(b_c08_qt45, 'Gianni', 2026).
```

```
says(b_c08_qt45, qt45_ribozyme, length_nt, 45).
claim(b_c08_qt45_unknown, book, 8, "Its discoverers still write that how
   the capacity for heredity first emerged is unknown.", settled,
   historical).
source(b_c08_qt45_unknown, 'Gianni', 2026).
claim(b_c08hs_qt45, 'book-hs', 8, "The closest anyone has come is the 45-
   letter RNA from chapter 4", settled, empirical).
source(b_c08hs_qt45, 'Gianni', 2026).
says(b_c08hs_qt45, qt45_ribozyme, length_nt, 45).
claim(b_c08_codon_usage, book, 8, "One line of criticism is specific and
   substantial: when the differing frequencies with which organisms
   actually use particular codons are taken into account, the apparent
   optimization shrinks considerably.", settled, empirical).
source(b_c08_codon_usage, 'Archetti', 2004).
event_year(b_freeland_hurst_comparison, 1998, 'Freeland & Hurst 1998').
event_year(b_codon_usage_criticism, 2004, 'Archetti 2004').
order_claim(b_c08_codon_usage, b_freeland_hurst_comparison,
   b_codon_usage_criticism).   % "argued over since"
claim(b_c08_held_up, book, 8, "The general finding — that the standard
   code sits far out in the tail rather than in the middle — has held
   up.", settled, empirical).
source(b_c08_held_up, 'Freeland', 1998).
depends_on(b_c08_held_up, c8_million).
depends_on(b_c08_held_up, b_c08_codon_usage).

claim(b_c08_three_readings, book, 8, "Three readings of that fact
   compete, and the differences matter:", contested, interpretive).
claim(b_c08hs_three_answers, 'book-hs', 8, "Why it's like that is
   genuinely unsettled, and there are three serious answers.", contested,
   interpretive).
says(b_c08_three_readings, genetic_code_origin_readings, count, 3).
says(b_c08hs_three_answers, genetic_code_origin_readings, count, 3).

claim(b_c08_frozen, book, 8, "Crick's 1968 proposal: the assignments were
   largely arbitrary, fixed very early, and became impossible to change
   afterwards", contested, interpretive).
source(b_c08_frozen, 'Crick', 1968).
claim(b_c08_neutral_interesting, book, 8, "The most interesting of the
   three for this book's purposes, and the one most easily missed.",
   settled, interpretive).   % the author's emphasis on the neutral-
   origin reading, not a verdict on it
claim(b_c08_neutral_models, book, 8, "Models of this kind produce codes
   nearly as error-tolerant as the real one.", settled, empirical).
source(b_c08_neutral_models, 'Massey', 2008).

claim(b_c08_resemblance, book, 8, "The two arrive at related behaviour by
   unrelated means — which is a resemblance, not a shared mechanism.",
   settled, interpretive).
```

Chapter 9

```
claim(b_c09_two_kinds, book, 9, "Two kinds of evidence are worth
   examining closely, because they point in opposite directions and are
   both informative.", settled, interpretive).
claim(b_c09hs_two_kinds, 'book-hs', 9, "Two kinds of evidence show this,
   and they point in opposite directions.", settled, interpretive).
says(b_c09_two_kinds, evolution_record_evidence_kinds, count, 2).
says(b_c09hs_two_kinds, evolution_record_evidence_kinds, count, 2).

claim(b_c09_eyes, book, 9, "Light-sensitive structures have arisen
   independently many times across the animal kingdom — a frequently
   cited survey put the number somewhere above 40", settled,
   quantitative).
source(b_c09_eyes, 'Salvini-Plawen', 1977).
says(b_c09_eyes, independent_eye_origins, count, more_than(40)).

claim(b_c09_flight, book, 9, "Powered flight evolved in insects, in
   pterosaurs, in birds, and in bats — four times, from four different
   starting anatomies.", settled, quantitative).
claim(b_c09hs_flight, 'book-hs', 9, "Powered flight evolved four separate
   times: insects, pterosaurs, birds, bats.", settled, quantitative).
claim(b_c09hs_bodies, 'book-hs', 9, "Four different starting bodies.",
   settled, quantitative).
says(b_c09_flight, powered_flight_origins, count, 4).
says(b_c09hs_flight, powered_flight_origins, count, 4).
says(b_c09_flight, powered_flight_origins, starting_anatomies, 4).
says(b_c09hs_bodies, powered_flight_origins, starting_anatomies, 4).
common_knowledge(b_c09_flight).
common_knowledge(b_c09hs_flight).
common_knowledge(b_c09hs_bodies).

claim(b_c09_physics, book, 9, "The physics of focusing light, of binding
   oxygen reversibly, and of generating lift are the same everywhere, and
   they permit only a limited range of workable solutions.", settled,
   generalization).
depends_on(b_c09_physics, b_c09_eyes).
depends_on(b_c09_physics, b_c09_flight).

claim(b_c09_ear_bones, book, 9, "Two of the three small bones that
   transmit sound from the eardrum to the inner ear are modified from
   bones that formed the jaw joint in earlier vertebrates", settled,
   empirical).
claim(b_c09hs_ear_bones, 'book-hs', 9, "Two of the three tiny bones that
   carry sound from your eardrum to your inner ear were, in your distant
   ancestors, part of the jaw joint.", settled, empirical).
source(b_c09_ear_bones, 'Shubin', 2008).
```

```
source(b_c09hs_ear_bones, 'Shubin', 2008).
says(b_c09_ear_bones, middle_ear_ossicles, count, 3).
says(b_c09hs_ear_bones, middle_ear_ossicles, count, 3).
says(b_c09_ear_bones, middle_ear_ossicles_from_jaw_joint, count, 2).
says(b_c09hs_ear_bones, middle_ear_ossicles_from_jaw_joint, count, 2).

claim(b_c09_two_joints, book, 9, "Some early mammal relatives had two
   working jaw joints at once: the older one, between quadrate and
   articular, and a new one between different bones alongside it.",
   settled, empirical).
claim(b_c09hs_two_joints, 'book-hs', 9, "Some early mammal relatives had
   two working jaw joints at the same time — the old one, and a new one
   right next to it made from different bones.", settled, empirical).
```

Luo 2011 (†) was these claims' source, but its text could not be checked, and chapter 9 does not cite it by name. Since 2026-11-03 they rest on Anthwal 2013, read in full (fix_cells.pl). Luo stays in the reading list, still marked unchecked.

```
says(b_c09_two_joints, early_mammal_relative_jaw_joints, working_at_once,
   2).
says(b_c09hs_two_joints, early_mammal_relative_jaw_joints,
   working_at_once, 2).

claim(b_c09_modify_only, book, 9, "Taken together these are the signature
   of a process that can only modify what is already present.", settled,
   interpretive).
depends_on(b_c09_modify_only, c9_giraffe).
depends_on(b_c09_modify_only, b_c09_ear_bones).
depends_on(b_c09_modify_only, b_c09_two_joints).

claim(b_c09_syncytin_lineages, book, 9, "Syncytins have been identified
   in primates, in several separate rodent groups, in carnivores, in
   ruminants, and in marsupials.", settled, empirical).
source(b_c09_syncytin_lineages, 'Lavialle', 2013).
source(b_c09_syncytin_lineages, 'Cornelis', 2015).
claim(b_c09_syncytin_independent, book, 9, "They come from unrelated
   retroviruses, captured independently at different times in different
   lineages, and co-opted for the same job each time.", settled,
   empirical).
source(b_c09_syncytin_independent, 'Lavialle', 2013).
claim(b_c09_borrowed_parts, book, 9, "They acquired it repeatedly, from
   whatever virus was available, which is convergent evolution running on
   borrowed parts.", settled, interpretive).
depends_on(b_c09_borrowed_parts, b_c09_syncytin_lineages).
depends_on(b_c09_borrowed_parts, b_c09_syncytin_independent).
```

```
claim(b_c09_two_complications, book, 9, "Two complications sit on top of
   that.", settled, interpretive).   % "ancestor" means a population; the
   early tree was web-like
says(b_c09_two_complications, luca_complications, count, 2).
```

Claims: chapters 10–13

From `factcheck/claims/c_ch10-13.pl`

```
:- discontiguous claim/6, says/4, source/3, depends_on/2, computes/3,
   event_year/3,
                order_claim/3, interval_claim/4, variant_ok/3, gap/3,
   rests_on_gap/2,
                arc_gap/3, arc_descriptive/3, cite_ok/3, about/2,
   drift_ok/3,
                prose_ok/2, common_knowledge/1, certainty_ok/2.
```

Structured claims for chapters 10-13 of both editions; schema in claims.pl. Claim ids and new event names carry the c_ prefix; subject keys do not. Several claims anchor sentences whose existing claim quotes fewer than 40 characters: checks.pl's covered/1 takes a 40-character prefix and fails on shorter quotes, though extract.py --uncovered counts them as covered.

Chapter 10

Five ways to make a spiral

```
claim(c_c10_fixed_ratio, book, 10, "A nautilus adds shell at the opening;
   the new chamber is larger than the last by a fixed ratio; the shape
   follows by necessity.", settled, empirical).
source(c_c10_fixed_ratio, 'Thompson', 1917).
says(c_c10_fixed_ratio, edge_growth_spiral_increment, size_vs_previous,
   fixed_ratio).
claim(c_c10hs_fixed_amount, 'book-hs', 10, "A snail adds shell only at
   the opening, each new section bigger than the last by a fixed
   proportion.", settled, empirical).
says(c_c10hs_fixed_amount, edge_growth_spiral_increment,
   size_vs_previous, fixed_ratio).
```

Corrected 2026-11-03: the text had "a fixed amount bigger", which the checker reported as a conflict. A fixed amount (a constant difference) is not a fixed ratio, and only a ratio gives the logarithmic spiral; the HS paragraph's own lead-in is "Grow at the

edge, in proportion." Encoded as each edition has it, so the checker reports the conflict.

Corrected on 2026-11-03 in both editions: "elongate at different rates" chose a side in an open question. Gerbode 2012 finds coiling by asymmetric contraction of a fibre ribbon, and Dilly 2026 by differential growth. Both editions now say only that the sides end up different lengths.

```
claim(c_c10_tendril, book, 10, "A tendril coils because its two sides
   change length by different amounts.", settled, empirical).
says(c_c10_tendril, tendril_coiling, cause,
   differential_side_length_change).
claim(c_c10_three_mechanisms, book, 10, "Same shape, three mechanisms
   with nothing in common.", settled, generalization).
says(c_c10_three_mechanisms, helices_from_rotation_and_advance,
   shared_mechanism, false).

claim(c_c10_fibonacci, book, 10, "Repeat that rule and the angle between
   successive units converges on about 137.5 degrees, which generates the
   interlocking spiral families and the Fibonacci counts that go with
   them.", settled, quantitative).
source(c_c10_fibonacci, 'Douady', 1992).
says(c_c10_fibonacci, phyllotaxis_spiral_counts, sequence,
   fibonacci).      % c10_golden quotes only 32 characters of this
   sentence

claim(c_c10_five_least, book, 10, "The least familiar of the five, and
   the one that makes a turning spiral with no material rotation at
   all.", settled, empirical).   % reworded on 2026-11-03: shells and
   seed heads need no material rotation either, so the old wording
   claimed a uniqueness the mechanism doesn't have
says(c_c10_five_least, excitable_medium_spiral_wave,
   needs_material_rotation, false).
says(c_c10_five_least, spiral_formation_mechanisms, count_described, 5).
claim(c_c10_five_mechanisms, book, 10, "Five conditions, five mechanisms,
   one family of shapes.", settled, generalization).
says(c_c10_five_mechanisms, spiral_formation_mechanisms, count_described,
   5).
```

What the recurrence shows

```
claim(c_c10_two_cautions, book, 10, "Two cautions keep this from running
   away.", settled, interpretive).
says(c_c10_two_cautions, spiral_argument_cautions, count, 2).
claim(c_c10_form_not_mechanism, book, 10, "The first is that similar form
   does not imply shared mechanism.", settled, generalization).
```

```
says(c_c10_form_not_mechanism, similar_form, implies_shared_mechanism,
   false).
claim(c_c10_galaxy_arms, book, 10, "Galaxy arms are not material
   structures that wind up over time; they are better understood as waves
   of compression passing through the disc, with stars moving in and out
   of them.", settled, empirical).
source(c_c10_galaxy_arms, 'Lin', 1964).
says(c_c10_galaxy_arms, galaxy_spiral_arms, nature, density_waves).
claim(c_c10_two_images, book, 10, "Putting those two images side by side
   invites an inference that the physics does not support.", settled,
   interpretive).
depends_on(c_c10_two_images, c_c10_galaxy_arms).
```

Coiled for storage

```
claim(c_c10_linking_number, book, 10, "For a strand whose ends cannot
   turn freely, a quantity called the linking number stays fixed, and it
   is the sum of two others: twist, how far the strand turns about its
   own axis, and writhe, how far that axis coils through space.",
   settled, quantitative).
claim(c_c10hs_twist_plus_coil, 'book-hs', 10, "That's a rule, not bad
   luck: for a long strand whose ends can't spin freely, the amount it
   twists plus the amount it coils stays the same.", settled,
   quantitative).
source(c_c10_linking_number, 'Fuller', 1971).
source(c_c10hs_twist_plus_coil, 'Fuller', 1971).
says(c_c10_linking_number, strand_twist_plus_writhe,
   conserved_when_ends_fixed, true).
says(c_c10hs_twist_plus_coil, strand_twist_plus_writhe,
   conserved_when_ends_fixed, true).
```

Was “Most of them hold 46”; by count most human cells are red blood cells, which have no nucleus.

```
claim(c_c10hs_46_strands, 'book-hs', 10, "Most kinds of cell in your body
   hold 46 very long strands of DNA, wound around tiny protein spools —
   about thirty million in each cell — and folded again.", settled,
   quantitative).
says(c_c10hs_46_strands, human_chromosomes_per_cell, count, 46).      %
   c10hs_nucleosomes quotes only 33 characters of this sentence

claim(c_c10_one_kind, book, 10, "Rope, cable, line, and chromosome are
   one kind of object: a one-dimensional sequence that has to be packed
   into three dimensions and later unpacked in the order it went in.",
   author, interpretive).
depends_on(c_c10_one_kind, c10_spiral_principle).       % "this chapter's
   principle, extension that keeps its own history, put to practical use"
```

```
depends_on(c_c10_one_kind, c_c10_linking_number).       % "one piece of
   accounting covers a hose and a chromosome"
```

Spirals people make

```
claim(c_c10_newgrange_two_kinds, book, 10, "Its best-known stones set the
   two kinds side by side.", settled, empirical).
```

O'Kelly 1982 (†) could not be read, and the text doesn't name it. Since 2026-11-03 the claim rests on the Discovery Programme's records of kerbstones K1 and K52 (fix_spirals.pl).

```
says(c_c10_newgrange_two_kinds, newgrange_motif_kinds, count, 2).
   % curved and straight-edged
claim(c_c10_pairing, book, 10, "The pairing reads as a record of the two
   ways a bounded space gets filled.", author, interpretive).
depends_on(c_c10_pairing, c_c10_newgrange_two_kinds).
depends_on(c_c10_pairing, c10_spiral_principle).
```

c10_newgrange_reading opens with "Here the author would go further.", which the splitter makes a 33-character sentence of its own, so that claim anchors no sentence; this one anchors the reading.

HS twins (added 2026-11-06: the author asked for Book 1's neolithic spiral art in the HS edition, "without more research, just simplify and use our same resources/references"). Same sources as the full edition's c10_newgrange and c_c10_newgrange_two_kinds.

```
claim(c_c10hs_newgrange, 'book-hs', 10, "Newgrange, a huge stone tomb in
   Ireland, was built around 3100 BCE, more than five thousand years
   ago.", settled, historical).
source(c_c10hs_newgrange, 'Smyth', 2025).
claim(c_c10hs_newgrange_two_kinds, 'book-hs', 10, "The carving at the
   entrance puts spirals and diamonds side by side, and so does the stone
   directly opposite, at the back of the mound.", settled, empirical).
source(c_c10hs_newgrange_two_kinds, 'Discovery Programme', 2014).
claim(c_c10hs_newgrange_families, 'book-hs', 10, "Many of its stones
   carry carvings that come in a few repeating families: curvy ones, like
   circles and spirals, and straight-edged ones, like zigzags, parallel
   lines and diamond shapes.", settled, empirical).
source(c_c10hs_newgrange_families, 'Discovery Programme', 2014).
says(c_c10hs_newgrange_families, newgrange_motif_kinds, count, 2).
   % curved and straight-edged
```

```
claim(c_c10_form_constants, book, 10, "Geometric hallucinations fall into
   four recurring families, two of which are lattices and spirals — the
   pair carved together at Newgrange — and mathematical models of the
   wiring between the eye and the visual cortex reproduce those families
   from the structure of the cortex itself.", settled, empirical).
source(c_c10_form_constants, 'Bressloff', 2001).
says(c_c10_form_constants, hallucination_form_constants, count, 4).
claim(c_c10_entoptic, book, 10, "Archaeologists have argued that Irish
   passage-tomb art, and much older cave art, draws on imagery of this
   kind.", contested, empirical).
source(c_c10_entoptic, 'Dronfield', 1995).
source(c_c10_entoptic, 'Lewis-Williams', 1988).
depends_on(c_c10_entoptic, c_c10_form_constants).
```

The quote stops before “and concluded that van Gogh must have observed real turbulent flow”, the conclusion the next two paragraphs take apart (c_c10_vase_test).

```
claim(c_c10_kolmogorov_slope, book, 10, "Treating the brightness of the
   paint as a stand-in for a quantity carried by a flow, they found that
   its spectrum — how much variation there is at each scale, from the
   largest swirl to the smallest — falls off with the same slope that
   Andrei Kolmogorov's 1941 theory predicts for fully developed
   turbulence", settled, empirical).
source(c_c10_kolmogorov_slope, 'Ma', 2024).
source(c_c10_kolmogorov_slope, 'Kolmogorov', 1941).
event_year(c_kolmogorov_turbulence_theory, 1941, 'Kolmogorov 1941').
event_year(c_starry_night_spectrum_study, 2024, 'Ma et al. 2024').
order_claim(c_c10_kolmogorov_slope, c_kolmogorov_turbulence_theory,
   c_starry_night_spectrum_study).
claim(c_c10_earlier_study, book, 10, "An earlier study had reported a
   related match in the statistics of brightness across several of his
   late paintings.", settled, empirical).
source(c_c10_earlier_study, 'Aragón', 2008).
event_year(c_van_gogh_luminance_study, 2008, 'Aragón et al. 2008').
order_claim(c_c10_earlier_study, c_van_gogh_luminance_study,
   c_starry_night_spectrum_study).
claim(c_c10_vase_test, book, 10, "A test that a vase of flowers also
   passes cannot be what identifies turbulence.", settled, interpretive).
depends_on(c_c10_vase_test, c10_degas).
```

Becoming rather than being: the full edition marks the whole section as the author’s reading. The HS edition states unboxed (“Spirals are shapes of still going”) what its later “My own idea” box restates.

```
claim(c_c10hs_still_going, 'book-hs', 10, "Those are shapes of having
   arrived. Spirals are shapes of still going.", settled,
   generalization).
claim(c_c10hs_two_groups, 'book-hs', 10, "Things sort into two groups:
   settled ones, shaped by the forces on them right now, and running ones
   — growing, circulating — whose shape also records the path they took.
   The spirals are all in the second group.", author, interpretive).
says(c_c10hs_still_going, spirals, system_class, still_running).
```

The box's own claim is the sorting of every system into settled and running, which the full edition also marks as the author's ("Becoming rather than being"). Spirals as shapes of still going is unmarked in both editions (full ch. 10, "What the recurrence shows").

```
says(c_c10hs_two_groups, all_systems, sorting, settled_or_running).
depends_on(c_c10hs_still_going, c10hs_spiral_principle).
depends_on(c_c10hs_two_groups, c10hs_spiral_principle).
```

Since 2026-11-03 this paragraph has its own heading ("Where the reading stops"), outside the author's section, so the developmental fact is settled, as in the HS edition.

```
claim(c_c10_hair_whorl, book, 10, "Hair whorls arise from the orientation
   of follicles during skin development", settled, empirical).
claim(c_c10hs_hair_whorl, 'book-hs', 10, "Hair whorls come from the
   direction follicles point as skin develops", settled, empirical).
source(c_c10_hair_whorl, 'Guo', 2004).
source(c_c10hs_hair_whorl, 'Guo', 2004).
says(c_c10_hair_whorl, hair_whorl, origin,
   follicle_orientation_in_development).
says(c_c10hs_hair_whorl, hair_whorl, origin,
   follicle_orientation_in_development).
```

Coda: a planet that hums

```
claim(c_c10_schumann_freq, book, 10, "The cavity has resonant
   frequencies, the lowest around 7.8 Hz, and they are measurable with
   the right instruments anywhere on Earth.", settled, quantitative).
source(c_c10_schumann_freq, 'Balser', 1960).
says(c_c10_schumann_freq, schumann_resonance_fundamental, hz,
   approx(7.8)).
```

Chapter 11

```
claim(c_c11_two_things, book, 11, "It is a way of holding two things at
   once — that a process repeats, and that repeating it changes what it
   repeats into.", settled, interpretive).
claim(c_c11_same_constraint, book, 11, "Chapter 10 described this in
   space — extension that cannot revisit its own past. This is the same
   constraint operating in time.", settled, interpretive).
depends_on(c_c11_same_constraint, c10_spiral_principle).

claim(c_c11_ice_cycles_co2, book, 11, "across eight glacial cycles, and
   carries a direct record of past carbon dioxide — not inferred from a
   proxy, but measured from the air itself", settled, empirical).
claim(c_c11hs_ice, 'book-hs', 11, "The longest Antarctic record goes back
   about 800,000 years, through eight ice ages.", settled, quantitative).
claim(c_c11hs_co2, 'book-hs', 11, "The carbon dioxide levels of 800,000
   years ago aren't estimated from something else.", settled, empirical).
source(c_c11_ice_cycles_co2, 'EPICA community members', 2004).
source(c_c11_ice_cycles_co2, 'Lüthi', 2008).
source(c_c11hs_ice, 'EPICA community members', 2004).
source(c_c11hs_co2, 'Lüthi', 2008).
says(c_c11hs_ice, longest_antarctic_ice_record, years, approx(800000)).
says(c_c11_ice_cycles_co2, longest_antarctic_ice_record, glacial_cycles,
   8).
says(c_c11hs_ice, longest_antarctic_ice_record, glacial_cycles, 8).
says(c_c11_ice_cycles_co2, ice_core_co2_record, measured_directly, true).
says(c_c11hs_co2, ice_core_co2_record, measured_directly, true).
says(c_c11_ice_cycles_co2, ice_core_co2_record, years, approx(800000)).
says(c_c11hs_co2, ice_core_co2_record, years, approx(800000)).
```

An aside on cycles at the largest scale (“None of it is established”). As with c11_smolin_proposal: who proposed what, and when, is settled; the proposals are not.

```
claim(c_c11_three_cyclic, book, 11, "The question of whether the universe
   had a beginning or a previous state has several live answers, and at
   least three of them take a cyclic shape.", settled, empirical).
says(c_c11_three_cyclic, cyclic_cosmological_proposals, count,
   at_least(3)).
claim(c_c11_ccc_history, book, 11, "Conformal cyclic cosmology, developed
   by Roger Penrose and set out for general readers in Cycles of Time
   (2010)", settled, historical).
source(c_c11_ccc_history, 'Penrose', 2010).
event_year(c_cycles_of_time_published, 2010, 'Penrose 2010').
```

```
claim(c_c11_hawking_points, book, 11, "Penrose and collaborators argued
   in 2020 that faint circular features in the cosmic microwave
   background are traces of black hole evaporation in a previous aeon.",
   contested, empirical).
source(c_c11_hawking_points, 'An', 2020).
event_year(c_hawking_points_claim, 2020, 'An et al. 2020').
claim(c_c11_cns_interesting, book, 11, "Cosmological natural selection,
   proposed by Lee Smolin in 1992, is the most structurally interesting
   of the three for this book's purposes, because it applies the
   vocabulary of chapter 9 to physics itself.", settled, interpretive).
says(c_c11_cns_interesting, cyclic_cosmological_proposals, count,
   3).      % c11_smolin_proposal quotes only 30 characters of this
   sentence
```

The text dates the statement 2004; the reference list has it only as Smolin 2007.

```
claim(c_c11_stakes_2004, book, 11, "He stated the stakes plainly in 2004:
   a neutron star heavier than about 1.5 solar masses would be troubling
   for the theory", settled, historical).
source(c_c11_stakes_2004, 'Smolin', 2007).
event_year(c_smolin_stakes_stated, 2004, 'the text: "in 2004"; listed as
   Smolin 2007').
order_claim(c_c11_stakes_2004, cns_proposed, c_smolin_stakes_stated).
order_claim(c_c11_stakes_2004, c_smolin_stakes_stated,
   first_two_solar_mass_pulsar).
```

```
claim(c_c11_attraction_three, book, 11, "The attraction of all three is
   obvious, and it is worth being explicit about why that is a reason for
   caution rather than confidence.", settled, interpretive).
says(c_c11_attraction_three, cyclic_cosmological_proposals, count, 3).
```

Chapter 12

```
claim(c_c12_giant_viruses, book, 12, "Giant viruses discovered in the
   last two decades have genomes larger than those of some free-living
   bacteria and encode components of the protein-synthesis machinery that
   viruses were supposed to lack entirely.", settled, empirical).
claim(c_c12hs_giant_viruses, 'book-hs', 12, "Giant viruses found in the
   last twenty years have more genes than some free-living bacteria, and
   carry bits of protein-building machinery that viruses were supposedly
   defined by not having.", settled, empirical).
source(c_c12_giant_viruses, 'Raoult', 2004).
source(c_c12hs_giant_viruses, 'Raoult', 2004).
says(c_c12_giant_viruses, giant_virus_discovery, years_before_manuscript,
   approx(20)).
says(c_c12hs_giant_viruses, giant_virus_discovery,
   years_before_manuscript, approx(20)).
```

```
says(c_c12_giant_viruses, giant_viruses,
   genome_exceeds_some_free_living_bacteria, true).
says(c_c12hs_giant_viruses, giant_viruses,
   genome_exceeds_some_free_living_bacteria, true).
says(c_c12_giant_viruses, giant_viruses,
   encode_protein_synthesis_components, true).
says(c_c12hs_giant_viruses, giant_viruses,
   encode_protein_synthesis_components, true).
computes(c_c12_giant_viruses, 2026 - 2004, approx(20)).        %
   Mimivirus genome (Raoult 2004) to the manuscript's date
computes(c_c12hs_giant_viruses, 2026 - 2004, approx(20)).
```

```
claim(c_c12_not_converged, book, 12, "A hundred and twenty attempts have
   not converged, and the failure to converge may not be a failure of
   effort.", settled, interpretive).
says(c_c12_not_converged, published_definitions_of_life, count,
   approx(120)).
says(c_c12_not_converged, definition_of_life, agreed, false).
depends_on(c_c12_not_converged, c12_definitions).
```

The HS "Still argued about" box holds a statement of fact (there is no agreed definition) that the full edition makes plainly; what is argued about is what life is.

```
claim(c_c12hs_no_agreed, 'book-hs', 12, "There's no agreed definition of
   life, and the disagreement isn't close to being resolved.", settled,
   interpretive).   % unboxed on 2026-11-03: a plain fact about the
   field, as the full edition states it
says(c_c12hs_no_agreed, definition_of_life, agreed, false).
claim(c_c12hs_not_pinned, 'book-hs', 12, "A thing with a hundred and
   twenty has not.", settled, interpretive).
says(c_c12hs_not_pinned, published_definitions_of_life, count,
   approx(120)).
says(c_c12hs_not_pinned, life_concept, pinned_down, false).
depends_on(c_c12hs_not_pinned, c12hs_definitions).
depends_on(c_c12hs_no_agreed, c_c12hs_not_pinned).      % the box ends
   "What follows is why."
```

Chapter 13

```
claim(c_c13_discard, book, 13, "It is a device that throws away about
   ninety-nine per cent of what it collects before transmission",
   settled, quantitative).
claim(c_c13hs_discard, 'book-hs', 13, "Your eye discards about ninety-
   nine per cent of what it collects before sending anything.", settled,
   quantitative).
says(c_c13_discard, eye_input_discarded_before_transmission, percent,
   approx(99)).
```

```
says(c_c13hs_discard, eye_input_discarded_before_transmission, percent,
   approx(99)).
computes(c_c13_discard, 100 * (1 - 1.0e6 / 9.6e7), approx(99)).
computes(c_c13hs_discard, 100 * (1 - 1.0e6 / 9.6e7), approx(99)).
depends_on(c_c13_discard, c13_retina).
depends_on(c_c13_discard, c13_fibres).
depends_on(c_c13hs_discard, c13hs_retina).
depends_on(c_c13hs_discard, c13hs_fibres).
common_knowledge(c_c13_discard).
common_knowledge(c_c13hs_discard).
```

Corrected in both editions on 2026-11-03. The text had five receptors per fibre at the centre of vision and thousands at the edge. Curcio & Allen 1990 found more ganglion cells than cones at the fovea, enough for each cone to have its own fibre and perhaps an ON and an OFF one. The average is about a hundred to one, so the periphery runs to hundreds.

```
claim(c_c13_convergence_range, book, 13, "the reduction is not uniform —
   at the centre of vision each receptor has an output fibre to itself,
   and may have two, while towards the edge a single fibre serves
   hundreds", settled, quantitative).
claim(c_c13hs_convergence_range, 'book-hs', 13, "(It isn't even — at the
   centre of your vision each sensor gets at least one wire to itself,
   maybe two, and at the edges one wire serves hundreds.", settled,
   quantitative).
source(c_c13_convergence_range, 'Curcio', 1990).
source(c_c13hs_convergence_range, 'Curcio', 1990).
says(c_c13_convergence_range, central_vision_receptors_per_fibre, count,
   at_most(1)).
says(c_c13hs_convergence_range, central_vision_receptors_per_fibre,
   count, at_most(1)).
says(c_c13_convergence_range, peripheral_receptors_per_fibre, count,
   at_least(100)).
says(c_c13hs_convergence_range, peripheral_receptors_per_fibre, count,
   at_least(100)).
```

```
claim(c_c13_humans_good, book, 13, "Humans are unusually good at
   transferring patterns across domains — noticing that two unlike
   systems seem to work alike — and this supports analogy, invention,
   scientific insight, and storytelling.", settled, generalization).
claim(c_c13hs_humans_good, 'book-hs', 13, "Humans are unusually good at
   noticing that two unrelated things seem to work alike — which gives us
   analogy, invention, science, and storytelling.", settled,
   generalization).
says(c_c13_humans_good, human_cross_domain_pattern_finding, ability,
   unusually_good).
```

```
says(c_c13hs_humans_good, human_cross_domain_pattern_finding, ability,
   unusually_good).
```

Corrected on 2026-11-03 in both editions. The text said there was no signal at all, and that the feeling of fit was "if anything stronger for the false ones". The experiments (Salvi 2016, Danek & Wiley 2017, Laukkonen 2020; fix_culture.pl) find a real but unreliable signal: insight solutions are more often right, the feeling is stronger for right ones, false insights occur, and an unrelated "aha" can make falsehoods feel true.

```
claim(c_c13_no_signal, book, 13, "The capacity has no reliable internal
   signal distinguishing the two cases.", settled, empirical).
says(c_c13_no_signal, pattern_finding_faculty,
   internal_signal_true_vs_false, unreliable).
claim(c_c13_only_correction, book, 13, "It is the only available
   correction for a faculty that cannot reliably audit itself.", settled,
   interpretive).
depends_on(c_c13_only_correction, c_c13_no_signal).
```

A fugue is an exercise. Rewritten on 2026-11-03 at the author's direction. The section used to state flatly that demanding exercises change the systems that perform them, and then call the direction of causation genuinely unsettled. The claim is now offered as a theory, beside an undisputed bodily case (calluses). The fugue "exercises" the capacity, and "builds" it only if the theory holds.

```
claim(c_c13_exercise_changes, book, 13, "That is a demanding exercise,
   and the theory is that demanding exercises change the systems that
   perform them.", contested, generalization).
claim(c_c13_calluses, book, 13, "a guitarist's fingertips harden into
   calluses where the strings press", settled, empirical).
common_knowledge(c_c13_calluses).
claim(c_c13_musicians, book, 13, "Trained musicians show measurable
   differences in auditory and motor regions relative to non-musicians,
   and the differences scale with training.", settled, empirical).
source(c_c13_musicians, 'Hyde', 2009).
claim(c_c13_causation_unsettled, book, 13, "The direction of causation is
   genuinely unsettled, and the honest version has to say so", contested,
   empirical).
claim(c_c13_fugue_trains, book, 13, "a cultural artefact built to be
   complex in a specific way, which exercises the capacity needed to
   handle it", settled, interpretive).
claim(c_c13_fugue_builds, book, 13, "which exercises the capacity needed
   to handle it and, if the theory holds, builds it.", contested,
   interpretive).
depends_on(c_c13_exercise_changes, c_c13_musicians).
```

```
depends_on(c_c13_exercise_changes, c_c13_causation_unsettled).
depends_on(c_c13_fugue_builds, c_c13_exercise_changes).
```

Claims: chapters 14–20

From `factcheck/claims/d_ch14-20.pl`

```
:- discontiguous claim/6, says/4, source/3, depends_on/2, computes/3,
   event_year/3,
                order_claim/3, interval_claim/4, variant_ok/3, gap/3,
   rests_on_gap/2,
                arc_gap/3, arc_descriptive/3, cite_ok/3, about/2,
   drift_ok/3,
                prose_ok/3, common_knowledge/1, certainty_ok/2.
:- multifile claim/6, says/4, source/3, depends_on/2, computes/3,
   event_year/3,
           order_claim/3, interval_claim/4, variant_ok/3, gap/3,
   rests_on_gap/2,
           arc_gap/3, arc_descriptive/3, cite_ok/3, about/2,
   drift_ok/3,
           prose_ok/3, common_knowledge/1, certainty_ok/2.
```

Structured claims for book chapters 14, 15, 16, 17 and 20, and book-hs chapters 14-16. Schema: the header of factcheck/claims.pl. HS chapter 16 restates material from full chapters 16, 17 and 20, so its claims sit beside their full-edition twins, and where the full edition cites a work for the same statement the HS claim carries that source too.

Chapter 14

```
claim(d_c14_protochemistry, book, 14, "What the two have in common is a
   combinatorial structure with rules about what may attach to what.",
   settled, interpretive).
says(d_c14_protochemistry, language_and_chemistry, shared_mechanism,
   false).

claim(d_c14_two_true, book, 14, "Two things are nonetheless true and
   worth separating out.", settled, interpretive).
claim(d_c14hs_two_true, 'book-hs', 14, "But there are two things
   underneath it that are true, and leaving them out would be its own
   kind of dishonesty.", settled, interpretive).
says(d_c14_two_true, snow_vocabulary_true_residue, count, 2).      %
   polysynthesis makes counting meaningless; the sea-ice surveys
```

```
says(d_c14hs_two_true, snow_vocabulary_true_residue, count, 2).
depends_on(d_c14_two_true, c14_seaice).
depends_on(d_c14hs_two_true, c14hs_seaice).

claim(d_c14hs_boas_modest, 'book-hs', 14, "He didn't suggest four was a
   lot or that it meant anything.", settled, historical).
source(d_c14hs_boas_modest, 'Boas', 1911).
says(d_c14hs_boas_modest, boas_snow_roots, count, 4).
```

c14_wcs (claims.pl) quotes the full-edition sentence but has no says/4; this one gives the HS "more than a hundred languages" a full-edition figure to be compared with.

```
says(c14_wcs, world_color_survey_languages, count, 110).
claim(d_c14hs_warm_colours, 'book-hs', 14, "Across more than a hundred
   languages, colours get named most precisely when they belong to things
   — a ripe fruit, a red flower — and those are mostly warm colours.",
   settled, empirical).
source(d_c14hs_warm_colours, 'Gibson', 2017).
says(d_c14hs_warm_colours, world_color_survey_languages, count,
   more_than(100)).
```

HS "All four are Arabic." is under 40 characters, so no quote can cover it; the HS claim anchors the sentence before it and carries the count.

```
claim(d_c14_desert_terms, book, 14, "What landscapes reliably multiply is
   vocabulary for the states of things that matter — the sea-ice terms
   above, and in deserts the kinds of ground, which geologists found
   useful enough to borrow: erg, hamada, sabkha, and wadi from Arabic",
   settled, generalization).
claim(d_c14hs_desert_terms, 'book-hs', 14, "Geologists describing deserts
   say erg for a sea of dunes, hamada for a bare rocky plateau, sabkha
   for a salt flat, and wadi for a riverbed that's dry until a storm
   fills it. All four are Arabic.", settled, empirical).
says(d_c14_desert_terms, arabic_desert_terms_in_geology, count, 4).
says(d_c14hs_desert_terms, arabic_desert_terms_in_geology, count, 4).
computes(d_c14hs_desert_terms, 1 + 1 + 1 + 1, 4).               % erg,
   hamada, sabkha, wadi
depends_on(d_c14_desert_terms, c14_seaice).
```

Full-edition twins of two readings HS ch. 14 now boxes "Still argued about" (c14hs_number_meaning, c14hs_narrow_version in claims.pl; the colour reading is c14_colour_meaning there). The full text states both unhedged, though its own line 68 says "anyone claiming the question is settled in either direction is ahead of the evidence". The two says/4 on c14hs_* annotate claims.pl's claims, which have none, so the checker can compare the tiers.

```
claim(d_c14_number_meaning, book, 14, "The researchers' reading is that
   the difficulty lies not in perceiving or matching exact quantity but
   in holding one in memory without a symbol for it.", contested,
   interpretive).   % attributed to the researchers on 2026-11-03,
   matching the HS box: an interpretation of the result, not the result
source(d_c14_number_meaning, 'Frank', 2008).
says(d_c14_number_meaning, exact_quantity_difficulty, located_in,
   memory_without_symbol).
says(c14hs_number_meaning, exact_quantity_difficulty, located_in,
   memory_without_symbol).
claim(d_c14_narrow_version, book, 14, "Habitually used linguistic
   categories appear to be recruited during cognitive tasks, and to
   affect speed and accuracy where the categories are relevant.",
   contested, interpretive).   % hedged in the text ("appear to be"); the
   HS box agrees
says(d_c14_narrow_version, habitual_category_recruitment, status,
   supported_narrowly).
says(c14hs_narrow_version, habitual_category_recruitment, status,
   supported_narrowly).
depends_on(d_c14_narrow_version, c14_colour_meaning).
depends_on(d_c14_narrow_version, d_c14_number_meaning).
depends_on(d_c14_narrow_version, c14_replication).

claim(d_c14_word_compression, book, 14,"A single short token stands for a
   category with indefinitely many members, no two identical, and makes
   that category available for combination with others.", settled,
   generalization).
claim(d_c14hs_word_compression, 'book-hs', 14, "One short sound stands in
   for a whole category of things, no two of which are identical.",
   settled, generalization).
says(d_c14_word_compression, word_as_category, lossy, true).
says(d_c14hs_word_compression, word_as_category, lossy, true).
```

Chapter 15

```
claim(d_c15_two_differences, book, 15, "This creates parallels with
   biological evolution, with two differences large enough that the
   parallel has to be held loosely.", settled, generalization).
claim(d_c15hs_two_differences, 'book-hs', 15, "That sounds like
   evolution, and it rhymes with it, with two differences big enough to
   matter.", settled, generalization).
says(d_c15_two_differences, culture_vs_genetic_evolution, differences,
   2).   % faster and not parent-to-offspring only; lower fidelity
says(d_c15hs_two_differences, culture_vs_genetic_evolution, differences,
   2).
```

```
says(d_c15_two_differences, cultural_transmission, fidelity_vs_genetic,
   lower).
says(d_c15hs_two_differences, cultural_transmission, fidelity_vs_genetic,
   lower).
```

Dawkins and the word “meme” (added 2026-11-06 at the author’s request; both editions). Checked in Wikipedia’s “Meme” and “The Selfish Gene” (offline copy, 2025-08): coined by the evolutionary biologist in The Selfish Gene (1976); his examples included melodies, catchphrases and fashion; “mimeme”, from the Greek for “imitated thing”, shortened to sound like “gene”.

```
claim(d_c15_meme_coined, book, 15, "The word was coined in 1976 by the
   evolutionary biologist Richard Dawkins, in The Selfish Gene, on the
   model of gene, for a tune, a catchphrase or a fashion copied from one
   mind to another as a gene is copied from parent to offspring.",
   settled, historical).
source(d_c15_meme_coined, 'Dawkins', 1976).
says(d_c15_meme_coined, meme_word, coined_by, dawkins).
says(d_c15_meme_coined, meme_word, year_coined, 1976).
claim(d_c15hs_meme_book, 'book-hs', 15, "In 1976 Richard Dawkins, a
   biologist who studies evolution, wrote a book called The Selfish
   Gene.", settled, historical).
source(d_c15hs_meme_book, 'Dawkins', 1976).
says(d_c15hs_meme_book, meme_word, year_coined, 1976).
claim(d_c15hs_meme_named, 'book-hs', 15, "In it he made up a word for a
   tune, a catchphrase or a fashion that spreads by being copied from one
   person to the next, the way a gene is copied from parent to child.",
   settled, historical).
source(d_c15hs_meme_named, 'Dawkins', 1976).
says(d_c15hs_meme_named, meme_word, coined_by, dawkins).
claim(d_c15hs_meme_greek, 'book-hs', 15, "He took mimeme, from an old
   Greek word meaning \"something imitated\", and cut it short so it
   would sound like gene.", settled, historical).
source(d_c15hs_meme_greek, 'Dawkins', 1976).
claim(d_c15hs_meme_online, 'book-hs', 15, "Today the word mostly means a
   funny picture that spreads online.", settled, generalization).
common_knowledge(d_c15hs_meme_online).
claim(d_c15hs_meme_argued, 'book-hs', 15, "Whether memes really explain
   how culture works is an open question.", contested, interpretive).
claim(d_c15hs_meme_caveats, 'book-hs', 15, "They get copied far less
   faithfully, and they can't be pulled apart from the people who hold
   them.", contested, interpretive).
```

```
claim(d_c15_two_kinds, book, 15, "How early is worth stating carefully,
   because the evidence comes in two kinds.", settled, historical).
source(d_c15_two_kinds, 'Santini', 2022).
```

```
says(d_c15_two_kinds, nixtamalization_dating_evidence, kinds, 2).   %
   equipment (c15_nixta_equipment); residues (c15_nixta_direct)
```

On 2026-11-03 the Staller & Carrasco volume was replaced by the chapter that carries the support, Green 2010 (fix_culture.pl).

The full edition's span is now "more than a thousand years, and perhaps for over three thousand". The direct evidence is eighth or ninth century CE (c15_nixta_direct). The equipment, which line 25 calls only "suggestive", is 1200–1500 BCE.

```
claim(d_c15_thousands, book, 15, "The practice worked for more than a
   thousand years, and perhaps for over three thousand, without anyone
   knowing why.", settled, historical).
```

The HS twin had the same unhedged "thousands of years" and was aligned on 2026-11-03.

```
claim(d_c15hs_thousands, 'book-hs', 15, "The practice worked for more
   than a thousand years, and maybe for over three thousand, with nobody
   knowing why.", settled, historical).
depends_on(d_c15hs_thousands, c15_nixta_equipment).
depends_on(d_c15hs_thousands, c15_nixta_direct).
says(d_c15_thousands, nixtamalization_unexplained_span, years,
   at_least(1000)).
says(d_c15hs_thousands, nixtamalization_unexplained_span, years,
   at_least(1000)).
depends_on(d_c15_thousands, c15_nixta_equipment).

claim(d_c15_mcount, book, 15, "In such chains, content that is mildly
   counterintuitive survives better than content that is either entirely
   ordinary or wildly strange.", settled, empirical).
source(d_c15_mcount, 'Barrett', 2001).
source(d_c15_mcount, 'Norenzayan', 2006).
says(d_c15_mcount, mildly_counterintuitive_content,
   transmission_retention, higher).

claim(d_c15_list_caveats, book, 15, "Two things need saying plainly about
   the list of functions, because lists of this kind are easy to
   misread.", settled, interpretive).
says(d_c15_list_caveats, religion_function_list_caveats, count, 2).  %
   function is not origin; no single origin or function

claim(d_c15_talk_past, book, 15, "Two people may be addressing genuinely
   overlapping concerns — why there is suffering, what obligations people
   have to one another, what happens at the end — in terms that do not
   map onto each other at any point.", settled, generalization).
```

Chapter 16

```
claim(d_c16_sixteen_in, book, 16, "Sixteen chapters in, the shape of the
   argument can be set out in one place.", settled, interpretive).

claim(d_c16_biology_stretch, book, 16, "Once templates are copied with
   variation in a population, standard evolutionary biology carries the
   account through multicellularity, specialization, nervous systems, and
   the capacity to build and use internal models.", settled,
   generalization).
claim(d_c16_language_weaker, book, 16, "That representation became
   externally transmissible is clear. How, and in what stages, is not,
   and this book says little about it that would survive a linguist's
   scrutiny.", settled, empirical).
says(d_c16_language_weaker, language_origin_stages, known, false).
claim(d_c16_above_language, book, 16, "Cultural selection, economic
   persistence, and religious transmission are described here with the
   same terms used for chemistry and biology, and the machinery is
   unrelated.", settled, interpretive).
claim(d_c16hs_above_language, 'book-hs', 16, "Cultural selection and
   biological selection get described with the same words, and the
   machinery is unrelated.", settled, interpretive).
says(d_c16_above_language, cultural_and_biological_selection,
   shared_mechanism, false).
says(d_c16hs_above_language, cultural_and_biological_selection,
   shared_mechanism, false).
depends_on(d_c16_above_language, d_c15_two_differences).          %
   "Chapter 15 makes it explicit for transmission fidelity."
```

“in principle” keeps the verdict off c16_prebiotic_open, so there is no edge to it.

```
claim(d_c16_verdict, book, 16, "An honest reader should take the
   progression as firm in principle in its lower half, broken at one
   specific joint, firm again through biology, and increasingly a matter
   of description rather than mechanism towards the top.", settled,
   interpretive).
depends_on(d_c16_verdict, c16_lower_half).
depends_on(d_c16_verdict, c16_no_new_principle).
depends_on(d_c16_verdict, c16_break).
depends_on(d_c16_verdict, d_c16_biology_stretch).
depends_on(d_c16_verdict, d_c16_language_weaker).
depends_on(d_c16_verdict, d_c16_above_language).
```

The full edition qualified this stretch on 2026-11-03 (“in principle”; c16_prebiotic_open). HS ch. 16 line 13 still states it flatly, while HS ch. 4 line 35 boxes the

premise “Still argued about”. c4_status is that box’s full-edition twin. The nearest HS claim is a_c04hs_theory_shown, but it covers the box’s mathematical sentence, not the “rich enough” clause; retarget the edge if an HS twin of c4_status is added. HS ch. 16 repeated the overstatement fixed in full ch. 16; corrected the same way on 2026-11-03.

```
claim(d_c16hs_lower_half, 'book-hs', 16, "Self-sustaining reactions are
   chemistry: chapter 4 shows them working, and small ones built in
   labs.", settled, empirical).
depends_on(d_c16hs_lower_half, c4_peptide).
claim(d_c16hs_prebiotic_open, 'book-hs', 16, "Whether early Earth's
   chemistry was rich enough to make them on its own is still argued
   about, as chapter 4 says.", settled, empirical).
says(d_c16hs_prebiotic_open, prebiotic_sufficiency_question, open, true).
says(c16_prebiotic_open, prebiotic_sufficiency_question, open, true).

claim(d_c16_four_recurring, book, 16, "Four things appear at nearly every
   level, and unlike the vocabulary they are structural rather than
   terminological.", settled, generalization).
says(d_c16_four_recurring, recurring_structural_features, count, 4).
computes(d_c16_four_recurring, 1 + 1 + 1 + 1, 4).   % boundary,
   throughput, differential persistence, something carried forward; the
   "fifth" is announced separately

claim(d_c16_two_routes, book, 16, "The two routes are the same process at
   different stages, and most relationships never finish the journey.",
   settled, generalization).
says(d_c16_two_routes, routes_to_organization, count, 2).          %
   nesting; relationship
```

Chapter 17

```
claim(d_c17_labelled, book, 17, "Every comparison in this book is
   supposed to be labelled as a comparison, and chapters 9, 10, and 14 go
   out of their way to show where surface similarity hides unrelated
   mechanisms.", settled, interpretive).
depends_on(d_c17_labelled, d_c14_protochemistry).
```

--- three claims, and what happened when they were checked

“each tied to a particular chapter”: the third is tied to two, chapters 14 and 15.

```
claim(d_c17_three_offered, book, 17, "An earlier draft of this chapter
   offered three, each tied to a particular chapter, and presented them
   as tests someone might run.", settled, interpretive).
```

```
claim(d_c16hs_three_listed, 'book-hs', 16, "An earlier version of this
   chapter listed three claims this framework makes that could be tested,
   and presented them as experiments somebody should run.", settled,
   interpretive).
says(d_c17_three_offered, framework_checkable_claims, count, 3).
says(d_c16hs_three_listed, framework_checkable_claims, count, 3).

claim(d_c17_cycling, book, 17, "Repeated wet-dry cycling drives
   nucleotides to polymerize into long chains without chemical
   activation, in conditions resembling those at volcanic hot springs",
   settled, empirical).
source(d_c17_cycling, 'Rajamani', 2008).
source(d_c17_cycling, 'Damer', 2020).
claim(d_c17_compartments, book, 17, "Ribozymes encapsulated in model
   protocells adapt faster under selection than the same sequences
   unencapsulated, and compartmentalization protects a cooperative
   replicating system from being overrun by parasitic sequences that
   would otherwise destroy it.", settled, empirical).
source(d_c17_compartments, 'Lai', 2021).
source(d_c17_compartments, 'Matsumura', 2016).
```

Line 48 used to say chapters 14 and 15 claim retention is biased toward what is easy to transmit "rather than what is accurate or useful". That contradicted chapter 14's colour vocabulary tracking usefulness and chapter 15's useful practice. Since 2026-11-03: "not only by".

```
claim(d_c17_retention_bias, book, 17, "Transmission-chain experiments —
   passing material along a line of participants and measuring what
   degrades — have identified several specific biases.", settled,
   empirical).
source(d_c17_retention_bias, 'Mesoudi', 2006).
source(d_c17_retention_bias, 'Barrett', 2001).
source(d_c17_retention_bias, 'Kalish', 2007).
source(d_c17_retention_bias, 'Bartlett', 1932).
says(d_c17_retention_bias, mildly_counterintuitive_content,
   transmission_retention, higher).
depends_on(d_c17_retention_bias, d_c15_mcount).

claim(d_c17_already_investigated, book, 17, "All three have already been
   investigated, by people who would not describe themselves as working
   on anything in this book.", settled, empirical).
claim(d_c16hs_already_run, 'book-hs', 16, "Then came the check: all three
   had already been run. Cycling really does help prebiotic chemistry
   accumulate. Enclosing things in bags really does help them persist and
   adapt. Cultural transmission really does have measurable biases in
   what survives.", settled, empirical).
source(d_c17_already_investigated, 'Rajamani', 2008).
```

```
source(d_c17_already_investigated, 'Lai', 2021).
source(d_c17_already_investigated, 'Mesoudi', 2006).
source(d_c16hs_already_run, 'Rajamani', 2008).
source(d_c16hs_already_run, 'Lai', 2021).
source(d_c16hs_already_run, 'Mesoudi', 2006).
says(d_c17_already_investigated, framework_checkable_claims,
   already_tested, true).
says(d_c16hs_already_run, framework_checkable_claims, already_tested,
   true).

claim(d_c17_all_held, book, 17, "So the framework's three checkable
   claims were checked, and all three held.", settled, interpretive).
claim(d_c16hs_all_favour, 'book-hs', 16, "All three came out in the
   book's favour. That's a worse result than it sounds, and here's why.",
   settled, interpretive).
says(d_c17_all_held, framework_checkable_claims, count, 3).
says(d_c17_all_held, framework_checkable_claims, held, 3).
says(d_c16hs_all_favour, framework_checkable_claims, held, 3).
computes(d_c17_all_held, 1 + 1 + 1, 3).          % one "survives" verdict
   under each of claims 1, 2 and 3
depends_on(d_c17_all_held, d_c17_cycling).
depends_on(d_c17_all_held, d_c17_compartments).
depends_on(d_c17_all_held, d_c17_retention_bias).
depends_on(d_c16hs_all_favour, d_c16hs_already_run).

claim(d_c17_not_predictions, book, 17, "A prediction reverse-engineered
   from results already in circulation is not a prediction.", settled,
   interpretive).
claim(d_c17_undiminished, book, 17, "Until it does, the objection in the
   previous section stands undiminished, and the three results above are
   better understood as the framework being compatible with the evidence
   than as the framework having earned anything.", settled,
   interpretive).
claim(d_c16hs_objection_stands, 'book-hs', 16, "What would count is a
   prediction this framework makes that a specialist wouldn't expect. I
   don't have one. Until I do, the objection above stands.", settled,
   interpretive).
says(d_c17_undiminished, strongest_objection, status, stands).
says(d_c16hs_objection_stands, strongest_objection, status, stands).
says(d_c17_undiminished, framework_novel_prediction, exists, false).  %
   "The chapter does not have one."
says(d_c16hs_objection_stands, framework_novel_prediction, exists,
   false).
depends_on(d_c17_undiminished, d_c17_not_predictions).
depends_on(d_c17_undiminished, d_c17_all_held).
depends_on(d_c16hs_objection_stands, d_c16hs_all_favour).
```

--- where the account stops

```
claim(d_c17_four_places, book, 17, "Four places where the framework
   identifies genuine uncertainty rather than papering over a gap.",
   settled, interpretive).
says(d_c17_four_places, acknowledged_open_questions, count, 4).
computes(d_c17_four_places, 1 + 1 + 1 + 1, 4).   % heredity,
   consciousness, direction of complexity, collective threshold

claim(d_c17_heredity_gap, book, 17, "How a chemical system first came to
   copy a pattern reliably enough for selection to accumulate across
   generations is not known, and nothing in this book narrows it.",
   settled, empirical).
claim(d_c16hs_copying_biggest, 'book-hs', 16, "How copying started. The
   biggest one. Named several times already, and not narrowed by anything
   in this book.", settled, empirical).
says(d_c17_heredity_gap, heredity_origin, known, false).
says(d_c16hs_copying_biggest, heredity_origin, known, false).
says(d_c17_heredity_gap, acknowledged_open_questions, largest,
   heredity_origin).
says(d_c16hs_copying_biggest, acknowledged_open_questions, largest,
   heredity_origin).
```

“Named twice already”: before this line the HS edition states the gap in ch. 1 (line 45), ch. 8 (line 59), ch. 12 (line 59) and earlier in this chapter (line 15).

```
says(d_c16hs_copying_biggest, hs_prior_statements_of_heredity_gap, count,
   at_least(3)).
computes(d_c16hs_copying_biggest, 1 + 1 + 1 + 1, at_least(3)).    %
   "several"; it said "twice" until 2026-11-03

claim(d_c17_two_statements, book, 17, "The relationship between those two
   statements needs more care than it has been given here.", settled,
   interpretive).
claim(d_c16hs_not_squared, 'book-hs', 16, "This book describes
   organization piling up. It also says evolution has no push towards
   complexity. Both are defensible and I haven't squared them.", settled,
   interpretive).
says(d_c17_two_statements, complexity_direction_tension, resolved,
   false).
says(d_c16hs_not_squared, complexity_direction_tension, resolved, false).

claim(d_c17_two_explanations, book, 17, "There are two explanations for
   an apparent increase.", settled, interpretive).
says(d_c17_two_explanations, complexity_increase_explanations, count,
   2).   % a drive; spread from a floor
```

```
claim(d_c17_passive_favoured, book, 17, "Where the question has been
   examined empirically, the tests have tended to favour the passive
   explanation and have not found evidence of a drive.", settled,
   empirical).
source(d_c17_passive_favoured, 'McShea', 1996).
says(d_c17_passive_favoured, complexity_trend_evidence, favours,
   passive_diffusion).
```

HS ch. 16 states this finding both inside and outside a box. The box (line 55) is a label between paragraphs, written "**Still argued about** —" without the full stop the other boxes use. By book-hs/00 ("each box covers what's inside it") the paragraphs after it are unboxed, i.e. settled. Encoded as presented. The two HS ch. 16 boxes used to sit beside the paragraphs they labelled; since 2026-11-03 the paragraphs are inside them, so what follows the box label carries the box's tier. Restructured on 2026-11-03. The box now holds only the open question. The evidence report is unboxed and settled, as in full ch. 17, and Marcot & McShea 2007 confirms it: phylogenetic tests mostly could not reject equal rates of increase and decrease.

```
claim(d_c16hs_complexity_box, 'book-hs', 16, "Whether anything pushes
   life towards complexity at all.", contested, interpretive).
claim(d_c16hs_passive_favoured, 'book-hs', 16, "Where this has been
   studied properly, the evidence has tended to favour that boring
   explanation over a drive towards complexity", settled, empirical).
source(d_c16hs_passive_favoured, 'McShea', 1996).
source(d_c16hs_passive_favoured, 'Marcot', 2007).
source(d_c17_passive_favoured, 'Marcot', 2007).
says(d_c16hs_passive_favoured, complexity_trend_evidence, favours,
   passive_diffusion).
```

```
claim(d_c17_wall, book, 17, "The honest version is that much of what the
   later chapters describe as accumulation may be a distribution
   spreading from a wall, observed by something that happens to be at the
   far end of it.", settled, interpretive).
depends_on(d_c17_wall, d_c17_passive_favoured).
depends_on(d_c17_wall, c17_gould).
```

The full edition flags this item in italics as "the author's speculation". The HS puts "**My own idea**, and a shaky one." in a box above the paragraph, not around it, so the paragraph is unboxed, i.e. settled by the HS rule. Encoded as presented.

```
claim(d_c17_collective, book, 17, "Whether a population can be said to
   model its own modelling in any sense beyond the metaphorical is
   genuinely unclear", author, interpretive).
```

```
claim(d_c16hs_collective, 'book-hs', 16, "Whether a whole society can be
   said to model its own modelling in any way that isn't merely a
   metaphor is unclear, and people treating it as settled in either
   direction haven't convinced me.", author, interpretive).
says(d_c17_collective, collective_self_modelling, status, unclear).
says(d_c16hs_collective, collective_self_modelling, status, unclear).
```

--- what would count against all of this

"for the reason given in the previous section": the previous section is "Where the account stops"; the reason (both candidates already checked) is given two sections back, in "Three claims, and what happened when they were checked".

```
claim(d_c17_two_struck, book, 17, "Two candidates have to be struck from
   this list, for the reason given in the section on the three tested
   claims.", settled, interpretive).   % pointed at "the previous
   section" until 2026-11-03
says(d_c17_two_struck, struck_falsifier_candidates, count, 2).
computes(d_c17_two_struck, 1 + 1, 2).            % the two italicised
   conditionals that follow
depends_on(d_c17_two_struck, d_c17_all_held).
```

The full sentence "This is the most likely of the three." is 37 characters, too short to cover.

```
claim(d_c17_most_likely, book, 17, "If the origin of heredity is
   explained by chemistry that owes nothing to search, selection, or
   compression, then the vocabulary was decoration over the interesting
   part. This is the most likely of the three.", settled, interpretive).
claim(d_c16hs_most_likely, 'book-hs', 16, "That last one is the most
   likely of the three.", settled, interpretive).
says(d_c17_most_likely, most_likely_falsifier, which,
   heredity_explained_without_framework).
says(d_c16hs_most_likely, most_likely_falsifier, which,
   heredity_explained_without_framework).
says(d_c17_most_likely, remaining_falsifiers, count, 3).
says(d_c16hs_most_likely, remaining_falsifiers, count, 3).
computes(d_c16hs_most_likely, 1 + 1 + 1, 3).     % the cheapest test; a
   single unifying mechanism; copying explained without the framework

claim(d_c17_three_falsifiers, book, 17, "Three falsifiers, one of them
   cheap and unrun.", settled, interpretive).
claim(d_c16hs_cheapest, 'book-hs', 16, "The cheapest test is one nobody
   has run, including me.", settled, interpretive).
says(d_c17_three_falsifiers, remaining_falsifiers, count, 3).
computes(d_c17_three_falsifiers, 1 + 1 + 1, 3).  % the three bullets
   under "What remains"
```

```
says(d_c17_three_falsifiers, cheapest_falsification_test, run, false).
says(d_c16hs_cheapest, cheapest_falsification_test, run, false).
depends_on(d_c17_three_falsifiers, d_c17_two_struck).
```

Chapter 20

```
claim(d_c20_summary, book, 20, "The honest summary of twenty chapters is
   that the recurrence is real and uneven.", settled, interpretive).
claim(d_c16hs_summary, 'book-hs', 16, "The honest summary is that the
   continuity is real and uneven.", settled, interpretive).
says(d_c20_summary, cross_level_continuity, character, real_and_uneven).
says(d_c16hs_summary, cross_level_continuity, character,
   real_and_uneven).
depends_on(d_c20_summary, d_c16_verdict).
depends_on(d_c16hs_summary, d_c16hs_lower_half).
depends_on(d_c16hs_summary, d_c16hs_above_language).
```

Line 23 was rewritten on 2026-11-03 at the author's direction. It used to say that vesicles (ch. 5) were "becoming" replicators (ch. 8), which crosses the heredity gap, and that the two were "the same process", which contradicted ch. 2's table (different rows, "share no machinery"). It now gives the measured coupling as settled, followed by the author's capture reading. That reading rests on the gap, which stays open.

```
claim(d_c20_different_rows, book, 20, "Selection among fatty acid
   vesicles in chapter 5 and selection among early replicators in chapter
   8 sit in different rows of chapter 2's table", settled, interpretive).
depends_on(d_c20_different_rows, a_c02_three_rows).
claim(d_c20_encloses, book, 20, "A vesicle encloses a sample of the water
   it forms in and whatever that water holds, RNA included.", settled,
   empirical).
source(d_c20_encloses, 'Hanczyc', 2003).
claim(d_c20_coupling, book, 20, "RNA inside a vesicle makes it grow at
   the expense of emptier neighbours", settled, empirical).
depends_on(d_c20_coupling, a_c05_osmotic).
claim(d_c20_protocells, book, 20, "ribozymes inside model protocells
   adapt faster under selection than the same sequences outside",
   settled, empirical).
depends_on(d_c20_protocells, d_c17_compartments).
claim(d_c20_capture, book, 20, "On this reading the first cell arose by
   combination, chapter 6's two evolutionary histories merging into one",
   author, interpretive).
depends_on(d_c20_capture, b_c06_combination).
depends_on(d_c20_capture, d_c20_coupling).
rests_on_gap(d_c20_capture, heredity_origin).
```

```
claim(d_c20_rate, book, 20, "Each level of organization in this book
   operates faster than the one beneath it. Geological and cosmic change
   proceeds over billions of years; biological evolution over millions;
   cultural change over thousands; technological change over decades.",
   settled, generalization).
says(d_c20_rate, rate_comparison_categories, count, 4).
computes(d_c20_rate, 1 + 1 + 1 + 1, 4).          % geological and cosmic;
   biological; cultural; technological
depends_on(d_c20_rate, d_c15_two_differences).   % line 35: transmission
   between contemporaries outpaces parent to offspring
claim(d_c20_incommensurable, book, 20, "The comparison is less clean than
   it sounds, because the four categories are not commensurable.",
   settled, interpretive).
says(d_c20_incommensurable, rate_comparison_categories, count, 4).

claim(d_c20_break, book, 20, "There is a break in this account, at
   heredity, and it is not a small one: how a chemical system first came
   to copy a pattern reliably is not known, and this book does not narrow
   it.", settled, empirical).
says(d_c20_break, heredity_origin, known, false).
depends_on(d_c20_break, c16_break).               % "chapters 16 and 17
   both say"
depends_on(d_c20_break, d_c17_heredity_gap).

claim(d_c20_two_senses, book, 20, "The two senses of the word have to be
   kept apart.", settled, interpretive).
claim(d_c16hs_two_meanings, 'book-hs', 16, "But there are two meanings of
   \"break\" and they have to be kept apart.", settled, interpretive).
says(d_c20_two_senses, break_senses, count, 2).
says(d_c16hs_two_meanings, break_senses, count, 2).
```

The third example used to be “a code tolerating error because of the order its parts arrived in”, which is chapter 8’s “neutral origin”, one of three readings that chapter calls open. Since 2026-11-03 it is the model result chapter 8 reports as shown (“Models of this kind produce codes nearly as error-tolerant as the real one”), not a claim about what happened.

```
claim(d_c20_no_sequence_break, book, 20, "The second has not been found
   anywhere, and every transition that has been worked out in detail —
   amphiphiles closing into vesicles, vesicles competing by osmotic
   pressure, a model code gaining its tolerance for error from nothing
   but the order its parts arrived in — turned out to need nothing beyond
   what was already there.", settled, generalization).
```

```
claim(d_c16hs_no_sequence_break, 'book-hs', 16, "A break in the sequence
   would be a place where something showed up that the previous step
   couldn't supply — and in every transition that has been worked out in
   detail, nothing like that has been found.", settled, generalization).
says(d_c20_no_sequence_break, sequence_break, found, false).
says(d_c16hs_no_sequence_break, sequence_break, found, false).
depends_on(d_c20_no_sequence_break, c16_no_new_principle).
```

“What I make of everything” (added 2026-11-06): the author’s own passage, in the author’s words, full edition only. Its lead-in marks the rest of the section as the author’s. The eye figure is the one chapter 13 states and checks, so the claim depends on that one. (A says/4 fact here would be reported as a tier mismatch: the same figure, settled there and the author’s here.)

```
claim(d_c20_rant_lead, book, 20, "What follows is the author's own rant
   about what it all means, and the author wants this read as prose
   rather than a scientific theory.", author, interpretive).
claim(d_c20_every_religion, book, 20, "So in a way, every religion is
   true — they're just incomplete in the way the human eye throws away
   ninety-nine per cent of what it takes in.", author, interpretive).
depends_on(d_c20_every_religion, c_c13_discard).
claim(d_c20_holobiont_guiding, book, 20, "Maybe it's the bacteria /
   holobiont guiding each and every one of us and we're just little
   carbon-based armour for an immensely complex and orchestrated system
   of life, energy, and cooperation.", author, interpretive).
```

Sources checked: origins

From `factcheck/claims/fix_origins.pl`

```
:- discontiguous claim/6, says/4, source/3, depends_on/2, computes/3,
   event_year/3,
                order_claim/3, interval_claim/4, variant_ok/3, gap/3,
   rests_on_gap/2,
                arc_gap/3, arc_descriptive/3, cite_ok/3, about/2,
   drift_ok/3,
                prose_ok/2, common_knowledge/1, certainty_ok/2.
```

Support for nine settled claims that had neither a source nor a premise: the five statements of the heredity gap (book chs. 1, 16, 17; HS chs. 1, 16), the open-endedness limit (book ch. 4), the open prebiotic question (HS ch. 16), the three energy sources (book ch. 3) and the open origin of language (book ch. 16). No tier or kind is changed; only source/3 and depends_on/2.

Checked 2026-11-03. Every work's metadata was checked against Crossref; content was read where each comment says, and quotations are verbatim from those texts. In book/19 already, without a dagger: Szathmáry 2015, Lincoln 2009, Kauffman 1986, Williams 2000. NEW, not yet in book/19, so each raises "not in the reference list" until it is added: Gianni 2026, Ameta 2021, Orgel 2008, Hauser 2014, Kleidon 2012.

Line comments only in this file: SWI-Prolog nests block comments.

The origin of heredity, stated five times

Sources sit where the gap is stated in its own right: book ch. 17's list of where the account stops, its HS twin in HS ch. 16, and HS ch. 1, whose pointer is to HS ch. 8, where nothing is encoded. Where the text points to an encoded claim, the restatement depends on that claim.

Szathmáry 2015, PNAS full text (PMC4547294), section "Phases of Transitions to Cells": "The Origin of the First Hereditary Replicators. This process is still an unsolved

problem. By itself, this transition is not an evolutionary one because, without hereditary replicators, no Darwinian evolution is possible."

```
source(d_c17_heredity_gap, 'Szathmáry', 2015).
```

Gianni et al. 2026 (NEW), Science author manuscript (PMC7618777), first paragraph: "How this capacity for heredity and evolution first emerged is unknown." The paper is the newest advance (QT45, a 45-nucleotide ribozyme that makes itself and its complement), and its own model finds the fidelity reached too low to keep a population of active variants; the gap survives it.

```
source(d_c17_heredity_gap, 'Gianni', 2026).
```

Book ch. 16 says "Chapters 8 and 17 say so"; ch. 8's statement is not encoded, ch. 17's is.

```
depends_on(c16_break, d_c17_heredity_gap).
```

Book ch. 1 previews the gap ch. 17 lists as "The largest one, named already at the end of chapter 8", and ends "Chapter 17 lists the others".

```
depends_on(a_c01_largest_gap, d_c17_heredity_gap).
```

HS ch. 16 is ch. 17's twin (both say acknowledged_open_questions/largest); HS twins carry the full edition's sources, as in d_ch14-20.pl. Szathmáry 2015, as above.

```
source(d_c16hs_copying_biggest, 'Szathmáry', 2015).
```

Gianni et al. 2026 (NEW), as above.

```
source(d_c16hs_copying_biggest, 'Gianni', 2026).
```

HS ch. 1 points to HS ch. 8, which has no encoded claim, so it takes the sources directly. Szathmáry 2015, as above.

```
source(a_c01hs_largest_gap, 'Szathmáry', 2015).
```

Gianni et al. 2026 (NEW), as above.

```
source(a_c01hs_largest_gap, 'Gianni', 2026).
```

Book ch. 4: nothing evolves open-endedly

Lincoln & Joyce 2009, the "pairs of RNA enzymes that make each other" of ch. 4; NIH author manuscript (PMC2652413). Recombinants did take over its population, but from a fixed menu: "only two genetic loci with, at present, only 12 alleles per locus". It ends: "The self-sustained evolution of RNA with open-ended opportunities for discovering novel function likely has not occurred on Earth since the time of the RNA world, and continues to present an intriguing research opportunity."

```
source(c4_open_ended, 'Lincoln', 2009).
```

Szathmáry 2015 (PMC4547294): for autocatalytic networks without templates, calculations show "limited evolvability, provided that reflexively autocatalytic networks are compartmentalized"; and "we still have no general RNA-based replicase that could replicate a great variety of sequences, including copies of its own."

```
source(c4_open_ended, 'Szathmáry', 2015).
```

Ameta et al. 2021 (NEW), a review judging experimental autocatalytic systems for Darwinian evolution; open full text (PMC8066523): "so far, there is no unambiguous experimental demonstration of the generation of such variation"; "The realization of Darwinian evolution in minimal chemical systems … will require intensive investigations".

```
source(c4_open_ended, 'Ameta', 2021).
```

Gianni et al. 2026 (NEW), the state this year: earlier polymerase ribozymes "have so far fallen short of self-replication"; QT45 makes its two strands only in separate reactions, "with further improvements in synthetic efficiency and fidelity likely required to overcome chemical degradation and achieve self-sustained replication".

```
source(c4_open_ended, 'Gianni', 2026).
```

HS ch. 16: was early chemistry rich enough?

One side: Kauffman 1986 (abstract, Europe PMC): such sets "may be an essentially inevitable collective property of any sufficiently complex set of polypeptides".

```
source(d_c16hs_prebiotic_open, 'Kauffman', 1986).
```

The other: Orgel 2008 (NEW), open full text (PMC2211548), answering that paper by name: that a cycle written down on paper "would self-organize spontaneously" is "unlikely because peptide molecules do not have the properties that Kauffman assigns to them"; Ghadiri's networks "illustrate 'intelligent design' of input peptides, not spontaneous self-organization".

```
source(d_c16hs_prebiotic_open, 'Orgel', 2008).
```

Still open: Ameta et al. 2021 (NEW; PMC8066523), on autocatalytic systems that "spontaneously emerge from an abiotic chemical mixture": "Although there are theoretical studies that discuss the conditions for such an emergence, there are no experimental demonstrations so far."

```
source(d_c16hs_prebiotic_open, 'Ameta', 2021).
```

Book ch. 3: three engines

Empirical: the Sun, the interior and the tides are separate, continuing energy supplies. The book's own: counting them as three engines, and "Each pushed in one direction". Kleidon 2012 (NEW), read in arXiv:1103.2014v2, the revised submission of August 2011 (the published PDF was not reachable): the drivers of free-energy generation in the Earth system are "solar heat engines", "solar photochemical engines", "gravitational engines: gravitational forces by the Moon and the Sun … in form of tides" and "interior heat engines: radioactive decay, crystallization of the core, and secular cooling of the interior". Four drivers from the book's three sources. It describes today's Earth, and its interior engine runs on "the depletion of the initial conditions of the planet", so "none of them was going to run out" holds over the chapter's timescales rather than forever.

```
source(a_c03_three_engines, 'Kleidon', 2012).
```

Williams 2000 (abstract, via Crossref): tidal rhythmites record strong tides through the Precambrian, at 620 Ma and 2450 Ma, with the Moon closer than now (relative distance 0.965 and 0.906), so the tidal supply ran on through deep time. Its record starts at 2450 Ma.

```
source(a_c03_three_engines, 'Williams', 2000).
```

Book ch. 16: how language arose is open

Hauser et al. 2014 (NEW), abstract (open access, Europe PMC): "essentially no explanation of how and why our linguistic computations and representations evolved", the evidence "leaving details of origins and selective pressure unresolved" among "the many open hypotheses". A deliberately sceptical survey; the sentence claims much less. Nothing in book/19 covers this: Szathmáry 2015 sketches a staged scenario for language but presents it as a scenario, not as settled.

```
source(d_c16_language_weaker, 'Hauser', 2014).
```

Christiansen & Kirby 2003 (added at the author's request, alongside Hauser), abstract checked in Europe PMC: how language "came to be this way" is among "old unsolved questions", with "major remaining controversies (such as gestural origins of language)".

```
source(d_c16_language_weaker, 'Christiansen', 2003).
```

Added during the merge (2026-11-03)

The full edition's twin of d_c16hs_prebiotic_open, re-encoded as empirical to match it; the same three sources apply.

```
source(c16_prebiotic_open, 'Kauffman', 1986).
source(c16_prebiotic_open, 'Orgel', 2008).
source(c16_prebiotic_open, 'Ameta', 2021).
```

c4_status is contested, so it raised no warning, but the same two works state its open question directly (quotations above).

```
source(c4_status, 'Orgel', 2008).
source(c4_status, 'Ameta', 2021).
```

Sources checked: cells

From `factcheck/claims/fix_cells.pl`

```
:- discontiguous claim/6, says/4, source/3, depends_on/2, computes/3,
   event_year/3,
                order_claim/3, interval_claim/4, variant_ok/3, gap/3,
   rests_on_gap/2,
                arc_gap/3, arc_descriptive/3, cite_ok/3, about/2,
   drift_ok/3,
                prose_ok/2, common_knowledge/1, certainty_ok/2.
```

Support for the warnings on cells, Margulis, the major-transitions critics, skin, the double jaw joint and nucleosomes: book chs. 6, 7, 9 and 10 and their HS twins. Only source/3 is added. No tier, kind or quote is changed, and every fact attaches to a claim defined in another file.

Checked 2026-11-03. Every new work's metadata was checked against Crossref. Content was read where each comment says, and quotations are verbatim from those texts. Already in book/19 without a dagger: Sagan 1967, Luger 1997. NEW, not yet in book/19, so each raises "not in the reference list" until it is added: Anthwal 2013, Eme 2011, Harper 2006, McFall-Ngai 2013, McShea 2011, Schwartz 2019, Sender 2016.

```
Not cleared here, deliberately:
- The dagger warnings (O'Malley 2016, Margulis 1995, Luo 2011) clear only
    when the dagger is
  removed in book/19. The evidence for each decision is in the comments
    below.
- Margulis 1995 does not support "since 1964" (b_c06_sagan, c6_margulis).
    Her account says
  "In 1966, I wrote a paper", and the JTB paper is marked "(Received 8
    June 1966)". No source
  for 1964 was found anywhere, so a text correction is proposed instead
    of a source.
- Two chapter 6 sentences are sourced here for the reading the evidence
    supports, and need a
  text correction to match it. The billion-year gap holds for animals,
    not for multicellularity
```

```
  in general. And most of HS's thirty trillion cells have no nucleus, so
   "Every one of them"
  cannot carry division machinery.
```

Line comments only in this file: SWI-Prolog nests block comments.

Chapter 6

The machinery's age. Eme et al. 2011, abstract (Europe PMC, PMID 21943402): "Taken together our analyses indicated that LECA had a complex and highly controlled modern-like cell cycle."

```
source(b_c06_billion_years, 'Eme', 2011).
```

The gap. McFall-Ngai et al. 2013, full text (PMC3587249), "Bacteria and the Origin of Animals": "Animals diverged from their protistan ancestors 700–800 Mya, some 3 billion years after bacterial life originated and as much as 1 billion years after the first appearance of eukaryotic cells". Cross-check: LECA at 1866–1679 Ma (Parfrey et al. 2011, abstract) to the Cryogenian, 850–635 Ma (Erwin et al. 2011, abstract), is 829–1231 Myr. About a billion years, but for ANIMALS. Aktipis 2015 (book/19) has red algae reaching complex multicellularity "around 1.2 billion years ago", only a few hundred million years after LECA.

```
source(b_c06_billion_years, 'McFall-Ngai', 2013).
```

HS twin of the above: same work, same caveat.

```
source(b_c06hs_billion_years, 'Eme', 2011).
```

HS twin of the above: same work, same caveat.

```
source(b_c06hs_billion_years, 'McFall-Ngai', 2013).
```

Sender, Fuchs & Milo 2016, full text (PMC4991899): "revise past estimates to 3.0·10^13 human cells" for the 70 kg reference man. It also finds "almost 90% of the cells are estimated to be enucleated cells (26·10^12 cells), mostly red blood cells and platelets", which is the reason for the correction to the HS sentence that follows this one.

```
source(b_c06hs_cells, 'Sender', 2016).
```

McShea & Simpson 2011, the critique itself, read in full (PDF on co-author C. Simpson's site, pp. 19–34): "In sum, we cannot find any theoretical unity in the

Maynard Smith and Szathmáry list of major transitions"; "a series of miscellaneous transitions with no obvious unity beyond their salience for us". O'Malley & Powell 2016 (dagger) stays attached in b_ch06-09.pl. Its abstract (Springer) says MTE "has been criticized for inconsistently combining different types of event", and its reference list includes McShea & Simpson 2011.

```
source(b_c06_critics, 'McShea', 2011).
```

For "in 1967": Sagan 1967 itself, read in a PDF reproduction of the paper: "J. Theoret. Biol. (1967) 14, 225-274", byline "LYNN SAGAN", "(Received 8 June 1966)". "Around fifteen" comes from Margulis 1995 (edge.org reproduction of The Third Culture, ch. 7): "The paper was rejected by about fifteen scientific journals ... Finally, James F. Danielli, the editor of The Journal of Theoretical Biology, accepted it". Nothing in this HS sentence depends on the 1964 date.

```
source(b_c06hs_rejected, 'Sagan', 1967).
```

Chapter 7

"That environment" is a world whose surfaces bacteria had occupied for billions of years. McFall-Ngai et al. 2013 (PMC3587249): animals appeared "some 3 billion years after bacterial life originated"; animal development "both originated and evolved in a microbe-rich environment"; "Since the dawn of metazoan evolution, the ecology of animals has depended on bacterial communities." All skin is younger than the first animals, so it evolved into an occupied world and was not colonized after the fact.

```
source(b_c07_into_occupied, 'McFall-Ngai', 2013).
```

Chapter 9

Anthwal, Joshi & Tucker 2013, full text (PMC3552421): "the first step in the process of formation of a mammalian-like ear and jaw was the development of a double jaw joint, that is, two side by side joints, one between the articular and the quadrate and the other between the dentary and the squamosal"; "A clear double jaw joint is evident in the prototypical basal mammaliform Morganucodon". A verified alternative to Luo 2011 (dagger), whose text could not be reached: Annual Reviews blocked access, and its abstract does not mention the double joint.

```
source(b_c09_two_joints, 'Anthwal', 2013).
```

HS twin (“the old one, and a new one right next to it made from different bones”): same work.

```
source(b_c09hs_two_joints, 'Anthwal', 2013).
```

Chapter 10

Schwartz et al. 2019, full text (PMC6379673), introduction: “the human genome is covered by ~30 million nucleosome cores, which are separated by DNA linkers whose length is cell-type specific and ranges between 20 and 75 bp”. Arithmetic: the diploid genome is 6.27 Gbp (male) or 6.37 Gbp (female) (Piovesan et al. 2019, abstract). Divided by a 147 bp core plus a 20–75 bp linker, that is 28–38 million, and 31–34 million at a typical 185–200 bp repeat.

```
source(c10hs_nucleosomes, 'Schwartz', 2019).
```

The full edition’s twin. Its only source, Luger 1997, gives 146 bp in its abstract but no per-cell count, so its “about thirty million” was unsupported too.

```
source(c10_nucleosomes, 'Schwartz', 2019).
```

“46 very long strands”: Harper 2006, abstract (Europe PMC, PMID 16463025): “The correct determination of the human diploid chromosome number as 46, by J-H Tjio and A Levan”.

```
source(c_c10hs_46_strands, 'Harper', 2006).
```

“about thirty million in each cell”: Schwartz et al. 2019, as above.

```
source(c_c10hs_46_strands, 'Schwartz', 2019).
```

“wound around tiny protein spools”: Luger 1997, abstract: “how the histone protein octamer is assembled and how 146 base pairs of DNA are organized into a superhelix around it”.

```
source(c_c10hs_46_strands, 'Luger', 1997).
```

Sources checked: spirals

From `factcheck/claims/fix_spirals.pl`

```
:- discontiguous claim/6, says/4, source/3, depends_on/2, computes/3,
   event_year/3,
                order_claim/3, interval_claim/4, variant_ok/3, gap/3,
   rests_on_gap/2,
                arc_gap/3, arc_descriptive/3, cite_ok/3, about/2,
   drift_ok/3,
                prose_ok/2, common_knowledge/1, certainty_ok/2.
```

Support for the [unsourced] and [unchecked_source] warnings on chapter 10 and 11 claims defined in claims.pl and claims/c_ch10-13.pl. Schema: the header of factcheck/claims.pl. Checked 2026-11-03. Every reference marked NEW below is not yet in book/19, so the checker reports it as "not in the reference list" until the entry proposed for it is added there. Several of these sources support a corrected sentence rather than the current one. Each comment says so, and the corrections are proposed separately. They are not settled here. The [unchecked_source] warnings (O'Kelly 1982, Ortiz 1947, ICOMOS 2006, Whitney 1986) clear only when a dagger is removed from book/19. Facts here cannot clear them.

c10_newgrange ("built around 3100 BCE"). NEW: Smyth et al. 2025, Antiquity 99(405), open access, read in the published version (White Rose eprint 228326). p. 672: radiocarbon dates from the 1960s excavations "demonstrated its construction and primary use in the later Neolithic, c. 3200-3000 cal BC (O'Kelly 1982; Lynch et al. 2014)". 3100 BCE is the midpoint. Guiry et al. 2025 (Proc. Prehist. Soc. 91) give c. 3300-3000 cal BC. O'Kelly 1982 itself was not reachable, so its dagger stays.

```
source(c10_newgrange, 'Smyth', 2025).
```

c_c10_newgrange_two_kinds ("Its best-known stones set the two kinds side by side"). NEW: the Discovery Programme's published descriptions of its 3D models (2014), read via the Sketchfab API. Entrance stone K1: "a series of clockwise and anti-clockwise spirals … There are also lozenges to the left side of the stone." Kerbstone 52, "at

the rear of the mound directly opposite kerbstone 1”: “The lefthand side is decorated with a double spiral, with 2 smaller spirals to its right. Below these a series of lozenges picked in relief.” This is an institutional record, not peer-reviewed. O’Kelly 1982 remains the better source once read, and if that happens this line can go.

```
source(c_c10_newgrange_two_kinds, 'Discovery Programme', 2014).
```

c10_ortiz: no new fact. The existing source is right. Ortiz 1947 was read in a scan of the FCE edition (Internet Archive), chapter “El enigma de unas figuras indocubanas”. p. 32: the S-shaped arms give “la idea inequivoca de una actitud rotatoria”, turning counterclockwise, and are “como dos espirales conjugadas en un solo foco comun”. p. 35: the storm can be figured “por una simple linea espiral”. p. 40: “Interpretacion meteorologica”, the figure read as the hurricane. p. 45: hurricanes north of the equator are “siempre levogiro”. This supports removing the dagger from Ortiz 1947.

c10_plazuelas: no new fact. ICOMOS 2006 was read in full (ICOMOS open archive PDF, Mexico section by Murray & Viramontes, p. 5). It gives “more than 1400 petroglyphs” and dates the centre, not the carvings, to 750-1200 A.D. It describes the petroglyphs as “principally spirals, dot figures and scale models of architectural features”, not predominantly spirals. INAH dates the city’s height to 450-700 CE. The sentence needs correcting before the dagger goes.

c10_rosse. NEW: Whitney 1992, a letter in Physics Today 45(8): 13, by the astronomer himself, read in the AIP-hosted scan. He and Boime “independently concluded” that the sky was observed. Flammarion “had reproduced the drawing of the Whirlpool nebula by Lord Rosse”, and “Van Gogh’s painting is remarkably similar to Lord Rosse’s drawing”. The letter does not give the year 1845 or mention the mistral. Whitney 1986 itself was not reachable, so its dagger stays.

```
source(c10_rosse, 'Whitney', 1992).
```

Added during the merge: Nasim 2010, British Journal for the History of Science 43(3), abstract via Crossref: M51 is followed “beginning with its discovery and first sketch made by the third Earl of Rosse in 1845”. This supplies the date that Whitney’s letter leaves out.

```
source(c10_rosse, 'Nasim', 2010).
```

c10_rope and c10hs_rope. NEW: Glory 2008, the excavator's own account, published posthumously (CNRS Editions, open access on OpenEdition), chapter "Lascaux", paragraphs 273-279. The cord was about 30 cm long and 7-8 mm thick, and "semblait avoir ete faconnee avec trois brins torsades vers la droite". It lay in contact with the reindeer-bone palaeosol, which paragraphs 299-300 date to GrN-1632, 17,190 +/- 135 BP ("environ 17 000 ans"). NEW: Ducasse & Langlais 2019, Paleo 30, open access. That date is "21-20.5 cal ka BP", and their new AMS dates on the occupation centre on 21.5-21 cal ka BP. Together these support the HS "at least 17,000 years" as it stands. They contradict the full edition's "two-ply" (Glory gives three strands) and its "about seventeen thousand years old", which is radiocarbon years read as calendar years. A correction is proposed.

```
source(c10_rope, 'Glory', 2008).
source(c10_rope, 'Ducasse', 2019).
source(c10hs_rope, 'Glory', 2008).
source(c10hs_rope, 'Ducasse', 2019).
```

c_c10_five_least (a spiral wave that needs no material rotation). NEW: Winfree 1972, Science 175: 634-636, abstract read on Crossref and Europe PMC: the reagent "propagates waves of chemical activity", and "If a wave is broken, then spiral waves ... appear, persist". Prigogine 1978, already in book/19, mentions chemical waves but not spirals, so it cannot carry this. "The one that" reads as unique, but by the chapter's own account growth spirals and phyllotaxis need no rotating material either. What is unique is a turning spiral with no material rotation. An optional precision is proposed.

```
source(c_c10_five_least, 'Winfree', 1972).
```

c_c10_tendril ("its two sides elongate at different rates"). NEW: Gerbode et al. 2012, Science, abstract (PubMed 22936777): coiling "occurs via asymmetric contraction of an internal fiber ribbon of specialized cells", which is contraction, not elongation. NEW: Dilly, Derr & Zanchi 2026, PNAS, abstract (Europe PMC; full text not reachable): "curvature arises from differential growth". The two disagree on the mechanism. Both support a neutral wording, that the two sides change length by different amounts, and a correction to that is proposed.

```
source(c_c10_tendril, 'Gerbode', 2012).
```

```
source(c_c10_tendril, 'Dilly', 2026).
```

c_c10hs_fixed_amount (a snail grows at the opening by a fixed proportion). Thompson 1917, already in book/19, which is the source of the full-edition twin c_c10_fixed_ratio. Read in the 1917 Cambridge edition (Internet Archive scan), chapter XI. p. 496: the snail-shell “has been conformed by successive and continuous increments”. p. 508: the shell grows “by growing at one end only”, and “this property of continued similarity is found only in the logarithmic spiral”. p. 509: the gnomon, an addition that “leaves the resultant figure similar to the original”.

```
source(c_c10hs_fixed_amount, 'Thompson', 1917).
```

c_c11_three_cyclic (at least three cyclic answers). All three are in book/19, none daggered. Ashtekar, Pawlowski & Singh 2006, abstract (arXiv gr-qc/0602086): “the big bang is replaced by a big bounce”. Smolin 1992, abstract (IOP): “all final singularities ‘bounce’ or tunnel to initial singularities of new universes”. Penrose 2010: conformal cyclic cosmology’s sequence of aeons, as chapter 11 states it; this book was not re-read here.

```
source(c_c11_three_cyclic, 'Ashtekar', 2006).
source(c_c11_three_cyclic, 'Penrose', 2010).
source(c_c11_three_cyclic, 'Smolin', 1992).
```

Sources checked: culture

From `factcheck/claims/fix_culture.pl`

```
:- discontiguous claim/6, says/4, source/3, depends_on/2, computes/3,
   event_year/3,
                order_claim/3, interval_claim/4, variant_ok/3, gap/3,
   rests_on_gap/2,
                arc_gap/3, arc_descriptive/3, cite_ok/3, about/2,
   drift_ok/3,
                prose_ok/2, common_knowledge/1, certainty_ok/2.
```

Support for the [unsourced] and [unchecked_source] warnings on chapters 13-15 of both editions: cognition (13), language (14) and culture (15). Schema: the header of factcheck/claims.pl. The claims are defined in claims.pl and the other files under claims/; this file only adds support to them, which works because checks.pl declares the predicates multifile. (SWI-Prolog nests block comments, so a glob written with a slash and an asterisk inside one of these comments swallows the rest of the file.)

Checked 2026-11-03, each against the work itself (abstract, full text, or the text of the original printing) unless a comment says otherwise.

An [unchecked_source] warning clears only when the † comes off the reference in book/19, or when the source fact is replaced. Seven references cited below are not in book/19 yet: Salvi 2016, Danek 2017, Laukkonen 2020, Everett 2008, Jackson 1997, Green 2010 and Cagnato 2024. Until they are added, each raises “source not in refs”.

Three of the sentences supported here are wrong or overstated as the text now stands. The sources below support the corrected wording, and the corrections must go in with them: c_c13_no_signal (and the sentence after it), c14_number/c14hs_number (“two quantity words”), and c15_nixta_direct (“chemical”; the century).

Chapter 13

c_c13_no_signal: “The capacity has no internal signal distinguishing the two cases.” The experimental work on the feeling of insight finds a signal that is real but unreli-

able, so it supports “no reliable internal signal”. It contradicts the next sentence, that the feeling of sudden fit is “if anything stronger for the false ones”. All three abstracts were read in Europe PMC or PubMed. Salvi et al. 2016 (PMC5035115), four experiments: “participants’ self-judged insight solutions were, on average, more accurate than their analytic ones”.

```
source(c_c13_no_signal, 'Salvi', 2016).
```

Danek & Wiley 2017 (PMC5247466), 70 people solving magic tricks: “false insights exist”, but correct solutions led to “stronger Aha! experiences”. “Strong Aha! experiences are clearly, but not exclusively linked to correct solutions.”

```
source(c_c13_no_signal, 'Danek', 2017).
```

Laukkonen et al. 2020, preregistered, n = 300: statements shown with a solved anagram got “higher truth ratings … even if the facts were false”, most of all after an Aha. Feelings of insight “can be overgeneralized and bias how true an idea or fact appears”.

```
source(c_c13_no_signal, 'Laukkonen', 2020).
```

Chapter 14

c14_hanunoo (claims.pl) already rests on Conklin 1955. Read in the OCR text of the original issue, Southwestern Journal of Anthropology 11(4), on the Internet Archive (item sim_journal-of-anthropological-research_winter-1955_11_4). p. 342: “there is an opposition between dryness or desiccation and wetness or freshness (succulence) in visible components of the natural environment which are reflected in the terms rara? and latuy respectively”; “A shiny, wet, brown-colored section of newly-cut bamboo is malatuy (not marara?).” The same page gives malatuy as “light green, and mixtures of green, yellow, and light brown”. The sentence is supported; the † can come off.

c14hs_number, and its full-edition twin c14_number (which already rests on Frank 2008). Everett 2008, read by full-text search of the 2008 Pantheon edition on the Internet Archive (item dontsleeptherear00ever; page not visible): “At first I thought that the Pirahãs had the numbers one, two, and ‘many’“, and later “what I and previous workers thought were numbers were only relative quantities”. So the glosses were first his, and “the same researcher” holds.

```
source(c14hs_number, 'Everett', 2008).
```

```
source(c14_number, 'Everett', 2008).
```

Frank et al. 2008, pp. 819-820 (PDF from the first author's site): Gordon 2004 glossed the words "one" (hói), "two" (hoí) and "many" (baagiso); in the elicitation every answer was "one of the three words hói, hoí, and baágiso", and in the decreasing order "hói was used to refer to quantities as large as six".

```
source(c14hs_number, 'Frank', 2008).
```

Everett 2005, read in the author's manuscript posted on his site (section "No numerals", examples 9-11, and Appendix Three): "There are three words in Pirahã that are easy to confuse with numerals": hói 'small size or amount', hoí 'somewhat larger size or amount', baágiso 'cause to come together (loosely "many")'. That supports the reanalysis. Together with Frank, it shows the language has three such words, not two, in both editions.

```
source(c14hs_number, 'Everett', 2005).
source(c14_number, 'Everett', 2005).
```

d_c14hs_desert_terms, and its full-edition twin d_c14_desert_terms, which until now rested only on c14_seaice and so had nothing behind the desert terms. Jackson 1997, the AGI Glossary of Geology (4th ed.), read by full-text search on the Internet Archive (item glossaryofgeolog0000unse_k9a5; the 1973, 1980 and 1987 editions agree). It has entries for erg ("a sand sea"), hammada/hamada, sabkha and wadi, which shows that geologists use the words. It gives sabkha "Etymol: Arabic. Also spelled: sabkhah; sebkha", wadi "Etymol: Arabic. Variant plurals: wadis ...", and hammada "Etymol: Arabic, hammadah" (1987 edition). For erg it says "Etymol: Hamitic. Pl: areg; ergs", though its own entry for areg reads "Arabic plural of erg". The American Heritage Dictionary (ahdictionary.com) gives erg as "[Colloquial Arabic (Maghrebi dialect) ʿarq, ʿarg, from Arabic ʿirq ...]", so "All four are Arabic" stands on the usual etymology.

```
source(d_c14hs_desert_terms, 'Jackson', 1997).
source(d_c14_desert_terms, 'Jackson', 1997).
```

Chapter 15

c15_nixta_equipment, d_c15_two_kinds, d_c15_thousands now rest on Staller & Carrasco 2010 (†), a whole edited volume. The support is in one chapter of it, Green 2010, pp. 315-343, note 3, which Springer's chapter page shows: "The culinary equipment

for the nixtamalization process was found archaeologically, suggesting its use in south coastal Guatemala between 1500 bc and 1200 bc." The note cites nothing for that sentence. The same note repeats the bound-niacin account that chapter 15 says is wrong. The wording follows Coe 1994 (America's First Cuisines, which Green cites for neighbouring points): "typical household equipment for making nixtamal out of maize is known on the south coast of Guatemala at dates between 1500 and 1200 b.c." Neither names the equipment or the excavation, which fits the text's "suggestive". Cite the chapter, not the volume, and delete the three source(_, 'Staller', 2010) facts in claims.pl and d_ch14-20.pl.

```
source(c15_nixta_equipment, 'Green', 2010).
source(d_c15_two_kinds, 'Green', 2010).
source(d_c15_thousands, 'Green', 2010).
```

"More than a thousand years" rests on the direct evidence, as the comment on the claim in d_ch14-20.pl says; c15_nixta_direct carries Santini 2022.

```
depends_on(d_c15_thousands, c15_nixta_direct).
```

c15_nixta_direct (claims.pl) rests on Santini et al. 2022. Its abstract (SSRN preprint via Crossref; the published abstract via AGRIS is identical) describes microbotanical remains, "maize starch spherulites, a unique byproduct of nixtamalization", from two chultuns "in a Late and Terminal Classic period residential group at the site of San Bartolo". So the identification is microscopic, not chemical. The lead author dated the deposits as "probably the eighth or ninth century CE" (Boston University release, Futurity). At merge (2026-11-03) the text was made to say what this first report shows: one Maya site, San Bartolo, Late and Terminal Classic, around the eighth and ninth centuries CE. Cagnato et al. 2024 (La Corona) was not added. No abstract could be reached, and support resting on a title alone falls short of the reference list's rule.

c15_pellagra and c15hs_pellagra (claims.pl) already rest on Bollet 1992. Read in the full text, PMC2589605. p. 211: "The epidemic of pellagra, which caused over 3 million cases and 100,000 deaths in the U.S."; p. 213: "the epidemic of pellagra, which lasted from about 1906 to 1940, resulted in roughly 3 million cases, with 100,000 deaths in the reporting states". p. 212: the reporting area left out Alabama, Arkansas, Texas and West Virginia, "and the true extent of the epidemic always was greatly underestimated". The years and the cases are as the text gives them. "Over a hundred thousand"

deaths is Bollet's 100,000 for the reporting states, which he says is an undercount. Supported; the † can come off.

About the author

Freddi Van Gogh grew up moving between England, Germany, and the United States, often left to themselves, out in nature or in an empty Oxbridge library, with a great deal of curiosity and a habit of getting lost in books.

Freddi lives a mostly private life, doing science by day and writing fantasy by night. They are not related to “the famous Van Goghs”, but have been taken for a lost cousin often enough that they stopped correcting people.

Freddi can be reached at freddi.van.gogh@proton.me, but values their privacy, and so writes under a changed first name.

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